

# Intangibles Investment and Asset Quality<sup>\*</sup>

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## Abstract

Treating expenditures that create intangible assets as investments yields a profitability measure that more accurately reflects asset quality. Firms with higher intangibles-adjusted profitability generate higher cash flows over longer periods, and their stock returns exhibit lower downside risk and reduced co-tail exposure to market shocks. A long-short factor portfolio constructed from this measure delivers a 2.9% annualized Fama-French alpha, driven by superior stock selection on the long side and effective market timing on the short side. Incorporating this asset quality factor improves the explanatory power of standard linear asset pricing models. The short side potentially captures mispricing from investor overoptimism.

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# 1 Introduction

The investment literature has long emphasized the importance of asset quality in guiding capital allocation decisions. This notion is also central to Warren Buffett’s investment philosophy, as reflected throughout his shareholder letters. A study by Frazzini et al. (2018) confirms that Buffett’s superior performance is largely attributable to his consistent exposure to “cheap, safe, high-quality stocks.” While there is no universal definition of quality, in either practice or academic research, it is often proxied by a combination of high profitability and low financial leverage, which together signal strong and stable operating fundamentals with reduced downside risk.

Recent empirical studies have formalized this intuition by identifying profitability as a distinct, priced factor in asset returns (Hou et al., 2015; Fama and French, 2015; Asness et al., 2019). However, conventional accounting-based profitability measures often treat expenditures on intangible assets—such as R&D, brand development, and organizational capital—as immediate expenses. This treatment may understate the economic profitability of firms that invest heavily in innovation, thereby obscuring important dimensions of asset quality.

We address this gap by creating an adjusted profitability measure that more accurately reflects asset quality and enhances its explanatory power for stock returns. We find that firms with higher adjusted profitability generate stronger and more persistent cash flows over longer horizons. Furthermore, their stock returns exhibit lower downside risk. A long-short factor portfolio constructed using this adjusted measure generates returns consistent with the quality premium documented in the literature, showing improved robustness in intangibles-intensive sectors. In this sense our approach aligns profitability measurement more closely with a firm’s underlying quality of economic performance, particularly in sectors where intangible investment is significant. While others have argued for capitalizing intangible expenditures (Lev and Sougiannis, 1996; Greenwald et al., 2001; Eisfeldt and Papanikolaou, 2013; Lev and Anup, 2022; Eisfeldt et al., 2022; Gulen et al., 2025), we believe we are the first to explicitly connect this adjustment to the importance of profitability as a quality factor in linear factor models of stock returns. Furthermore, we only adjust earnings (the numerator) and not book values (the

denominator) while computing intangibles investment-adjusted measure of profitability, which results in a better measure of profitability as a proxy for quality.

Fama and French (1993) and Fama and French (2015) construct three-factor (FF3) and five-factor (FF5) models, respectively, by using the present value relation between asset valuation and future payouts (the dividend discount model) to identify characteristics plausibly conditionally related to cross-sectional variation in expected returns. The FF5 model augments their original three-factor model in Fama and French (1993) (FF3) with factors based on profitability (robust-minus-weak (RMW)) and asset growth (conservative-minus-aggressive (CMA)).

The very name of the RMW factor suggests that it is a metric for firm quality: robust firms are of higher quality than weak firms. They perform better in downturns than weak firms.<sup>1</sup> In fact, a wide array of earnings/revenue-based factors have negative market betas and can be thought of as quality metrics (e.g., Novy-Marx (2013), Ball et al. (2015), and Gulen et al. (2025)). There are other quality factors that incorporate profitability as well as other metrics Asness et al. (2019) and other non-factor measures of quality (Piotroski (2000), Monahanram (2005)).

We show that the earnings measure in the RMW factor works better than an array of GAAP-based earnings measures in predicting the cross-section of future operating profits up to a ten-year horizon, which is consistent with its ability to predict higher returns.

We then propose a modification of the FF5 profitability factor which takes into account the fact that profitability based on U.S. generally accepted accounting principles (GAAP) includes currently expensed costs that should rightly be considered investments in intangible assets (such as research and development (R&D) and portions of sales general and Administrative (SG&A) expenses used to build a brand or organizational capital). GAAP profitability reflects investments in physical capital by capitalizing and depreciating those expenditures over time. However, GAAP immediately expenses R&D and SG&A expenditures. Investments in intellectual capital, brand, and organizational capital are, therefore, treated very differently from investments in physical capital, under GAAP. Fama and French use EBTDA divided by book

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<sup>1</sup>Jagannathan and Zhang (2020) propose a method for separating higher quality stocks from lower quality stocks based on performance during the month with the worst market return in a calendar year.

equity (BE) as the measure of profitability. Our modified profitability measure leads to better performance in predicting future operating profitability.

The correction we propose modifies only the numerator. When there are intangible assets that do not appear in the balance sheet of a firm, our denominator BE will not include intangible assets created in-house through R&D and SG&A etc. We intentionally do not modify the denominator. That is because of our view that the ability of firms to create intangible assets through in-house expenditures varies significantly across firms. Hence, modifying the denominator BE by adding in-house created intangible assets would add substantial measurement and specification errors. We find that using BE plus model-based estimates of in-house created intangible assets adversely affects the factor performance relative to modifying only the numerator.

We find that the intangible-adjusted profitability factor has positive annualized alpha of 3% ( $t$ -statistic=6.18) against the FF5 model when our factor portfolios are rebalanced annually. Substituting the original FF profitability with the intangibles-adjusted profitability factor leads to a model that substantially reduces some of well-known asset pricing anomalies, such as momentum (Jegadeesh and Titman (1993)), gross profitability (Novy-Marx (2013)), and monthly-rebalanced ROE (Hou et al. (2015)).

The performance of our factor comes from both the long and short sides of the factor portfolio, but the nature of the alphas are quite different. The alpha long side of the strategy comes from positive security selection alphas while the alpha on the short side comes from market timing, that is, from shorting stocks that do particularly poorly in down markets.

While the exposure to our profitability factor offers improved explanatory power for the cross-section of stock returns, the question of whether this factor reflects rational or irrational pricing remains difficult to answer definitively.<sup>2</sup> Nevertheless, we present evidence suggesting that the short leg of the long-short intangible-adjusted profitability factor is likely influenced by biased investor beliefs, implying that mispricing plays a role in explaining the low return on the portfolio of low profitability stocks.

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<sup>2</sup>Fama and French (2006) provide a compelling argument for why this is the case.

## 2 Characteristics Sorted Portfolios As Factors

Fama and French (2006, 2015) use the dividend discount model to relate a stock's firm characteristics like market-to-book equity ratio, profitability, and asset growth to its (long-run) expected return - also referred to as the internal rate of return, discount rate, stock yield (Jagannathan et al. (2001)), and implied cost of capital (Gebhardt et al. (2001)). They then sort stocks based on firm characteristics and construct long-short portfolios of stocks, which they use as factors in their linear factor-beta pricing model. In this section, we argue that characteristics sorted portfolios are better proxies for unobserved latent economy-wide pervasive factors than factors estimated using statistical methods when factor betas change over time.

### 2.1 The Present Value Relation

The present value model equates the market price of a firm's share to the present value of the expected stream of future dividends per share:

$$P_t = \sum_{\tau=1}^{\infty} \frac{\mathbb{E}(D_{t+\tau})}{(1+r)^\tau}, \quad (1)$$

where  $P_t$  is the share price,  $\mathbb{E}(D_{t+\tau})$  is the expected dividend at time  $t + \tau$ , and  $r$  is the implied cost of capital (ICC). The dividend at any future time  $t + \tau$  can be written as the earnings minus retained earnings:

$$D_{t+\tau} = Y_{t+\tau} - (Y_{t+\tau} - D_{t+\tau})$$

where the  $Y_{t+\tau}$  is the earnings. When  $Y_{t+\tau}$  is after-tax earnings, "clean surplus accounting" implies that the net change of book equity is equal to the retained earnings:

$$dB_{t+\tau} = Y_{t+\tau} - D_{t+\tau} \quad (2)$$

where  $dB_{t+\tau} \equiv B_{t+\tau} - B_{t+\tau-1}$  is the net change of book equity in year  $t + \tau$ .

Replacing  $D_{t+\tau}$  with  $Y_{t+\tau} - dB_{t+\tau}$  and dividing both sides of (1) by the current book equity  $B_t$ , the dividend discount model becomes

$$\frac{P_t}{B_t} = \sum_{\tau=1}^{\infty} \mathbb{E} \left( \frac{Y_{t+\tau}}{B_t} - \frac{dB_{t+\tau}}{B_t} \right) / (1+r)^\tau. \quad (3)$$

There are several variables in the above equation in addition to expected return: the market price,  $P_t$ , the book value,  $B_t$ , (and the market-to-book ratio  $P_t/B_t$ ), the stream of expected profitability  $\mathbb{E}(Y_{t+\tau})/B_t$ , and the stream of expected equity growths  $\mathbb{E}(dB_{t+\tau})/B_t$ , where  $\tau = 1, \dots, \infty$ . While the present value relation is a mathematical identity, it takes on economic content when the expectations in equation (3) represent the expectations of the investor determining the market price of the share at the margin.

Fama and French (2015) observe that, according to the present value relation, *ceteris paribus*, firms with 1) higher market-to-book ratios should have lower expected returns; 2) higher expected profitability should have higher expected returns; 3) higher expected growth in book value should have lower expected returns; and 4) higher market value (holding book value constant) should have lower expected returns. Fama and French (2015) use this observation to motivate their use of the returns on characteristics sorted portfolios as proxies for economy-wide pervasive factors. The *ceteris paribus* condition is not innocuous since  $P_t$  and  $P_t/B_t$  are endogenous. In particular, the relations between  $r$  and the variables in (3) need not hold unconditionally.

In a simulation analysis, we provide further evidence supporting the use of long-short portfolio returns, sorted by firm characteristics, as proxies for pervasive economy-wide factors. We consider a model economy where the expected growth rate and one-period expected returns are constant over time but vary across firms, consistent with the Gordon Growth Model. Expected excess returns are assumed to be driven by four latent factors, and the expected excess return of a stock is a linear function of its factor betas, implying that a linear four-beta model explains the cross-section of expected excess returns. Additionally, we assume that investors rationally price securities. We find that factors constructed by sorting stocks based on firm characteristics – such as book-to-market, ROE, and asset growth – along with the equally weighted market index factor, proxy well for the four unobserved factors. When factor betas are constant over time, factors derived using the asymptotic principal component method proposed by Connor and Korajczyk (1986) provide better proxies than the characteristic-sorted factor portfolios. However, when factor betas vary over time, as discussed in Kim et al. (2021), long-short

characteristics-sorted portfolios exhibit superior performance as proxies for the latent factors. A detailed description of the simulation setup and results are available in Section B of the Internet Appendix.

Any set of variables that help explain the cross-section of expected profitability and expected asset growth would be candidates for constructing characteristics-sorted portfolios. These include lagged values of profitability and asset growth, which form the basis of some FF5 factors. The factors in FF5 model are the excess return on the market portfolio (clearly motivated by the Capital Asset Pricing Model (CAPM)) and four long-short portfolios, all of which can be motivated by the present value relation, (3). SMB (Small Minus Big) is a portfolio that is long small capitalization stocks and short large capitalization stocks. HML (High Minus Low) is a portfolio that is long high book-to-market stocks and short low book-to-market stocks. RMW (Robust Minus Weak) is a portfolio that is long high expected profitability stocks and short low expected profitability stocks. Finally, CMA (Conservative Minus Aggressive) is a portfolio that is long low expected asset growth stocks and short high expected asset growth stocks. Holding  $B_t$ ,  $\mathbb{E}(Y_{t+\tau})/B_t$ , and  $\mathbb{E}(dB_{t+\tau})/B_t$  constant the present value relation suggests that SMB should have a positive expected return, as argued by Berk (1995). By similar logic, HML, RMW, and CMA should have positive average returns.

## 2.2 Profitability and Future Cash Flows

Our focus is on the profitability factor. Fama and French (2006) use net income divided by the book equity (which we call bottom-line earnings hereafter) as a proxy for expected profitability. They find a significant relation between future returns and current profitability. However, they find no significant relation between expected returns and expected profitability measured by fitted values from cross-sectional regression of future profitability on current profitability and additional variables. They offer two possible explanations for the puzzling results: (1) the measurement error in the first-stage cross-section profitability regressions and (2) the collinearity problem since the book-to-market ratio enters both the first-stage and the second-stage regressions.

Measurement error in profitability has been discussed extensively in the literature. Several alternative measures of profitability have been suggested as alternatives to bottom-line earnings since they may be less susceptible to manipulation by management. Using Compustat definitions (in parentheses), we move from the top-line of the income statement (revenue) to bottom-line earnings, as follows:<sup>3</sup>

$$\begin{aligned}
& \text{Revenue (REVT)} \\
& - \text{Cost of goods sold (COGS)} = \text{Gross profit (GP)} \\
& - \text{Selling, general and administrative expenses (XSGA)} = \text{Operating income before depreciation (OIBDP)} \\
& - \text{Depreciation and amortization (DP)} = \text{Operating income after depreciation (OIADP)} \\
& - \text{Interest and related expense (XINT)} \\
& + \text{Special items (SPI)} \\
& + \text{Non-operating income (NOPI)} = \text{Pretax income (PI)} \\
& - \text{Income taxes (TXT)} \\
& - \text{Minority interest (MII)} = \text{Income before extraordinary items (IB)}
\end{aligned}$$

To avoid largely non-recurring items, Li and Mohanram (2014) measure earnings as income before extraordinary items minus special items, IB-SPI. Novy-Marx (2013) uses gross profit (GP) as the proxy for profitability and argues that *“The farther down the income statement one goes, the more polluted profitability measures become, and the less related they are to true economic profitability.”* To measure earnings in their profitability factor, Fama and French (2015) subtract the selling, general, and administrative expenses and interest expenses from the gross profit ( $\text{EBTDA} = \text{GP} - \text{XSGA} - \text{XINT}$ ).

We assess the ability of various earnings measures to predict future cash flows (years  $t + 1$  through  $t + 10$ ) by regressing future cash flows (measured by the Compustat Cash Flow from Operations, OANCF) on current earnings metrics (Table 1). The cross-sectional regressions are for 50 portfolios sorted by 5 industry groups and the Fama-French measure of profitability. Of the measures that do not adjust for intangibles, we use (moving from bottom-line to top-line measures) net income (Fama and French (2006)), earnings before extraordinary items

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<sup>3</sup>All the items have quarterly counterparts.



(Li and Mohanram (2014)), earnings before taxes (EBT), earnings before taxes, depreciation, and amortization (EBTDA, Fama and French (2015)), gross profit, (Novy-Marx (2013)), and revenue. Since the Compustat variable for cash flows from operations (OANCF) is available starting in 1988, our predictive regressions are for the period 1988-2022. For all horizons, the average  $R^2$  values of the regression increase from net income through the Fama and French (2015) EBTDA measure and then decline as we move to more top-line measures. The results for a subset of these measures, net income; earnings before taxes, depreciation, and amortization (EBTDA); and gross profit; are reported in Table 1. Across all horizons, the gross profit measure of profitability has higher average values of  $R^2$  than net income. Additionally, the FF5 measure of profitability has higher values of  $R^2$  than gross profitability. Because of this result, we follow Fama and French (2015) and measure unadjusted profitability using EBTDA instead of the more bottom-line measure used in Fama and French (2006) or the more top-line measure of gross profitability in Novy-Marx (2013).

### 2.3 Earnings, Value, and Growth Factors with Investments in Intangibles

Analysts have long recognized that the accounting treatment of investments in physical assets and intangible assets has been asymmetric (for example, Bloomberg (1934), Graham et al. (1962), Graham (1973), Greenwald et al. (2001)).<sup>4</sup> Investments in physical capital are capitalized and depreciated over time while investments in intangible capital, which build intellectual property, brands, and organizational capital (reported in R&D and SG&A), are expensed immediately unless they are purchased intangibles. Treating investments in intangible capital similarly to physical capital has implications for the measurement of earnings and book value and influences three of the variables in the present value relation (3),  $P_t/B_t$ ,  $dB_{t+\tau}/B_t$ , and

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<sup>4</sup>In the 2010 CEO’s letter to Amazon shareholders, Jeffrey Bezos emphasizes that Amazon’s continuing investment in intangible assets has played a determinant role in advancing the company and the ultimate payoff to shareholders: “*The advances in data management developed by Amazon engineers have been the starting point for the architectures underneath the cloud storage and data management services offered by Amazon Web Services (AWS). ... in my opinion, these techniques are not idly pursued – they lead directly to free cash flow.*”

In the 1997 annual report Bezos illustrates his principle of investing: “*We will make bold rather than timid investment decisions where we see a sufficient probability of gaining market leadership advantages. ... When forced to choose between optimizing the appearance of our GAAP accounting and maximizing the present value of future cash flows, we’ll take the cash flows.*”

$Y_{t+\tau}/B_t$ .

Similarly, incorporating investment in intangible capital might affect the way we define three of the factors in the FF5 model, RMW, HML, and CMA. Arnott et al. (2021) and Eisfeldt et al. (2022) adjust HML for intangible capital and argue that the intangible-adjusted HML has significant advantages, in terms of pricing the cross-section, relative to the original HML. Wang (2023) proposes an intangibles-adjusted asset growth factor, similar to CMA, and argues it outperforms the original CMA factor. Gulen et al. (2025) construct variants of the FF5 factors that incorporate the effects of off-balance-sheet intangible assets and find that some outperform the FF5 factors.

Fundamental analysts often adjust earnings for investment in intangibles, e.g., Greenwald et al. (2001) (see Table 7.7 in the chapter “Inside Intel”) add 25% of both the R&D and SG&A expenses back to Intel’s earnings to project its future earnings. Even earlier, Graham (1973) highlights the danger of earnings “smoothing” and the choice between fully charging off R&D costs or amortizing them.<sup>5</sup>

The academic literature has advocated several approaches to account for investment in intangibles when explaining equity valuation. For example, Lev and Sougiannis (1996) capitalize R&D expense and adjust bottom-line earnings for R&D expense. Peters and Taylor (2017) and Crouzet and Eberly (2023) consider all R&D expense as investment in knowledge capital and 30% of the SG&A expense, net of R&D expense, as the investment in organizational capital. Belo et al. (2022) treat R&D and advertising expenses and the change in the number of employees as separate components of investment in intangibles. Eisfeldt and Papanikolaou (2013) capitalize the sum of R&D and SG&A expenses as measure of firms’ organizational capital. Eisfeldt et al. (2022) suggest: “*Without better estimates of the fraction of SG&A spending that is investment in intangible assets, ... it is best to use 100% of SG&A ... to avoid introducing noise.*”

Because of its superior ability to forecast future cash flows relative to other unadjusted

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<sup>5</sup>In chapter 12 “Things to consider about per-share earnings”: “*Still another factor, important at times, is the choice between charging off research and development costs in the year they are incurred or amortizing them over a period of years.*”

earnings measures, our earnings factors are intangibles-adjusted versions of the FF5 RMW factor. Since the FF5 RMW factor corresponds to earnings before taxes, depreciation, and amortization (EBTDA), our corresponding adjustment to earnings is the equivalent of EBTDA with intangibles investments. That is, we add back the expense of intangibles investment to the FF5 version of earnings. We focus on three alternative measures of intangibles investment a) one that adds to earnings R&D expenses only as in Lev and Sougiannis (1996) (noted with a superscript RD below), b) one that adds to earnings R&D expenses and 30% of SG&A expenses as in Peters and Taylor (2017) and Crouzet and Eberly (2023) (noted with a superscript RD+.3SG below), and c) one that adds to earnings R&D expenses and 100% of SG&A expenses as in Eisfeldt et al. (2022) (noted with a superscript RD+SG below). The first of these measures was included in early versions of Fama and French (2018) but not the published version.

We first study the ability of intangibles-adjusted profitability measures to predict future cash flows, a requirement for them to be proxies for expected cash flows in (3). There are fewer individual firms available at longer horizons due to firms entering and dropping out of the sample (the number of firms with cash flow data at a horizon of ten years is 60% of those with data at a horizon of one year). Because of this, we run regressions using 50 portfolios sorted by the FF5 profitability measure  $(Y(\text{FF})/B_t)$  and Fama-French industry classification (details given below). Portfolios are rebalanced as firms exit the sample. In the profitability regressions (Table 1),  $Y(\text{FF})/B_t$  is the single best predictor of profitability among the unadjusted profitability measures. A subset of these is reported in the table. The gross profitability measure of Novy-Marx (2013) has a higher average  $R^2$  than net income (IB), for all horizons (Panels A and B of the table). The difference in average  $R^2$  values is significant for horizons 3 and 10 (t-statistics for the difference are reported in Panel B). The EBTDA measure of Fama and French (2015) has a higher average  $R^2$  relative to gross profitability for all horizons (Panels A and B of the table), and the differences are significant at horizons of 1 through 7 years.

When we include the intangibles adjustments (Panels C and D)  $Y(\text{RD})/B_t$  has less predictive power (measured by  $R^2$ ) than  $Y(\text{FF})/B_t$  at horizons of 1 to 2 years and greater at horizons of 3 to 10 years, with horizons 4 to 10 being significantly higher. This makes intuitive sense

since R&D expenditures may take some time to impact cash flows.  $Y(\text{RD} + .3\text{SG})/B_t$  has lower predictive power (measured by  $R^2$ ) than  $Y(\text{FF})/B_t$  at horizons of one year through three years and higher average  $R^2$ s for longer horizons (significant for years six through 10).  $Y(\text{RD} + \text{SG})/B_t$  has lower predictive power (measured by average  $R^2$ ) than  $Y(\text{FF})/B_t$  from year one to year eight.

In Figure A1 of the Internet Appendix, we plot the time series of the difference in  $R^2$  between the profitability measures  $Y(\text{NM})/B$  versus  $Y(\text{IB})/B$ ,  $Y(\text{FF})/B$  versus  $Y(\text{NM})/B$ , and  $Y(\text{RD})/B$  versus  $Y(\text{FF})/B$  across three different forecasting horizons  $\tau = 1, 5, 10$ . Though  $Y(\text{NM})/B$  does not always outperform  $Y(\text{IB})/B$  in forecasting future profitability,  $Y(\text{FF})/B$  clearly performs better than  $Y(\text{NM})/B$  at 1- and 5-year horizons.  $Y(\text{RD})/B$  forecasts future profitability better than  $Y(\text{FF})/B$  for longer-horizons (5- and 10-year ahead) but worse for short-horizons (1-year ahead).

Among measures that do not adjust for intangibles, the results in Table 1 provide empirical support for the ability of the earnings measure used in Fama and French (2015) to predict future cash flows (Panels A and B). The results also provide support for increased predictive power at longer horizons when using intangibles-adjusted profitability (Panels C and D).

Table 2 performs a similar portfolio-level cross-sectional regression in which the FF5 profitability measure,  $\text{RD}_t/B_t$ ,  $\text{SGA}_t/B_t$ , asset growth, and book-to-market are included as explanatory variables together. Both the Fama-French profitability measure and R&D adjustment are significant at all horizons. The SG&A adjustment is also positive at all horizons except for year one. The slope coefficients of R&D and SG&A adjustments generally increase with horizon. The results in Table 2 confirm the intuition from Table 1 that R&D takes some time to influence future cash flows. SG&A kicks in by year 1 and also grows in importance at longer horizons, consistent with portions of SG&A being investments in intangible capital that payoff in future cash flows. Thus the Fama and French (2015) EBTDA profitability measure as well as measures of EBTDA that adjust for investment in intangibles seem to pass the test of being able to predict future cash flows.

We report the predictive regression results replacing operating cash flows with the cash

dividends in Tables A2 and A3 of the Internet Appendix. In these tables, the dependent variables are future cash dividends in year  $t + \tau$  divided by the book equity in year  $t$  and the independent variables remain the same as in the operating cash flows regressions. The portfolio-level regressions largely overcome the problem that many firms do not pay dividends. Differences between the results of forecasting cash dividends and the results of forecasting operating cash flows are that dividends are generally less forecastable than operating cash flows and  $Y(NM)/B$  does not appear better than  $Y(IB)/B$  when forecasting cash dividends while  $Y(RD)/B$  outperforms  $Y(FF)/B$  even in the short horizon (1-year ahead). The importance of the intangibles adjustment increases (both in magnitude and t-statistics of the slope coefficients) as the forecasting horizon grows.

Now that the profitability measures seem to pass the minimum requirement for being useful for being tied to future expected returns (or the ICC), that is, having the ability to forecast future cash flows, we can move on to asking whether factors derived from them are useful in pricing the cross-section of asset returns. Before that, we provide a more detailed discussion of the underlying data.

### 3 R&D Disclosure and Compustat Data

The Financial Accounting Standards Board Statement number 2 (FASBS2, FASB (1974)) mandated all material R&D costs encompassed by the statement to be charged to expense when incurred and the statement became effective for fiscal years beginning on or after January 1, 1975. The accounting treatment of R&D costs varies across firms before 1975. Moreover, FASB (1974) required “... *research and development costs be charged to expense when incurred shall be applied retroactively by prior period adjustment, ... The prior period adjustment shall recognize any related income tax effect.*” Therefore, the recorded R&D expense in restated financial statements prior to the fiscal year of 1975 may not have been available to the public at the time of original publication. Therefore, the performance of portfolios constructed using those restated data may contain look-ahead biases. For this reason, we restrict our analysis to the sample period from 1975 to 2021 when using any accounting information.

Moreover, even post-FASB2, not all firms report (positive) R&D expenses. In the calendar year 1975 40% of firms report R&D expenses, experiencing a slight decline (but always above 35%) before 2000, and then gradually increasing to 52% in 2021. Koh and Reeb (2015) state that the materiality threshold is typically defined as amounts exceeding one percent of sales. Since R&D expense is an item of mandatory disclosure, empty R&D fields should imply zero or immaterial R&D expense. However, Koh and Reeb (2015) find that 10.5% of the missing R&D firms file and receive patents, which is 14 times greater than zero R&D firms. Koh and Reeb (2015), therefore, argue that a blank R&D field could also represent a firm’s conscious decision not to separate R&D expense from other reported expenses, rather than implying zero R&D expense. Without a clean approach to impute missing R&D expense, we set it as zero if missing in this paper. The consequence is that our R&D measure is subject to an errors-in-variables (EIV) problem. The EIV issue should weaken our results and lead to a bias toward failing to find a significant effect of the intangibles adjustment. Goyal and Wahal (2023) hand-collect data on R&D for firms with missing data to either supplement or impute missing R&D data. They find that, while the number of firms with non-zero R&D increases significantly, aggregate industry R&D changes imperceptibly for most industries at the two-digit SIC code level (see their Table 1).

For the firms reporting R&D expense, the expense can be shown either as (1) a separate line item on the income statement or (2) separately stated in notes to their financial statements.<sup>6</sup> It is Compustat’s standard to collect R&D expense (Compustat data item XRD) from either line items or notes to financial statements.

Since we construct both annual- and monthly-rebalanced profitability factors using annual and quarterly filings data, respectively, we also examine the R&D data recorded in quarterly Computat. First, we find that no firms report R&D in the quarterly filings (quarterly Compustat data item XRDQ) before 1989 and only on average 41% of the firms have XRDQ record in the quarterly Compustat data between 1989 and 2021.<sup>7</sup> Moreover, among all firms that have XRDQ

<sup>6</sup><https://www.irs.gov/businesses/corporations/faqs-irc-41-qres-and-asc-730-lbi-directive>.

<sup>7</sup>98% of the firms reporting annual R&D expense report quarterly R&D expense, though not necessarily for all quarters.

records, there are about 25% of the firms report only once a year and the amount is the same as in the XRD recorded in annual Compustat data. We verify that firms reporting their R&D expense as a separate line item in the income statement typically have multiple XRDQ records in a year while the firms integrating R&D expense into other line items and only reporting in the notes to financial statement typically have only one XRDQ record each year.<sup>8</sup> From 1989 through 2021, the aggregate R&D expense for all firms with more than one XRDQ record is 8.8% of their total sales which is a much larger fraction relative to the 1.8% for firms with only one XRDQ record.

As an example, consider General Electric (GE). GE reports R&D expenses only in the notes to financial statements in the annual filings before the fiscal year ending in December 31, 2019. In their notes to financial statements in the 2019 10-K, GE explains “R&D expenses are classified in cost of goods and services sold in our consolidated Statement of Earnings (Loss).” After 2019, GE changed this practice and reports R&D expense as a line item in both quarterly and annual filings. This change is also reflected in the quarterly Compustat data.

GE integrates R&D expense in the cost of goods sold (Compustat data item COGS), though most firms include R&D expense in the selling, general and administrative expense (Compustat data item XSGA). Compustat does not provide any information to identify into which item the firm integrates R&D expense. This practice has implications for our analysis since we need to separate SG&A and R&D expenses.

We follow the procedure in Peters and Taylor (2017) and assume that R&D is included in reported SG&A. We, therefore, set our SG&A measure equal to reported SG&A minus reported R&D (XSGA minus XRD minus RDIP<sup>9</sup>) We add an additional screen: when XRD exceeds XSGA but is less than COGS, or when XSGA is missing, SG&A is set equal to XSGA with no further adjustments or zero if XSGA is missing. We recognize that this approach is not flawless as it does not work for firms with higher SG&A than R&D but integrating R&D in the cost of goods sold. We denote firms’ ex-R&D values of SG&A as SGA and use XSGA to refer

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<sup>8</sup>64% of the firms reporting quarterly R&D expense report four times a year.

<sup>9</sup>RDIP represents the portion of R&D considered to be “purchased” and written off immediately upon acquisition if the R&D items are deemed not to have an alternative use.

to the cum-R&D Compustat data item.

## 4 Data

Monthly stock returns, dividends, stock prices, and numbers of common shares outstanding are from the Center for Research in Security Prices (CRSP), and annual and quarterly accounting information is from Compustat.<sup>10</sup> To avoid look-ahead bias in using R&D data, the accounting information is from 1975 to 2022. We include all firms with share codes 10 and 11 and traded on NYSE, Amex, and Nasdaq. Table 18 contains a comprehensive list of variable definitions.

The profitability of firms is measured as earnings divided by total book equity. We define annual total book equity as Compustat data item CEQ<sup>11</sup> and quarterly book equity following Hou et al. (2015).<sup>12</sup> As mentioned above, we perform analyses using several measures of earnings but only report a subset of results to conserve space. Our earnings measures include (moving from top-line to bottom-line measures) (1) total revenue,  $Y(\text{REVT})$ , (2) gross profit,  $Y(\text{NM})$  following Novy-Marx (2013), (3) gross profit minus interest expense,  $Y(\text{RD} + \text{SG})$  as in Eisfeldt et al. (2022) for investment in intangibles, (4) gross profit minus interest expense and 70% of SG&A expense,  $Y(\text{RD} + .3\text{SG})$  following the definition in Peters and Taylor (2017) for investment in intangibles, (5) gross profit minus interest expense and 100% of SG&A expense,  $Y(\text{RD})$ , (6) gross profit minus interest expense, 100% of SG&A expense, and 100% of R&D expense,  $Y(\text{FF})$  following definition in Fama and French (2015) for profitability, (7) gross profit minus interest expense, 100% of SG&A expense, 100% of R&D expense, and 100% depreciation,  $Y(\text{EBT})$ , (8) income before extraordinary items minus special items,  $Y(\text{SPI})$ , and (9) income

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<sup>10</sup>The results presented in this paper are based on the CRSP and Compustat data downloaded on December 2, 2023.

<sup>11</sup>The book equity computation in Fama and French (2018) has broader coverage than the Compustat data item CEQ before 1968. Since our sample period starts in 1975, using CEQ or computed book equity does not alter our results.

<sup>12</sup>Quarterly book equity is shareholders' equity, plus balance-sheet deferred taxes and investment tax credit (item TXDITCQ) if available, minus the book value of the preferred stock. Depending on availability, we use stockholders' equity (item SEQQ), or common equity (item CEQQ) plus the carrying value of the preferred stock (item PSTKQ), or total assets (item ATQ) minus total liabilities (item LTQ) in that order as shareholders' equity. We use redemption value (item PSTKRQ) if available, or carrying value for the book value of the preferred stock.



before extraordinary items,  $Y(\text{IB})$ .

Due to the quarterly R&D data issues described in the earlier section, we focus on the annual-rebalanced profitability factors using annual Compustat data in most of our analyses. To be included in the profitability forecasting regression for  $t + \tau$  where  $\tau = 1, 2, \dots, 10$  in Tables 1 and 2, a firm must have Compustat data for calendar year  $t$  on book equity and revenue or income before extraordinary items, following Fama and French (2006). For the firms reporting accounting information more than once in a given calendar year, potentially due to the change of the fiscal year-end month, we use the most recently released information. In addition, to ensure that the accounting variables are known at the month of portfolio formation and return regressions, we follow Fama and French (2006) and include a firm in the portfolio formation and return regression for July of  $t+1$  to June of  $t+2$  if it has available book equity, revenue or income before extraordinary items in the calendar year  $t$ . This procedure guarantees, at a minimum, a six-month period for the information to be released.<sup>13</sup>

We exclude firms with negative book equity, missing share price, or missing shares outstanding when they are used in the explanatory variables (that is, at time  $t$ ). Following Fama and French (2006), we remove firms with total assets less than \$25 million or book equity less than \$12.5 million in year  $t$  to avoid the influence of small firms.

Other data, including various competing asset pricing factors, are taken from the authors' websites. In particular, the factors in the Fama-French five factor model (FF5) and the momentum (UMD and UMD(d)) are taken from Ken French's data library ([http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data\\_library.html](http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html)). The same data library is also the source of the five industry portfolios that we use in the ROE prediction regressions. The factors in the Hou-Xue-Zhang four factor model (HXZ4) and their expected growth factor (R<sub>EG</sub>) are from <https://global-q.org/factors.html>. The factors in the Daniel et al. (2020) behavioral three factor model (DHS3) are from <https://sites.google.com/view/linsunhom>. The factors in the Stambaugh and Yuan (2017) four factor model (SY4)

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<sup>13</sup>This conservative implementation could include some stale accounting information. For example, for a firm with a fiscal year-end in February 2021 with announced earnings in April 2021, the accounting information is already fourteen months ago when we first use it in the portfolio and return regression analysis in July 2022. This will tend to bias against finding significant results.

are from <https://finance.wharton.upenn.edu/~stambaug/>. The intangibles-adjusted HML factor, HML(EKP) is from Eisfeldt et al. (2022) and available from <https://github.com/edwardtkim/intangiblevalue>,. The factors in the adjusted FF5 and HXZ4 models in Gulen et al. (2025) (GLPZ7 and GLPZ5) are from <https://sites.google.com/site/rheathpeters>. The monthly rebalanced version of HML (HML(Devil)) of Asness and Frazzini (2013) is from AQR.<sup>14</sup> The data for GLPZ7 and GLPZ5 were downloaded on August 30, 2024, and all other factor data were downloaded on December 2, 2023. SY4 is available until December 2016, HML(EKP) are available until March 2022, GLPZ5 and GLPZ7 data are available until June 2022, HXZ4, R.EG, and DHS3 data are available until December 2022, FF5, UMD, and HML(Devil) are available until June 2023.

Since firm-level cross-section predictive regressions with horizons of one to ten years are subject to severe survivorship biases, we forecast the future operating cash flows of fifty managed portfolios. In Table A1 of the Internet Appendix, we report the number of firms that remain available after  $\tau$  years. 17%, 28%, and 47% of firms drop out of the sample after 3, 5, and 10 years respectively. To deal with the severe survivorship bias in the firm-level profitability regressions, we assign firms into 5-by-10 industry and  $Y(\text{FF})/B$  groups. The portfolios are rebalanced at the end of each month in the next ten years to ensure that the portfolio weights on each firm are proportional to their respective market size. The portfolio-level profitability regressions take the portfolio-level profitability (earnings in year  $t + \tau$  divided by book equity in year  $t$ ) in year  $t + \tau$  as the dependent variables and the current profitability, and other variables, in year  $t$  as the independent variables.

In the regressions reported in Tables 1 and 2, the earnings of portfolio  $k$  in the year  $\tau$  after its formation as its net worth times the value-weighted earnings-to-market ratio, i.e.,

$$Y_{k,t+\tau} = \underbrace{w_{k,t+\tau}}_{\text{portfolio net worth}} \sum_{i \in k} \underbrace{\frac{M_{i,t+\tau}}{\sum_{j \in k} M_{j,t+\tau}}}_{\text{weight on firm } i} \frac{Y_{i,t+\tau}}{M_{i,t+\tau}} \quad \text{where } \tau = \{1, 2, \dots, 10\} \quad (4)$$

where  $w_{k,t+\tau} = \prod_{s=\{1,2,\dots,\tau \times 12\}} \left(1 + \sum_{i \in k} \frac{M_{i,s-1}}{\sum_{j \in k} M_{j,s-1}} r_{i,s}\right)$  is one plus the cumulative ex-dividend return (CRSP data item RETX) of the portfolio from its formation to  $\tau$  years later, where  $r_{i,s}$

<sup>14</sup><https://www.aqr.com/Insights/Datasets/The-Devil-in-HMLs-Details-Factors-Monthly>.

is the monthly return of firm  $i$  in month  $s$  after the portfolio formation.

Following Fama and French (2015), the portfolios are formed at the end of June in year  $t$  based on the accounting information in calendar year  $t - 1$  from Compustat, which leaves at least six months between the fiscal year ends and the use of the accounting information. All the firms are sorted independently into 5-by-10 industry and  $Y(\text{FF})/B$  groups. We keep tracking the portfolios' returns at the end of each month, earnings, and cash flows at the end of each June for the next ten years after their formation, based on the rebalanced scheme described earlier.

## 5 Profitability Factors with an R&D Adjustment

Though FF5 contains only annual-rebalanced factors, many other factor models are based on monthly-rebalanced portfolios. Therefore, we construct monthly-rebalanced profitability factors as well, treating missing R&D expenses as being zeros as in the annual rebalancing case.

### 5.1 The Performance of Profitability in Pricing Assets

Table 3 reports mean returns and (t-statistics) of portfolios ranked independently on market capitalization (based on the median market equity for NYSE firms) and various measures of profitability (three groups based on the 30<sup>th</sup> and 70<sup>th</sup> NYSE percentiles of profitability). The table also provides the difference in average returns in high- versus low-profitability firms for each size category. The last two columns show the mean profitability factor return spread and its alpha, relative to a one-factor model using the FF RMW factor. The return spreads are generally significant for all unadjusted profitability measures with t-statistics increasing from bottom-line profitability to top-line, except for revenue, which is less significant than gross profitability.

The largest average return spreads and t-statistics are for the three intangibles-adjusted measures (results in bold in the table), followed by the gross profitability measure of Novy-Marx (2013). Similarly, the only measures with significantly positive alpha versus RMW are the intangibles-adjusted measures and gross profitability.

Panel A and B in Figure 1 present the cumulative return of four selected annual- and monthly-rebalanced profitability factors in three equally divided subsamples. For annual-rebalanced profitability factors,  $Y(IB)/B$  underperforms in all three subsamples,  $Y(RD)/B$  performs the best in the later two subsamples, and  $Y(FF)/B$  leads the performance in the first subsample. For the monthly-rebalanced factors, the R&D-adjusted profitability factor has the best performance in all subsamples. Interestingly,  $Y(FF)/B$  and  $Y(IB)/B$  consistently deliver very similar results.  $Y(NM)/B$  does not do as well before 2008 but has strong performance after 2008 for both annual- and monthly-rebalanced factors.

It is common to test alternative pricing models by choosing some test assets and asking whether one model provides lower estimated abnormal returns (alphas) than a competing model. The alphas are typically measured as the intercept in a time series regression of the excess return on each test asset on the factor returns (or factor-mimicking portfolios). Barillas and Shanken (2017) and Barillas et al. (2020) show that, when factors are traded portfolios, alternative models can be compared (in terms of which one leads to higher squared Sharpe ratios) by asking if each model can price the factors in the alternative model. The left-hand side “test assets” drop out in this case (they also show that looking at test asset alphas may be misleading).

We study the mispricing (or alphas) of the profitability factors studied in Table 3 by regressing their excess returns on eleven important multi-factor asset pricing models in the literature. The profitability alphas are reported in Table 4. The models that we use are (1) the FF5 model (Fama and French (2015)); (2) the FF5(C) model with RMW replaced by the cash-based operating profitability factor (Fama and French (2018)); (3) the FF6 model which adds the momentum factor, UMD, from Carhart (1997) to FF5; (4) FF6/Devil which is a version of the FF6 model in which the value factor, HML is replaced with a version that is updated on a monthly basis (Asness and Frazzini (2013)); (5) FF6/EKP which is a version of the FF6 model in which the value factor, HML is replaced with the intangibles-adjusted value factor from Eisfeldt et al. (2022); (6) the four-factor model of Hou et al. (2015); (7) the five-factor model from Hou et al. (2021); (8) the long- and short-horizon-mispricing three-factor model of Daniel et al. (2020); (9) the mispricing four-factor model of Stambaugh and Yuan (2017); (10) the seven-factor model of

Gulen et al. (2025); and (11) the five-factor model of Gulen et al. (2025). To conserve space, we report results with FF5, HMXZ5, DHS3, and GLPZ7 in the main text and relegate additional results to Table A4 of the Internet Appendix.

Our main results are the estimates,  $\hat{\alpha}$ , for the array of profitability factors, and their associated t-statistics, relative to the eleven competing models, reported in Table 4, Panels A-1 and A-2. For each measure of profitability, we do 2-by-3 sorts on size and profitability. For the annual rebalanced factors, the size sort breakpoint is the median size for NYSE firms. The profitability-sort breakpoints are the 30<sup>th</sup> and 70<sup>th</sup> NYSE percentiles of profitability. Using the Fama-French terminology “Small Robust” is the value-weighted portfolio of small-cap firms with high profitability and “Small Weak” is the portfolio of small-cap firms with low profitability (with equivalents of “Big Robust” and “Big Weak” for large-cap firms). Our factors are constructed as  $1/2$  (Small Robust + Big Robust) -  $1/2$  (Small Weak + Big Weak).

The first three columns of results show the profitability factors moving from bottom-line to top-line measures: net income (IB), income before special items (SPI), earnings before taxes (EBT), and earnings before taxes depreciation and amortization (EBTDA), used in Fama and French (2015) (FF), have insignificant values of  $\hat{\alpha}$ . This is not surprising since our predictive regressions for operating cash show that IB, SPI, and EBT all have less predictive power than (EBTDA). The results for FF in column 4 are also not surprising since the RMW is one of the factors in the first 4 models. The fact that  $\hat{\alpha}$  for our version of the FF5 profitability factor is not identically zero is just an indication that our construction of RMW differs slightly from the version of RMW on Ken French’s data library, which is the source of RMW on the RHS of the regression.

The next three columns (in bold font) include versions of our intangibles-adjusted profitability factors. The factor  $r_{ROE}^{RD}$  uses the FF5 measure of profitability plus R&D expenditures. The factor  $r_{ROE}^{RD+.3SG}$  also adds 30% of SG&A and  $r_{ROE}^{RD+SG}$  also adds 100% of SG&A. For  $r_{ROE}^{RD}$  and  $r_{ROE}^{RD+.3SG}$  the values of  $\hat{\alpha}$  are positive, statistically are significant, and the largest across all profitability measures and all eleven benchmark models. Adding these R&D-adjusted factors to the benchmark factors significantly increases the squared Sharpe ratio of the portfolio space

spanned by the factors. The  $t$ -statistics range from 2.24 to 7.05 and exceed 3.21 for all models except the HMXZ5 model. The factor which adds R&D and all of SG&A,  $r_{ROE}^{RD+SG}$ , has slightly smaller alphas and has significantly positive  $\hat{\alpha}$  for all models except HMXZ5, SY4, and GLPZ7. The eight significant alphas for  $r_{ROE}^{RD+SG}$  have  $t$ -statistics ranging from 2.12 to 4.42. The gross profitability measure,  $r_{ROE}^{NM}$ , has significantly positive alphas for six of the eleven models.

The results of Table 4 are consistent with a significant premium to the intangibles-adjusted profitability factors, particularly  $r_{ROE}^{RD}$  and  $r_{ROE}^{RD+.3SG}$ . Our concern is that these abnormal returns come mainly from the short side of the factor portfolio. An investor attempting to implement a smart beta portfolio based on a given factor might have concerns about potentially large asset borrowing costs on the short side. We study the source of abnormal returns by looking at the values of  $\hat{\alpha}$  for the long and short sides separately.

Panel B-1 and C-1 of Table 4 show our results for annual rebalancing. Panel B-1 shows the long side. The factors  $r_{ROE}^{RD}$  and  $r_{ROE}^{RD+.3SG}$  have significant long-side alphas for each benchmark model with the exception of GLPZ7 ( $r_{ROE}^{RD}$ ) and SY4 and GLPZ7 ( $r_{ROE}^{RD+.3SG}$ ). Panel C-1 shows the short side for annual-rebalanced factors. The values of  $\hat{\alpha}$  for  $r_{ROE}^{RD}$  and  $r_{ROE}^{RD+.3SG}$  are statistically significant for all models, but three (HMXZ5, DHS3, and GLPZ7). While the long-side alphas were not very strong against SY4, the short-side alphas are. Similarly, while the short-side alphas were not very strong against HMXZ5 and DHS3, the long-side alphas are.

In summary, the intangibles-adjusted profitability measures add explanatory power for the cross-section of assets beyond that captured by the eleven alternate models studied here.  $r_{ROE}^{RD}$  and  $r_{ROE}^{RD+.3SG}$  have abnormal returns coming from both sides of the position for eight of the eleven models, from the long side only for two models, and from the neither side for one model (although the long-short strategy is significant for that model). Therefore, the significant intangibles-adjusted factor alphas are not solely coming from the short side.

Panel A-2, B-2, and C-2 of Table 4 show the results for the monthly-rebalanced version of the factors. The results tend to be stronger with monthly rebalancing. In Panel A-2, the  $t$ -statistics on alphas range from 4.00 to 9.34 for  $r_{ROE}^{RD}$  and from 2.90 to 8.72 for  $r_{ROE}^{RD+.3SG}$ . Panels B-2 and C-2 report the results for the long- and short-sides of the monthly rebalanced

factors. The long-side alphas of both  $r_{\text{ROE}}^{\text{RD}}$  and  $r_{\text{ROE}}^{\text{RD}+.3\text{SG}}$  are significant for all eleven models. The short-side alphas of  $r_{\text{ROE}}^{\text{RD}}$  are significant for ten of the eleven alternative models while the alphas of  $r_{\text{ROE}}^{\text{RD}+.3\text{SG}}$  are significant for eight of the eleven models. Again, the significant intangibles-adjusted factor alphas are not solely coming from the short side.

## 5.2 Intangibles-Adjusted Profitability and Downside Protection

We discussed two characteristics of higher-quality assets: consistently higher cash flows in future years and lower downside risk associated with asset returns. In Section 2, we verified that higher intangibles-adjusted profitability is positively related to higher future cash flows in the cross section (for up to ten years in the future). In this subsection, we examine downside risk in three different ways: maximum drawdown; assets’ return co-tail risk with the market; and downside protection of asset returns using the market timing measure of Henriksson and Merton (1981).

The results are reported in Table 5. While all profitability factors exhibit non-trivial downside risk, the intangibles-adjusted profitability factors consistently show the smallest realized maximal drawdowns, highlighting their lower tail risk.

Table 6 looks at the behavior of the profitability factors during market downturns. One might expect from their construction (and even the name “Robust Minus Weak”) that these factors might provide some downside protection since weak firms might be expected to suffer more from economic downturns. That is born out in the table. The factors have higher returns when the excess return on the market is negative than when it is positive. This is true for all of the profitability factors, not just those adjusted for intangibles.

The resilient performance during the down markets could be attributed to a “protective” put option feature embedded in the profitability factors (see, e.g., Henriksson and Merton (1981) and Jagannathan and Korajczyk (1986)). We can test this conjecture by having an additional term to capture the put option features in a Henriksson-Merton-style time-series regression.

$$r_{\text{ROE}}^{i,t} = \alpha_i^s + \beta_i^U \text{RMRF} + \beta_i^{U-D} \max[-\text{RMRF}, 0] + \epsilon_{i,t}. \quad (5)$$

The intercept term,  $\alpha_i^s$ , measures the abnormal return from pure security selection, that is, returns independent of any put-like feature. The coefficient  $\beta_i^U$  measures the portfolio's up-market beta and the coefficient  $\beta_i^{U-D}$  measures the difference between the portfolio's up-market and down-market betas. If the factor portfolio has a put-like feature, then we should see that the estimated value of  $\beta_i^{U-D}$  is significantly positive. That is, a strategy with an embedded put will have a higher (lower) beta when the excess return on the market is positive (negative).  $\beta_i^{U-D}$  measures the number of put options needed to replicate the strategy's put feature. In a managed portfolio, this put feature could be ascribed to market timing ability. For the factor portfolios the put-like feature could be from the underlying put-like properties of the assets held or dynamic rebalanced inherent in the portfolio strategy. Jagannathan and Korajczyk (1986) argue that a measure which combines both sources of performance is

$$\frac{\alpha_i^s}{1 + r_f} + \beta_i^{U-D} \times p. \quad (6)$$

Here  $p$  is the value of a put option on the market index normalized to unity, with one period to expiration and an exercise price of  $1 + r_f$ .

Table 7 reports the regression results from Equation 5 for five profitability factors,  $Y(\text{IB})$ ,  $\text{RMW}$  from  $\text{FF5}$  ( $\text{FF}$ ); our measure that adds R&D to earnings ( $\text{RD}$ ); gross profitability ( $\text{NM}$ ); and  $\text{RMW}(\text{GLPZ})$  from Gulen et al. (2025).

For our factor portfolio,  $r_{\text{ROE}}^{\text{RD}}$ , the intercept, 0.24, and the coefficient of the put feature, 0.14, are both significant. We estimate the combined measure of performance assuming the monthly interest rate is zero, and the monthly standard deviation of the market is 5%. The Black-Scholes value of the  $p$  is \$0.02 per month, or 2% of the amount invested in the index. Hence the option adjusted total alpha of  $r_{\text{ROE}}^{\text{RD}}$  is  $0.24\% + 0.14 \times 2\% = 0.24 + 0.28 = 0.52\%$ . It is interesting to have both  $\alpha_i^s$  and  $\beta_i^{U-D}$  positive and significant since it suggests that investors get paid a premium for holding a “protective” put.

The four other profitability factors,  $\text{IB}$ ,  $\text{FF}$ ,  $\text{NM}$ , and  $\text{RMW}(\text{GLPZ})$ , also have positive



estimates of  $\alpha_i^s$  and  $\beta_i^{U-D}$ , but only the value of  $\alpha_i^s$  for FF and RMW(GLPZ) is statistically significant.

We report the regression results of the long side and short side of the profitability factors separately in the middle and right columns of Table 7.<sup>15</sup> All four factor portfolios have statistically significant estimates of  $\alpha_i^s$  for the long side of the portfolio. None of the four factor portfolios have significant estimated values of  $\beta_i^{U-D}$  on the long side. In fact, they are all negative. On the short side, the numbers in the table are for a long position on the short side of the portfolio. Shorting the short side would yield coefficients with the opposite sign. There is no indication that the short side of the factor portfolios provide significant security selection,  $\alpha_i^s$ . The short side leads to smaller  $\alpha_i^s$  for the long-short portfolio. Three out of four factors have significantly negative values for  $\beta_i^{U-D}$ . This implies that the assets in the short side decline much more in a down market than they rebound in an up market. Taking a short position in these assets yields positive values of  $\beta_i^{U-D}$  of the long-short portfolio. However, the net effect is only statistically significant for our R&D-adjusted earnings factor.

Overall, intangibles-adjusted profitability exhibits stronger return predictability, lower downside risk, and resilience in market downturns, partly due to a put-like feature. These findings suggest a more robust measure of quality, but also raise the question of whether the observed performance persists under alternative specifications and broader testing, which we explore next.

### 5.3 What Fraction of SG&A?

The evidence thus far highlights the importance of adjusting earnings for R&D expenses, however, it remains unclear how much of SG&A should be added back to earnings. The results in Table 4, panel A-1, suggest that adding 30% of SG&A can improve the explanatory power of the intangibles-adjusted factors. Alpha increases in magnitude, relative to only adjusting for R&D, for six of the eleven models, stays the same for three, and decreases for two, although the t-statistics only increase for four of the eleven models. Adding 100% of SG&A always reduces

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<sup>15</sup>We exclude RMW(GLPZ) for the separate analysis on the long side and short side since those are not available on their web site.

both the magnitude of the estimated  $\alpha$ s and the t-statistics.

In Table A5 of the Internet Appendix, we repeat the analysis in Table 3 by varying the fraction of SG&A added back to earnings from 0% to 100% in increments of 10% while constructing the profitability factor. The highest levels of  $\alpha$  are always for fractions between 0% and 30% (with 20% being the highest most often). The t-statistics for the estimated  $\alpha$  are also the largest for fractions between 0% and 20%. This suggests that R&D expense plays a crucial role in strengthening the relationship between profitability measure and expected returns while a fraction of SG&A expense is important. This is consistent with our predictive regressions in Tables 1.

#### 5.4 Subsample Performance

The cumulative returns shown in Figure 1, Panel A, suggest a potential regime shift in the returns to both the FF5 factor, RMW, and the intangibles-adjusted profitability factors, around the year 2000. We wish to determine if the significance of these factors is a post-2000 phenomenon. We report the time-series average monthly percent returns and t-statistics of several related factors/anomalies for the full sample and two subperiods (using December 1999 as the breakpoint) in Table 8. All the factors considered here are significant in the full sample. Most of the factor t-statistics fall in the second subsample, except for three factors,  $r_{ROA}^{NM}$ ,  $r_{ROA}^{BGLN}$ , and R&D-adjusted profitability (in the last two columns and in boldface). The monthly-rebalanced ROE factor from Hou et al. (2015) (R\_ROE) and the momentum factor becomes insignificant in the second subsample. The two R&D-adjusted profitability factors,  $r_{ROE}^{RD}$  and  $r_{ROEQ}^{RD}$ , have the highest t-statistics in the second subsample. Importantly, the two R&D-adjusted profitability factors are not solely post-2000 phenomena, with both being statistically significant in each subperiod.

#### 5.5 Performance versus Profitability/Intangibles-Related Factors

Many existing factor models include a profitability or earnings factor, e.g., RMW in FF5, R\_ROE in HXZ4, PEAD in DHS3, and PERF in SY4. The (price) momentum factor is often

related to the earnings momentum phenomenon. For example, Chan et al. (1996) argue that price momentum is largely due to the market’s underreaction to earnings-related information. Even though these factors are all connected to the earnings or post-earnings market reactions, they manifest very different properties of some asset price anomalies. For example, the monthly-rebalanced PEAD and R\_ROE factors can subsume the (price) momentum factor while Fama and French (2016) show that a momentum factor is required to take away the significance of the momentum phenomenon, though they use an annual-rebalanced profitability factor, RMW, in FF5. We study whether our R&D-adjusted measure adds alpha to the performance of these factors and, conversely, whether these factors add alpha to the performance of our R&D-adjusted measure.

In Table 9 we ask three separate, but related, questions. In Panel A we study whether various earnings/quality factors in the literature add alpha relative to RMW from Fama and French (2015). Panel A-1 asks if these factors add alpha to RMW in isolation while Panel A-2 asks if these factors add alpha to the FF5 model, which includes RMW. This criterion seems to be table stakes for any factor seeking to improve on RMW.

In Panel B we study whether the various factors add alpha relative to our R&D-adjusted version of RMW,  $r_{ROE}^{RD}$ . Panel B-1 asks if these factors add alpha to  $r_{ROE}^{RD}$  in isolation while Panel B-2 asks if these factors add alpha to a version of the FF5 model in which RMW is replaced by our measure,  $r_{ROE}^{RD}$ .

In Panel C we study if our R&D-adjusted version of RMW,  $r_{ROE}^{RD}$ , adds alpha to the various earnings/quality factors in the literature. Panel C-1 asks if our R&D-adjusted version of RMW,  $r_{ROE}^{RD}$ , adds alpha to these alternate factors in isolation while Panel C-2 asks if our R&D-adjusted version of RMW,  $r_{ROE}^{RD}$ , adds value to these factors when they each are appended to the FF5 model.

We study thirteen (in panels A and C) and twelve (in Panel B) factors in the literature: RMW from Fama and French (2015) (in Panel B only since RMW is on the RHS in Panels A and C); cash operating profitability in Fama and French (2018) ( $r_{ROE}^{FF}$ ); gross profitability ( $r_{ROA}^{NM}$ ) in Novy-Marx (2013); two profitability factors in Ball et al. (2015) ( $r_{ROA}^{BGLN}$ ) and Ball et al. (2016)

( $r_{CROA}^{BGLN}$ ); R\_ROE in HXZ4; the performance factor from Stambaugh and Yuan (2017) (PERF); the momentum factor in Carhart (1997) (UMD); the quality-minus-junk factor in Asness et al. (2019) (QMJ); the post-earnings announcement drift factor in Daniel et al. (2020) (PEAD); the intangibles-adjusted value factor (HML(EKP)) in Eisfeldt et al. (2022); and the two earnings factors used in GLPZ7 and GLPZ5 from Gulen et al. (2025) (RMW(GLPZ) and ROE(GLPZ), respectively.

Panel A-1 and A-2 in Table 9 report the regression results of the related factors on RMW and the full Fama-French five-factor model respectively. All included factors are significant against RMW and FF5 except for ROE(GLPZ) (in the univariate regression of RMW). Thus, these factors seem to add alpha relative to the FF5 model.

Panel B-1 and B-2 report results of similar regressions but with RMW replaced by our R&D-adjusted earnings factor,  $r_{ROE}^{RD}$ . In the univariate regressions,  $r_{ROE}^{RD}$  takes away the positive significance of most of the factors except for PEAD, and HML(EKP). RMW(GLPZ) has a significant alpha, but it is negative rather than positive. In the multiple regressions, however, seven of the thirteen related factors turn significant again due to their negative relation with the value factor (HML) in FF5. However, their alphas experience a marked drop (relative to Panel A-2) when RMW is replaced with  $r_{ROE}^{RD}$ . The percent drop in the alphas are listed in the last row of Panel B-2. Comparing the results on panels A and B, it is clear that the R&D adjustment is important for improving the performance of the profitability factor.

Panel C-1 and C-2 regress  $r_{ROE}^{RD}$  on the related factors. The alphas of  $r_{ROE}^{RD}$  are significant against all included factors in both univariate and multiple regressions. Therefore, none of these factors subsume the performance of our R&D-adjusted earnings factor.

Gulen et al. (2025) base their factors on an economically plausible model in which a) investments in R&D and organizational capital (in the form of SG&A) depreciate over time at rates that differ across R&D and SG&A and across industries and b) not all of SG&A should be capitalized in organizational capital. They postulate a capitalization rate for R&D of 100% and for a depreciation rate for organizational capital of 20%. The capitalization rate for SG&A and the depreciation rate for R&D are estimated, by industry. They adjust both earnings in

the numerator and book value in the denominator of the ROE measure to reflect investments in intangibles.

Interestingly, the two intangibles-adjusted profitability factors introduced by Gulen et al. (2025), RMW(GLPZ) and ROE(GLPZ), exhibit strong performance relative to the original FF5 model but become insignificant when evaluated against the adjusted FF5 model incorporating  $r_{ROE}^{RD}$ . In contrast,  $r_{ROE}^{RD}$  demonstrates a significant alpha relative to the original RMW factor in both univariate and multivariate regressions and relative to the GLPZ factors (Table 4 and Table 9, Panel C). We hypothesize that the intangibles adjustment applied to the denominator of the valuation ratio may introduce additional noise into the profitability measures and substantially drive the underperformance of these earnings measures relative to our simpler adjustment. We provide supporting evidence for this hypothesis in Section 5.7.

## 5.6 Within Industry Performance

Figure A4 in the Internet Appendix shows that firms within HighTech and Health industries are highly R&D intensive and their R&D expenses as a fraction of total sales have been growing through the entire sample period. The R&D adjustment might, therefore, have a stronger impact on the firms within these two industries, although the insight from the dividend discount model (section 2.1) does not imply differences across industries. We examine the performance of the profitability factors within each of the Fama-French five industries and report them in Table 10. Specifically, we assign firms into two size groups based on median within-industry NYSE market equity and three profitability groups based on 30<sup>th</sup> and 70<sup>th</sup> percentile within-industry NYSE profitability at the June end of year  $t$ . The firms are held until the June end of the year  $t + 1$ . The within-industry profitability factors are  $1/2(\text{Small Robust} + \text{Big Robust}) - 1/2(\text{Small Weak} + \text{Big Weak})$  within each industry. As before, we select the earnings measures from the bottom-line earnings to the top-line, including IB, SPI, EBT, FF, RD, RD+.3SG, NM, and REVT.

Panel A shows that the High Tech and Health industries have the highest fraction of firms reporting positive R&D expenses (column  $RD > 0$ ) and the highest R&D expenses as fractions

of both book equity (column  $RD/B$ ) and total sales (column  $RD/REVT$ ). Panel B reports the mean excess return of the profitability factors and the associated t-statistics. The only factor portfolios that have a statistically significant mean return across all industries are our three factors  $RD$ ,  $RD+.3SG$ , and  $RD+SG$ ; and the Novy-Marx factor,  $NM$ . For the intangibles-adjusted earnings measure the point estimate of the factor risk premium is relatively stable across industries, as one might expect from the dividend discount model.

## 5.7 Alternative Scaling Factors

The valuation equation (3) uses the book value of equity as the scaling factor. The use of book equity makes our results directly comparable to those of Fama and French (2006, 2015). However, whether there are better scaling factors is an empirical question. We study profitability factors with an array of alternative scalings: revenue ( $REVT$ ), book value of equity ( $B$ ), total assets ( $AT$ ), gross PP&E plus intangible capital (as in Peters and Taylor (2017),  $B^{PT}$ ), and market equity. Table 11 reports the time-series average of the monthly value-weighted strategy returns and the alpha against FF5 (along with t-statistics). For all scaling factors either  $Y(RD)$  or  $Y(RD + .3SG)$  deliver the highest mean return and either  $Y(RD)$ ,  $Y(RD + .3SG)$ , or  $Y(RD + SG)$  deliver the highest t-statistics for the mean return. Similarly, either  $Y(RD)$ ,  $Y(RD + .3SG)$ , or  $Y(RD + SG)$  deliver the highest alphas (relative to FF5) and t-statistics for alpha.

For the three intangibles-adjusted definitions of earnings, using market value of equity ( $ME$ ) almost universally yields the smallest t-statistics for the alphas against FF5 across alternative scaling factors. Using intangibles-adjusted book value ( $B^{PT}$ ) almost universally yields the second-smallest t-statistics for the alphas against FF5 across alternative scaling factors. While scaling by intangibles-adjusted book value has some intuitive appeal, the adjustment for intangibles might yield a noisy estimate of adjusted book value. Gulen et al. (2025) propose intangibles-adjusted profitability factors that adjust both the numerator and denominator, with industry-specific capitalization rates for SG&A and depreciation rates for R&D. These industry-specific parameters are intended to reduce bias in the estimation of intangibles-adjusted book

values. However, the observation that RMW(GLPZ) and ROE(GLPZ) fail to produce significant alphas against the adjusted FF5 incorporating  $r_{\text{ROE}}^{\text{RD}}$  in Table 9 suggests that the loss of degrees of freedom from estimating a larger number of parameters may outweigh the gains in precision from bias reduction.

To test this hypothesis, we construct two measures of intangibles-adjusted ROE, one that incorporates their adjustment only in the numerator (that is, using their numerator and our denominator) and one that incorporates their adjustment only in the denominator (that is, using our numerator and their denominator). We conjecture that the errors in variables (EIV) problem is more damaging in the denominator due to Jensen’s inequality. The factor incorporating their numerator adjustment has an estimated alpha, relative to the FF5 model, of 0.26% ( $t$ -statistic equal to 4.26). The factor only incorporating their denominator adjustment has an estimated alpha, relative to the FF5 model, of 0.00% ( $t$ -statistic equal to -0.01). This is consistent with our conjecture and in line with the arguments favoring more top-line measures in Novy-Marx (2013).

## 5.8 Model Comparison Tests

We benchmark our R&D-adjusted profitability factor against leading models using the Barillas et al. (2020) squared Sharpe-ratio tests. Across the full sample and both pre-2000 and post-2000 subsamples, augmenting FF5, FF6, HMXZ5, or DHS3 with our factor (annual or monthly as appropriate) significantly raises the squared Sharpe ratio. The largest p-value across all augmented models and subsamples is 0.012. Replacing the benchmark profitability factor with our measure also improves FF5 and FF6 over the full sample and in each subsample; for FF5(C) the improvement is insignificant pre-2000 but significant post-2000, and for HMXZ5 the replacement is not significant pre-2000 and only marginally significant post-2000. Full magnitudes and p-values are reported in the In Table A6 of the Internet Appendix.

## 6 Rational or Biased Beliefs?

We used perturbation analysis of the present value relation, under the assumption that the clean surplus accounting relation holds Fama and French (2006), to motivate our use of the intangible adjusted profitability measure. We showed that adjusting for intangible investments that are expensed under GAAP when measuring profitability strengthens the positive association between future operating cash flows and past profitability, especially over longer horizons.<sup>16</sup> With rational expectations, the ability past profitability to predict future cash flows in the cross section of firms, controlling for the other variables in the present value relation, implies stocks with high profitability will earn higher future returns, on average. Although our finding that more profitable firms earned higher returns on average is consistent with rational expectations on the part of investors regarding the cross-section of future cash flows, it does not imply rationality on the part of investors. For example, investors may be optimistic (pessimistic) and irrationally expect higher (lower) future cash flows for low-profitability (high-profitability) stocks, resulting in their overvaluation (undervaluation). Investors are, subsequently, unpleasantly (pleasantly) surprised, leading to lower (higher) future returns on such stocks.

We examine such a possibility in the following ways. In the first examination, we assume that analysts' earnings forecasts are a reasonable proxy for investors' expected future cash flows. If investors are irrational, analysts would also be optimistic (pessimistic) about future earnings of low-profitability (high-profitability) firms, and their earnings forecast errors (forecast minus actual) would be more positive (negative). That is what we find, particularly for low-profitability firms.

In the second empirical exercise, we compare stocks' implied costs of capital (ICC) to ex post returns. Under rationality, a stock's implied cost of capital (ICC) represents a rational investor's buy-and-hold expected return on the stock, i.e., the discount rate for that stock. If the term structure of expected returns on the stock is flat, then the realized average returns on the stock would be an unbiased measure of what investors expected to get. In particular, we would expect the ICC for low-profitability stocks to be lower than that for high-profitability

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<sup>16</sup>See Tables 1 and 2.



stocks. Interestingly, this is not what we find.

Third, we examine whether there are return spikes during earnings announcement day windows with the returns being significantly lower for low-profitability stocks relative to high-profitability stocks. Interestingly, we do not find any significant difference. We also find that the cumulative returns on the high- and low-profitability stock portfolios accrue uniformly over time during the 12-month post-portfolio formation holding period. This is consistent with analysts revising their expectations smoothly over time.

Finally, using Fama-Macbeth cross-sectional regressions, we show that there exists a significant positive association between the intangibles-adjusted profitability characteristic and future stock returns after controlling for size, market-to-book, past returns, and asset growth. Consistent with the view that profitability’s association with future return is, in large part, due to its ability to predict analysts’ earnings forecast errors (a measure of over-optimism on the part of investors), profitability loses its significance when ex-post realized earnings forecast errors are introduced as another control variable. This is consistent with the profitability factor’s performance being driven by expectational errors.

## 6.1 Analysts’ Earnings Forecast Errors

We begin by examining whether profitability is systematically related to analysts’ earnings forecast errors (FEs). If the profitability characteristic’s ability to predict the cross-section of future stock returns stems from investors’ biased beliefs, and analysts’ earnings forecasts is a reasonable proxy for investors’ expectations, we should observe that analysts are overly optimistic (pessimistic) about the future earnings of low-profitability (high-profitability) firms. This implies a negative association between lagged profitability and subsequent FEs, defined as the difference between forecasted and realized earnings.

To test this, we estimate the following cross-sectional regression over the 1986 to 2023 period for which we have analysts’ forecast data for earnings (for a subset of firms):

$$FE_{i,t} = \delta_{0,t} + \delta_{1,t} ROE_{i,t-1} + Controls_{i,t-1} + \epsilon_{i,t} \quad (7)$$

FEs are constructed as the difference between the most recent consensus analyst forecast of fu-

ture earnings per share (IBES item MEANEST) and the subsequently realized earnings per share (IBES item ACTUAL), multiplied by the number of shares outstanding (IBES item SHOUT), and scaled by lagged total assets. Regressions are conducted at the end of June each year. At the time of estimation, analyst forecasts and shares outstanding are available, while actual earnings are realized ex post. We examine forecast errors (FEs) over the subsequent four fiscal quarters (EQ1-EQ4).

Table 12 presents the results. Both  $Y(\text{FF})/B$  and  $Y(\text{RD})/B$  are significantly negatively associated with future forecast errors, consistent with greater optimism (Pessimism) for low-profitability (high-profitability) firms. In contrast,  $Y(\text{NM})/B$  shows little significant association with FEs.

Motivated by the asymmetry between the long and short sides of the profitability portfolios documented in Table 7, we further investigate how the long and short sides predict FEs differently. Figure 2 complements this analysis by plotting the time-series averages of FEs over the 12 months before and after portfolio formation. FEs are computed at the end of June each year and are value-weighted by market capitalization at the time of portfolio formation.<sup>17</sup> The figure provides strong evidence that analysts systematically overestimate the earnings of low-profitability firms.

Regression results in Table 13 offer further confirmation, using dummy variables based on profitability measures. These regressions follow the same specification as Equation (7), but replace the continuous profitability measure with two dummies: ROE-L and ROE-H, which equal one for firms below the bottom 30th and above the top 70th percentiles of NYSE profitability, respectively, for a given measure. The results show that, for both  $Y(\text{FF})/B$  and  $Y(\text{RD})/B$ , only firms in the low-profitability group (ROE-L) are significantly and positively associated with future FEs. In contrast, there is little evidence of a systematic relationship between high profitability (ROE-H) and future FEs.

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<sup>17</sup>At the end of June in year  $t$ , we use analyst forecasts issued 12-23 months prior to the fiscal year-end and calculate forecast errors for the 12 months before and after portfolio formation.

## 6.2 Implied Cost of Capital (ICC)

We next examine whether differences in ICC are related to differences in their profitability. ICC is computed using the present value equation that equated the price of a stock to its expected future payouts discounted at its implied cost of capital. It is based on observed stock price, analyst earnings forecasts, and assumptions regarding future payout ratios. We use firm-level ICC estimates from Eskildsen et al. (2024), averaging across four standard estimation methods.

Table 14 reports average ICCs for portfolios sorted by size and profitability, using both equal- and value-weighting. Due to data limitations, the ICC sample is smaller, particularly among small and low-profitability firms. We confirm that the ex-post return spreads in this restricted sample remain robust and comparable to those in the full sample, suggesting no significant sample selection bias.

Turning to ICC differences at the portfolio formation date ( $ICC[0]$ ), we find that among small firms, high-profitability stocks exhibit higher ICCs, consistent with rational pricing. However, among large firms, the pattern reverses: high-profitability firms have lower ICCs, inconsistent with the idea that they earn higher returns due to higher expected discount rates. Averaging across size groups, ICC spreads are statistically insignificant. To examine whether return predictability reflects declining discount rates, we track ICCs from portfolio formation time to 12 months after portfolio formation. Table 14 show that ICCs decline more for high-profitability firms post-formation, which could also be a contributing factor to the observed realized return differentials between high and low profitability stocks.

Overall, ICCs of high and low profitability portfolios are not sufficiently different to explain the higher return on high profitability stocks.

## 6.3 Returns Around Earnings Announcements

We next examine whether the returns occur mostly around earnings announcement dates (EADs). If the return accumulation of profitability-sorted portfolios are concentrated in a short window around EADs, that would suggest that the earnings of the firms in the portfolios were

relatively more difficult to anticipate by market participants. Investors in such stocks may have more diverse views about the future earnings of such stocks, and prices may reflect the views of more optimistic investors for reasons discussed in Diether et al. (2002).

For each firm, we identify all EADs using Compustat data item RDQ and compute the three-day buy-and-hold return from one trading day before to one trading day after the announcement, denoted as  $R_{i,s}^{\text{EAD}}$ . For each portfolio, we calculate the equal-weighted average of firm-level  $R_{i,s}^{\text{EAD}}$  returns for each year. We then compute the time-series averages of these portfolio-level EAD returns and the high-minus-low return spreads, along with their associated t-statistics.

These results are presented under “Quarterly,  $R_{i,s}^{\text{EAD}}$ ” in Panel A of Table 15. Portfolios are rebalanced annually based on profitability measures:  $Y(\text{FF})/B$ ,  $Y(\text{RD})/B$ , and  $Y(\text{NM})/B$ . For each portfolio year  $t$ , EAD returns are aggregated over the interval from the end of June in year  $t$  to the end of June in year  $t + 1$ .

Panel A of Table 15 reports the average EAD returns. The H-L spread is positive for all profitability measures but statistically significant only for ROE(NM). EAD returns explain roughly 30% of the annual return spread for ROE(NM), but only 6% for ROE(RD). This suggests that the returns accrue steadily over time and not just around earnings announcement windows for the high and low profitability stocks in the profitability sorted portfolios. Panel B replicates the analysis using excess EAD returns (net of market), yielding consistent results.

Further, we examine whether EADs cluster in calendar time and whether return spikes coincide. We plot the average number of EADs by calendar day alongside cumulative returns over the same window (reported in Figures A5 and A6 of the Internet Appendix). Returns appear smoothly distributed, suggesting that clustering of EADs has only a limited impact on our results.

Overall, these findings suggest that the superior performance of intangibles-adjusted profitability measures is not concentrated around EADs. While investors being overoptimistic about the future of low profitability firms may be contributing to the abnormal returns on the profitability sorted portfolio, the underlying mechanism through which such optimism affects stock prices may be different than the one discussed in Diether et al. (2002).

## 6.4 Earnings Forecast Errors and Stock Returns

In this section, we provide empirical support for the view that investor-optimism plays a role in stocks of more profitable firms having higher future returns on average, using Fama-MacBeth regressions, following Wang (2025). We construct FEs as the difference between analysts' forecasted and realized earnings scaled by lagged assets, and assess their role using cross-sectional regressions. We begin with a baseline return regression:

$$r_{i,t} = b_{0,t} + b_{1,t} ROE_{i,t-1} + Controls_{i,t-1} + \eta_{i,t} \quad (8)$$

As shown in Table 16, the slope coefficients for  $Y(RD)/B$  and  $Y(NM)/B$  are significant when controls are included. However, when FEs which are known ex post by the time returns (dependent variables) are realized, are included as additional explanatory variables, as given in the equation below, profitability becomes insignificant.

$$r_{i,t} = b'_{0,t} + b'_{1,t} ROE_{i,t-1} + b'_{2,t} FE_{i,t-1} + Controls_{i,t-1} + \eta'_{i,t} \quad (9)$$

To understand why this is what we should expect to find when profitability predicts future returns through its ability to predict FEs, consider the following three regression equations where we omit control variables for expositional convenience:

$$r_{i,t} = b'_{0,t} + b'_{1,t} ROE_{i,t-1} + b'_{2,t} FE_{i,t-1} + \eta'_{i,t} \quad (10)$$

$$r_{i,t} = b_{0,t} + b_{1,t} ROE_{i,t-1} + \eta_{i,t} \quad (11)$$

$$FE_{i,t-1} = \delta_{0,t} + \delta_{1,t} ROE_{i,t-1} + \epsilon_{i,t} \quad (12)$$

where equations (10), (11), and (12) correspond to regressions (8), (9), and (7), respectively but without the controls. The coefficients  $b_{1,t}$  and  $b'_{1,t}$  are related by the identity:

$$b_{1,t} = b'_{1,t} + b'_{2,t} \delta_{1,t} \quad (13)$$

Equation (13) decomposes the total return predictability of ROE, as captured by  $b_{1,t}$ , into two components: (i) its direct effect on returns,  $b'_{1,t}$ , and (ii) its indirect effect through its correlation with investors' biased beliefs,  $b'_{2,t} \delta_{1,t}$ . To the extent that the relationship between ROE and FE reflects systematic belief biases, then a significant difference between  $b_{1,t}$  and  $b'_{1,t}$  implies that a substantial portion of ROE's return predictability stems from its ability to predict FE and

FE's association with returns.

Table 16 reports that the average slope coefficients of regressions (8) and (9). After controlling for FEs, the slope coefficient  $Y(RD)/B$  and  $Y(NM)/B$  become insignificant. To rule out the possibility that this is driven by reduced sample size, we re-estimate regression (8) using the same restricted sample. The coefficient on  $Y(RD)/B$  falls by one-third in magnitude when FEs are included, indicating that FEs captures a significant portion of its predictive power.  $Y(NM)/B$  becomes insignificant in the restricted sample. These results lend support for the view that the abnormal returns to the profitability factor portfolio are in part due to investors' optimistic (biased) views about the future.

Given the earlier finding that FEs are concentrated among low-profitability firms, we re-estimate regressions (8) and (9) using dummy variables for profitability. The results, reported in Table 17, provide additional insight. Consistent with Table 4, where the alphas of  $Y(RD)/B$  appears to arise from both long and short sides, we find that both ROE-L and ROE-H are statistically significant predictors of returns for  $Y(RD)/B$ , even after controlling for other variables.

In contrast, ROE-L is not significant for  $Y(FF)/B$ , and ROE-H is not significant for  $Y(NM)/B$ . Consistent with the results in Table 16, both  $Y(FF)/B$  and  $Y(NM)/B$  lose significance when using the restricted sample, whereas both ROE-L and ROE-H remain significant for  $Y(RD)/B$ . Interestingly, once FEs are controlled for, the coefficient on ROE-L becomes insignificant, while the coefficient on ROE-H becomes even more significant. This pattern aligns with the evidence in Table 13 and Figure 2, which show that FEs are concentrated on the low-profitability side.

These results suggest that the underperformance of low-profitability firms is likely driven by investor biased beliefs, as captured by FEs. In contrast, the outperformance of high-profitability firms appears unrelated to FEs.

## 7 Conclusions

Fama and French (2015) develop a clever method for constructing an economy-wide pervasive risk factor that captures shocks to firms' profitability, denoted RMW (robust-minus-weak).

Fama and French (2015) use earnings before taxes and depreciation (EBTD) to measure earnings. Even though R&D and some components of SG&A expenditures should rightly be treated as investments in intangible assets, GAAP requires full expensing of such expenditures over our sample period. We use the Fama-French definition of earnings (EBTD) but adjust it to reflect investments in intangibles by adding the R&D expenditures (or R&D and a fraction of SG&A). We call the resulting earnings intangibles-adjusted earnings.

Intangibles-adjusted earnings forecast the cross-section of future operating cash flows and cash dividends of firms better than unadjusted earnings at longer horizons. Our intangibles-adjusted factors have statistically and economically significant alphas against an array of existing factor models. Replacing the profitability factor in FF5, RMW, by our R&D-adjusted profitability factor provides an intangibles-adjusted model that can subsume various anomalies, like price momentum. We intentionally do not adjust book equity for intangible assets in order to minimize specification and measurement errors. Our concern is that firms' differing abilities to create intangible assets through in-house expenditures plus estimation errors of the associated parameters may lead to worse performance when adjusting the denominator. Our empirical analysis is consistent with this conjecture.

The long side of our intangibles-adjusted factor portfolios picks stocks that outperform the market, on a risk-adjusted basis, independently of whether the market has positive or negative excess returns. The short side of our intangibles-adjusted factor portfolios does well by shorting stocks that do particularly poorly in down markets, thus giving the strategy a put-like, or quality, feature. Thus, the superior performance of the intangibles-adjusted factors comes from security selection on the long side and the put-like feature on the short side.

Our factors have negative market betas and a put feature. These features would normally lead one to expect them to have a negative risk premium, but they have a positive risk premium, thus yielding alpha relative to an array of extant models. The intangibles-adjusted factors add significant explanatory power for the cross-section of asset returns.

We also provide evidence supportive of the view that some of the abnormal returns on portfolios constructed by sorting on profitability may be due to investors' optimistic views of

the future, particularly for the underperformance of low-profitability firms's stocks.

## References

- Arnott, Robert D, Campbell R Harvey, Vitali Kalesnik, Juhani T Linnainmaa. 2021. Reports of value's death may be greatly exaggerated. *Financial Analysts Journal* **77**(1) 44–67.
- Asness, Clifford, Andrea Frazzini. 2013. The devil in hml's details. *The Journal of Portfolio Management* **39**(4) 49–68. URL <https://doi.org/10.3905/jpm.2013.39.4.049>.
- Asness, Clifford S, Andrea Frazzini, Lasse Heje Pedersen. 2019. Quality minus junk. *Review of Accounting Studies* **57** 34–112. URL <https://doi.org/10.1007/s11142-018-9470-2>.
- Ball, Ray, Joseph Gerakos, Juhani T Linnainmaa, Valeri Nikolaev. 2016. Accruals, cash flows, and operating profitability in the cross section of stock returns. *Journal of Financial Economics* **121**(1) 28–45.
- Ball, Ray, Joseph Gerakos, Juhani T Linnainmaa, Valeri V Nikolaev. 2015. Deflating profitability. *Journal of Financial Economics* **117**(2) 225–248.
- Barillas, Francisco, Raymond Kan, Cesare Robotti, Jay Shanken. 2020. Model comparison with sharpe ratios. *Journal of Financial and Quantitative Analysis* **55**(6) 1840–1874.
- Barillas, Francisco, Jay Shanken. 2017. Which alpha? *The Review of Financial Studies* **30**(4) 1316–1338.
- Belo, Frederico, Vito D Gala, Juliana Salomao, Maria Ana Vitorino. 2022. Decomposing firm value. *Journal of Financial Economics* **143**(2) 619–639.
- Berk, Jonathan B. 1995. A critique of size-related anomalies. *The Review of Financial Studies* **8**(2) 275–286. URL <http://www.jstor.org/stable/2962273>.
- Bloomberg, Lawrence N. 1934. The investment value of goodwill. Ph.D. thesis, Johns Hopkins University, Baltimore, MD.
- Carhart, Mark M. 1997. On persistence in mutual fund performance. *The Journal of Finance* **52**(1) 57–82. URL <http://www.jstor.org/stable/2329556>.
- Chan, Louis KC, Narasimhan Jegadeesh, Josef Lakonishok. 1996. Momentum strategies. *The Journal of Finance* **51**(5) 1681–1713.
- Connor, Gregory, Robert A. Korajczyk. 1986. Performance measurement with the arbitrage pricing theory: A new framework for analysis. *Journal of Financial Economics*



- 15**(3) 373–394. doi:[https://doi.org/10.1016/0304-405X\(86\)90027-9](https://doi.org/10.1016/0304-405X(86)90027-9). URL <https://www.sciencedirect.com/science/article/pii/0304405X86900279>.
- Crouzet, Nicolas, Janice Eberly. 2023. Rents and intangible capital: A q+ framework. *The Journal of Finance* **78**(4) 1873–1916.
- Daniel, Kent, David Hirshleifer, Lin Sun. 2020. Short-and long-horizon behavioral factors. *The Review of Financial Studies* **33**(4) 1673–1736.
- Diether, Karl B, Christopher J Malloy, Anna Scherbina. 2002. Differences of opinion and the cross section of stock returns. *The journal of finance* **57**(5) 2113–2141.
- Eisfeldt, Andrea L., Edward T. Kim, Dimitris Papanikolaou. 2022. Intangible value. *Critical Finance Review* **11**(2) 299–332.
- Eisfeldt, Andrea L, Dimitris Papanikolaou. 2013. Organization capital and the cross-section of expected returns. *The Journal of Finance* **68**(4) 1365–1406.
- Eskildsen, Marc, Markus Ibert, Theis Ingerslev Jensen, Lasse Heje Pedersen. 2024. In search of the true greenium. *Available at SSRN* .
- Fama, Eugene F, Kenneth R French. 1993. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics* **33**(1) 3–56.
- Fama, Eugene F, Kenneth R French. 2006. Profitability, investment and average returns. *Journal of Financial Economics* **82**(3) 491–518.
- Fama, Eugene F, Kenneth R French. 2015. A five-factor asset pricing model. *Journal of Financial Economics* **116**(1) 1–22.
- Fama, Eugene F, Kenneth R French. 2016. Dissecting anomalies with a five-factor model. *The Review of Financial Studies* **29**(1) 69–103.
- Fama, Eugene F, Kenneth R French. 2018. Choosing factors. *Journal of Financial Economics* **128**(2) 234–252.
- FASB. 1974. Statement of financial accounting concepts no. 2, accounting for research and development costs. *Financial Accounting Standards Board, Norwalk, CT* .
- Frazzini, Andrea, David Kabiller, Lasse Heje Pedersen. 2018. Buffett’s alpha. *Financial Analysts Journal* **74**(4) 35–55.
- Gebhardt, William R, Charles MC Lee, Bhaskaran Swaminathan. 2001. Toward an implied cost of capital. *Journal of Accounting Research* **39**(1) 135–176.

- Goyal, Amit, Sunil Wahal. 2023. R&d, innovation, and the stock market. *Available at SSRN 4568392* .
- Graham, Benjamin. 1973. The intelligent investor, a book of practical counsel, 4th rev. ed. *New York: Harper & Brothers Publishers* .
- Graham, Benjamin, David L. Dodd, Sidney Cottle. 1962. *Security Analysis: Principles and Technique*. McGraw-Hill Book Company, New York, NY.
- Greenwald, Bruce C, Judd Kahn, Paul Sonkin, Michael van Biema. 2001. Value investing: from graham to buffett and beyond. *John Wiley & Sons* 107–137.
- Gulen, Huseyin, Dongmei Li, Ryan H. Peters, Morad Zekhnini. 2025. Intangible capital in factor models. *Management Science* **71**(2) 1756–1778. URL <https://doi.org/10.1287/mnsc.2022.01261>.
- Henriksson, Roy D., Robert C. Merton. 1981. On market timing and investment performance. ii. statistical procedures for evaluating forecasting skills. *The Journal of Business* **54**(4) 513–533. URL <http://www.jstor.org/stable/2352722>.
- Hou, Kewei, Haitao Mo, Chen Xue, Lu Zhang. 2021. An augmented q-factor model with expected growth. *Review of Finance* **25**(1) 1–41.
- Hou, Kewei, Chen Xue, Lu Zhang. 2015. Digesting anomalies: An investment approach. *The Review of Financial Studies* **28**(3) 650–705.
- Jagannathan, Ravi, Robert A. Korajczyk. 1986. Assessing the market timing performance of managed portfolios. *The Journal of Business* **59**(2) 217–235. URL <http://www.jstor.org/stable/2353018>.
- Jagannathan, Ravi, Ellen McGrattan, Anna Scherbina. 2001. The declining us equity premium.
- Jagannathan, Ravi, Yang Zhang. 2020. A return based measure of firm quality. Tech. rep., National Bureau of Economic Research.
- Jegadeesh, Narasimhan, Sheridan Titman. 1993. Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance* **48**(1) 65–91.
- Kim, Soohun, Robert A. Korajczyk, Andreas Neuhierl. 2021. Arbitrage Portfolios. *The Review of Financial Studies* **34**(6) 2813–2856. doi:10.1093/rfs/hhaa102. URL <https://doi.org/10.1093/rfs/hhaa102>.
- Koh, Ping-Sheng, David M Reeb. 2015. Missing r&d. *Journal of Accounting and Economics* **60**(1) 73–94.

- Lev, Baruch, Srivastava Anup. 2022. Explaining the recent failure of value investing. *Critical Finance Review* **11**(2) 333–360.
- Lev, Baruch, Theodore Sougiannis. 1996. The capitalization, amortization, and value-relevance of r&d. *Journal of Accounting and Economics* **21**(1) 107–138.
- Li, Kevin K, Partha Mohanram. 2014. Evaluating cross-sectional forecasting models for implied cost of capital. *Review of Accounting Studies* **19**(3) 1152–1185.
- Monhanram, Partha S. 2005. Separating winners from losers among lowbook-to-market stocks using financial statement analysis. *Review of Accounting Studies* **10** 133–170.
- Novy-Marx, Robert. 2013. The other side of value: The gross profitability premium. *Journal of Financial Economics* **108**(1) 1–28.
- Peters, Ryan H, Lucian A Taylor. 2017. Intangible capital and the investment-q relation. *Journal of Financial Economics* **123**(2) 251–272.
- Piotroski, Joseph D. 2000. Value investing: The use of historical financial statement information to separate winners from losers. *Journal of Accounting Research* **38** 1–41. URL <http://www.jstor.org/stable/2672906>.
- Stambaugh, Robert F, Yu Yuan. 2017. Mispricing factors. *The Review of Financial Studies* **30**(4) 1270–1315.
- Wang, Kai. 2023. Organizational capital, tobin’s q, and the cross section of stock returns. *Tobin’s Q, and the Cross Section of Stock Returns (February 19, 2023)* URL <http://dx.doi.org/10.2139/ssrn.4364379>.
- Wang, Kai. 2025. Stock market anamolies: Mispricing or risk premia? insights from a new methodology. *manuscript* .

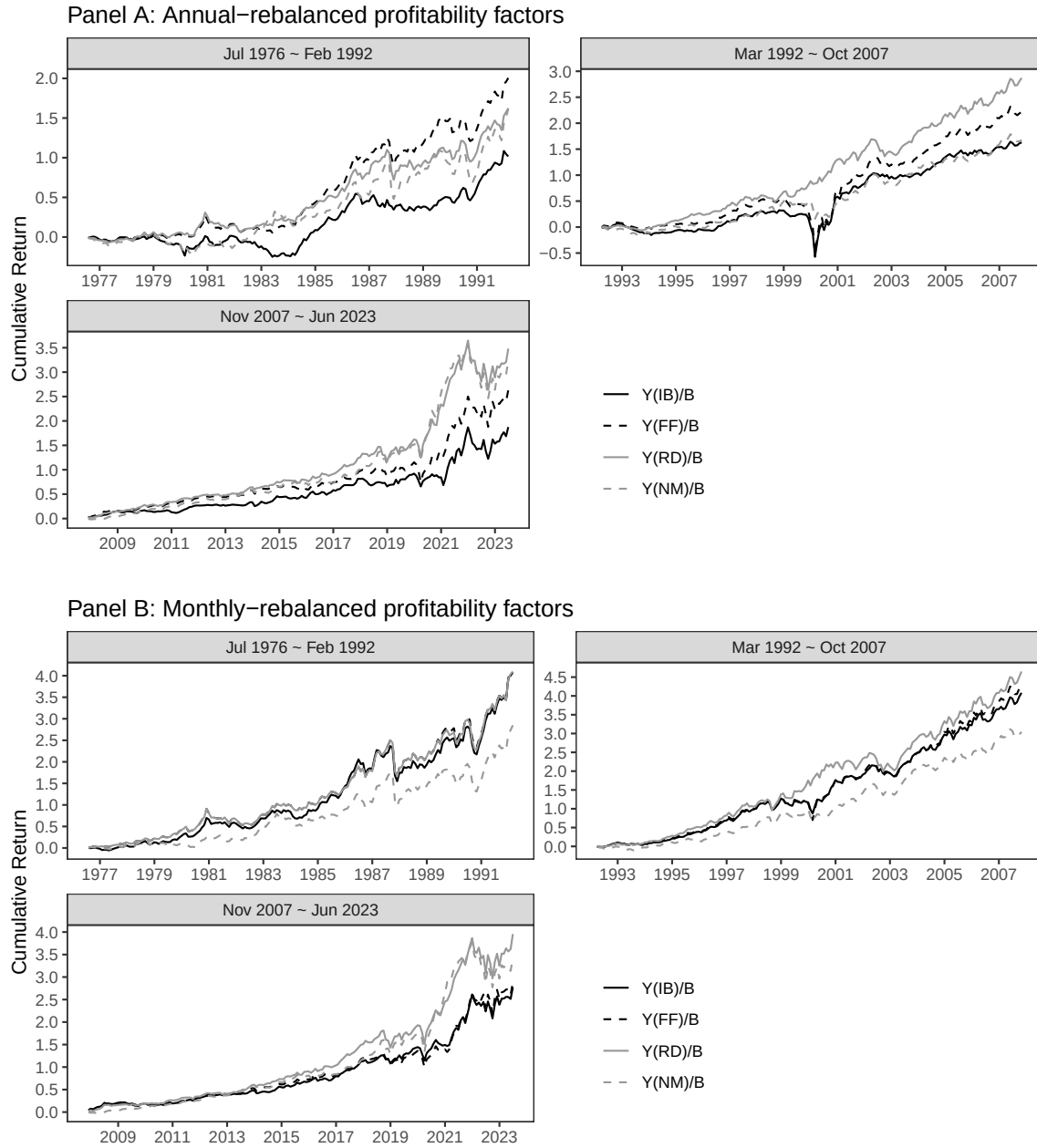


Figure 1: Cumulative return of annual- and monthly-rebalanced profitability factors, July 1976 to June 2023. Cumulative returns for each subsample are compounded long-side of the factor minus the compounded short-side of the factor.

ALT TEXT: The figures show the cumulative returns of profitability factors across three subsamples: July 1976 – Feb 1992, March 1992 – Oct 2007, and Nov 2007 – June 2023. For annual-rebalanced factors,  $Y(FF)/B$  performs best in the first subsample, while  $Y(RD)/B$  leads in the latter two. For monthly-rebalanced factors,  $Y(RD)/B$  consistently outperforms across all periods.

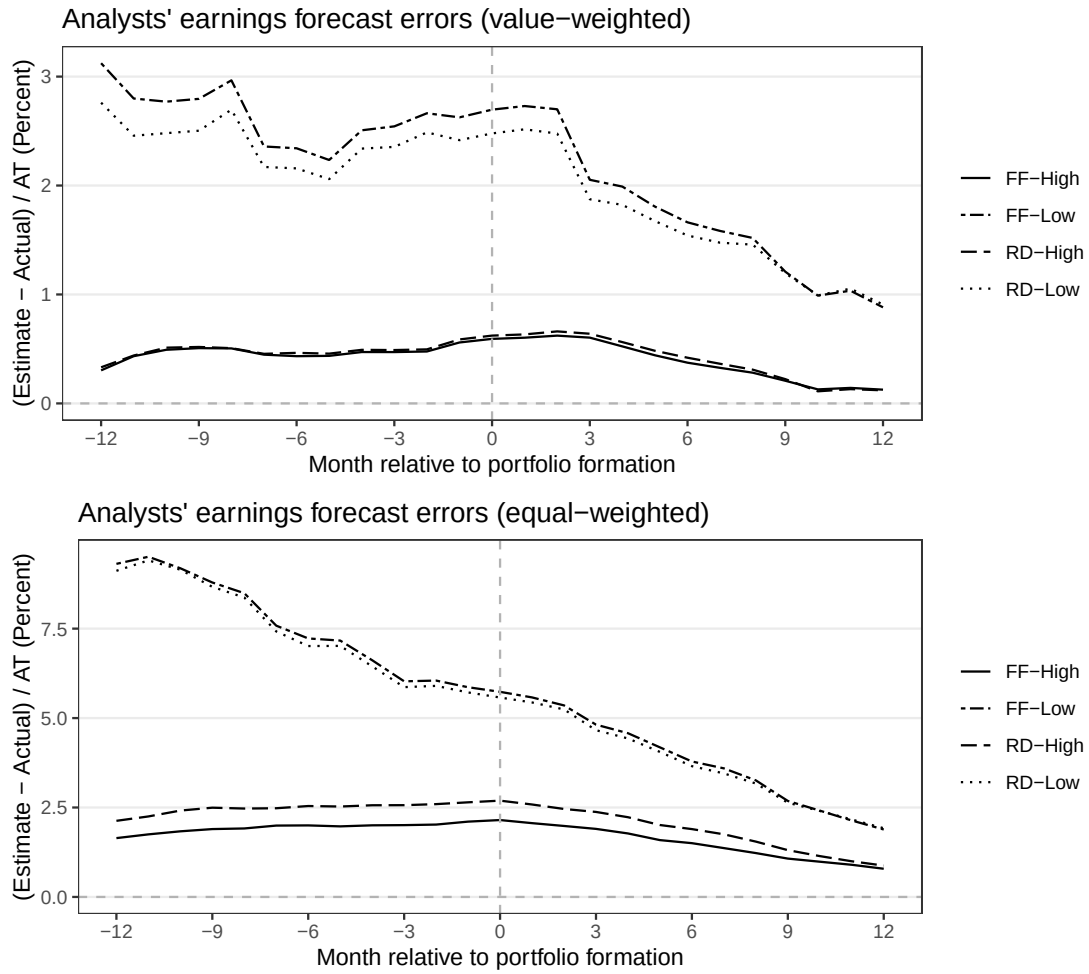


Figure 2: Average of earnings forecast errors 12-month before and after portfolio formation for high and low profitability portfolios, 1986 to 2023.

For a portfolio, earnings forecast error is the aggregate earnings forecast error divided by the aggregate total assets. At the time of portfolio formation (month 0), aggregate earnings forecast error is the sum of the product of firm-level EPS forecast errors, that are made 11 to 23 months prior to forecast period ends, and the total number of common shares across all firms in the portfolio. The portfolio earnings forecast error that are  $t$  months after (before) the portfolio formation time is computed following the same procedure except that the firm-level EPS forecast errors are  $t$ -month forward (lagged) values. The considered profitability measures include  $Y(\text{FF})/B$  and  $Y(\text{RD})/B$ .

The difference in value-weighted forecast errors between high and low  $Y(\text{FF})/B$  and  $Y(\text{RD})/B$  portfolios is -2.10% (t-statistic=-2.01) and -1.86% (t-statistic=-1.95), respectively. The difference in equal-weighted forecast errors between high and low  $Y(\text{FF})/B$  and  $Y(\text{RD})/B$  portfolios is -3.58% (t-statistic=-4.58) and -2.88% (t-statistic=-3.60), respectively.

ALT TEXT: The figures plot the average one-year-ahead analyst forecast errors at the time of portfolio formation and their evolution from 12 months before to 12 months after formation. Analysts substantially overestimate earnings for low-profitability firms, while forecasts for high-profitability firms are more accurate.

Table 1: Cross-section univariate regression of future profitability (operating cash flows) at the portfolio level (1978 - 2022).

This table reports the time-series averages of the slope coefficients, t-statistics, and r-squared for the cross-section regressions of future operating cash flows (divided by current book value of equity) at the portfolio level. The operating cash flow is the Compustat data item OANCF which becomes available on Compustat after 1988. Firms are assigned into 5-by-10 industry and ROE(FF) groups in year  $t$ . Industry groups are based on Fama-French five industry classification and ROE(FF) cut points are NYSE ROE(FF) decile within each industry group. For notation convenience, we denote  $Z_{t+\tau}^X$  as  $Y_{t+\tau}^X/B_t$  where  $X$  includes IB, NM, FF, RD, RD+.3SG, and RD+SG. The portfolio-level earnings and book equity are the value-weighted earnings and book equity for a given portfolio in year  $t$ . The portfolio-level future cash flows is computed following equation (4). We also test the null hypothesis that the difference of the r-squared time series between  $Z_{t+\tau}^{\text{NM}}$ ,  $Z_{t+\tau}^{\text{FF}}$ ,  $Z_{t+\tau}^{\text{RD}}$ ,  $Z_{t+\tau}^{\text{RD+.3SG}}$ ,  $Z_{t+\tau}^{\text{RD+SG}}$  and  $Z_{t+\tau}^{\text{IB}}$ ,  $Z_{t+\tau}^{\text{NM}}$ ,  $Z_{t+\tau}^{\text{FF}}$ ,  $Z_{t+\tau}^{\text{RD}}$ ,  $Z_{t+\tau}^{\text{RD+.3SG}}$  is zero and report the t-statistics in the panel B and D. We remove firms with total assets less than \$12.5 million or book equity less than \$25 million in year  $t$ .

| Panel A: Average slopes for $Z_t^{\text{IB}}$ , $Z_t^{\text{NM}}$ , and $Z_t^{\text{FF}}$         |       |                   |                   |                         |       |                   |                        |                         |       |                   |                      |
|---------------------------------------------------------------------------------------------------|-------|-------------------|-------------------|-------------------------|-------|-------------------|------------------------|-------------------------|-------|-------------------|----------------------|
|                                                                                                   | $R^2$ | Intc              | $Z_t^{\text{IB}}$ |                         | $R^2$ | Intc              | $Z_t^{\text{NM}}$      |                         | $R^2$ | Intc              | $Z_t^{\text{FF}}$    |
| $Z_{t+1}^{\text{OCF}}$                                                                            | 0.56  | 0.15              | 1.03              | $Z_{t+1}^{\text{OCF}}$  | 0.58  | 0.04              | 0.31                   | $Z_{t+1}^{\text{OCF}}$  | 0.74  | 0.06              | 0.71                 |
| $Z_{t+2}^{\text{OCF}}$                                                                            | 0.50  | 0.17              | 1.05              | $Z_{t+2}^{\text{OCF}}$  | 0.55  | 0.04              | 0.32                   | $Z_{t+2}^{\text{OCF}}$  | 0.67  | 0.08              | 0.72                 |
| $Z_{t+3}^{\text{OCF}}$                                                                            | 0.47  | 0.19              | 1.07              | $Z_{t+3}^{\text{OCF}}$  | 0.53  | 0.05              | 0.34                   | $Z_{t+3}^{\text{OCF}}$  | 0.62  | 0.10              | 0.73                 |
| $Z_{t+4}^{\text{OCF}}$                                                                            | 0.45  | 0.21              | 1.15              | $Z_{t+4}^{\text{OCF}}$  | 0.51  | 0.06              | 0.36                   | $Z_{t+4}^{\text{OCF}}$  | 0.58  | 0.11              | 0.77                 |
| $Z_{t+5}^{\text{OCF}}$                                                                            | 0.41  | 0.23              | 1.18              | $Z_{t+5}^{\text{OCF}}$  | 0.47  | 0.08              | 0.36                   | $Z_{t+5}^{\text{OCF}}$  | 0.53  | 0.13              | 0.78                 |
| $Z_{t+6}^{\text{OCF}}$                                                                            | 0.38  | 0.26              | 1.21              | $Z_{t+6}^{\text{OCF}}$  | 0.44  | 0.10              | 0.38                   | $Z_{t+6}^{\text{OCF}}$  | 0.49  | 0.16              | 0.79                 |
| $Z_{t+7}^{\text{OCF}}$                                                                            | 0.35  | 0.28              | 1.26              | $Z_{t+7}^{\text{OCF}}$  | 0.40  | 0.12              | 0.39                   | $Z_{t+7}^{\text{OCF}}$  | 0.44  | 0.18              | 0.83                 |
| $Z_{t+8}^{\text{OCF}}$                                                                            | 0.30  | 0.30              | 1.32              | $Z_{t+8}^{\text{OCF}}$  | 0.35  | 0.14              | 0.40                   | $Z_{t+8}^{\text{OCF}}$  | 0.37  | 0.21              | 0.84                 |
| $Z_{t+9}^{\text{OCF}}$                                                                            | 0.28  | 0.33              | 1.41              | $Z_{t+9}^{\text{OCF}}$  | 0.34  | 0.16              | 0.43                   | $Z_{t+9}^{\text{OCF}}$  | 0.35  | 0.24              | 0.87                 |
| $Z_{t+10}^{\text{OCF}}$                                                                           | 0.25  | 0.36              | 1.51              | $Z_{t+10}^{\text{OCF}}$ | 0.31  | 0.18              | 0.46                   | $Z_{t+10}^{\text{OCF}}$ | 0.31  | 0.27              | 0.90                 |
| Panel B: t-statistics for Panel A                                                                 |       |                   |                   |                         |       |                   |                        |                         |       |                   |                      |
|                                                                                                   | Intc  | $Z_t^{\text{IB}}$ |                   | $\Delta R^2$<br>(NM-IB) | Intc  | $Z_t^{\text{NM}}$ |                        | $\Delta R^2$<br>(FF-NM) | Intc  | $Z_t^{\text{FF}}$ |                      |
| 1                                                                                                 | 11.09 | 13.67             | 1                 | 0.55                    | 3.87  | 19.25             | 1                      | 9.33                    | 4.66  | 24.66             |                      |
| 2                                                                                                 | 10.74 | 15.41             | 2                 | 1.64                    | 4.34  | 17.99             | 2                      | 6.83                    | 5.96  | 25.76             |                      |
| 3                                                                                                 | 12.55 | 12.62             | 3                 | 2.64                    | 5.27  | 20.39             | 3                      | 5.36                    | 7.16  | 22.07             |                      |
| 4                                                                                                 | 14.64 | 13.32             | 4                 | 2.49                    | 5.89  | 21.22             | 4                      | 4.19                    | 7.60  | 20.80             |                      |
| 5                                                                                                 | 21.14 | 13.84             | 5                 | 2.46                    | 7.02  | 17.58             | 5                      | 3.65                    | 10.36 | 18.72             |                      |
| 6                                                                                                 | 25.98 | 16.06             | 6                 | 2.60                    | 7.62  | 17.82             | 6                      | 2.64                    | 12.13 | 18.55             |                      |
| 7                                                                                                 | 30.84 | 24.00             | 7                 | 1.85                    | 5.33  | 12.30             | 7                      | 2.21                    | 13.11 | 17.48             |                      |
| 8                                                                                                 | 32.62 | 23.43             | 8                 | 2.19                    | 4.41  | 8.76              | 8                      | 1.66                    | 12.52 | 13.78             |                      |
| 9                                                                                                 | 37.63 | 28.63             | 9                 | 2.61                    | 5.83  | 11.01             | 9                      | 0.72                    | 13.64 | 16.65             |                      |
| 10                                                                                                | 48.36 | 28.94             | 10                | 3.08                    | 6.54  | 13.84             | 10                     | -0.29                   | 10.73 | 13.24             |                      |
| Panel C: Average slopes for $Z_t^{\text{RD}}$ , $Z_t^{\text{RD+.3SG}}$ , and $Z_t^{\text{RD+SG}}$ |       |                   |                   |                         |       |                   |                        |                         |       |                   |                      |
|                                                                                                   | $R^2$ | Intc              | $Z_t^{\text{RD}}$ |                         | $R^2$ | Intc              | $Z_t^{\text{RD+.3SG}}$ |                         | $R^2$ | Intc              | $Z_t^{\text{RD+SG}}$ |
| $Z_{t+1}^{\text{OCF}}$                                                                            | 0.71  | 0.04              | 0.64              | $Z_{t+1}^{\text{OCF}}$  | 0.68  | 0.03              | 0.52                   | $Z_{t+1}^{\text{OCF}}$  | 0.55  | 0.05              | 0.31                 |
| $Z_{t+2}^{\text{OCF}}$                                                                            | 0.66  | 0.05              | 0.66              | $Z_{t+2}^{\text{OCF}}$  | 0.65  | 0.04              | 0.55                   | $Z_{t+2}^{\text{OCF}}$  | 0.53  | 0.06              | 0.33                 |
| $Z_{t+3}^{\text{OCF}}$                                                                            | 0.62  | 0.07              | 0.68              | $Z_{t+3}^{\text{OCF}}$  | 0.61  | 0.05              | 0.56                   | $Z_{t+3}^{\text{OCF}}$  | 0.51  | 0.07              | 0.34                 |
| $Z_{t+4}^{\text{OCF}}$                                                                            | 0.59  | 0.08              | 0.72              | $Z_{t+4}^{\text{OCF}}$  | 0.59  | 0.06              | 0.59                   | $Z_{t+4}^{\text{OCF}}$  | 0.49  | 0.09              | 0.36                 |
| $Z_{t+5}^{\text{OCF}}$                                                                            | 0.56  | 0.10              | 0.74              | $Z_{t+5}^{\text{OCF}}$  | 0.55  | 0.08              | 0.61                   | $Z_{t+5}^{\text{OCF}}$  | 0.46  | 0.10              | 0.37                 |
| $Z_{t+6}^{\text{OCF}}$                                                                            | 0.53  | 0.13              | 0.76              | $Z_{t+6}^{\text{OCF}}$  | 0.53  | 0.10              | 0.63                   | $Z_{t+6}^{\text{OCF}}$  | 0.44  | 0.13              | 0.38                 |
| $Z_{t+7}^{\text{OCF}}$                                                                            | 0.48  | 0.14              | 0.80              | $Z_{t+7}^{\text{OCF}}$  | 0.48  | 0.12              | 0.67                   | $Z_{t+7}^{\text{OCF}}$  | 0.40  | 0.14              | 0.40                 |
| $Z_{t+8}^{\text{OCF}}$                                                                            | 0.42  | 0.17              | 0.83              | $Z_{t+8}^{\text{OCF}}$  | 0.42  | 0.14              | 0.70                   | $Z_{t+8}^{\text{OCF}}$  | 0.36  | 0.16              | 0.42                 |
| $Z_{t+9}^{\text{OCF}}$                                                                            | 0.40  | 0.19              | 0.86              | $Z_{t+9}^{\text{OCF}}$  | 0.41  | 0.16              | 0.73                   | $Z_{t+9}^{\text{OCF}}$  | 0.35  | 0.18              | 0.44                 |

|                                      |                         |      |                   |                         |                               |      |                                 |                         |                            |      |                             |
|--------------------------------------|-------------------------|------|-------------------|-------------------------|-------------------------------|------|---------------------------------|-------------------------|----------------------------|------|-----------------------------|
| $Z_{t+10}^{\text{OCF}}$              | 0.35                    | 0.22 | 0.90              | $Z_{t+10}^{\text{OCF}}$ | 0.37                          | 0.18 | 0.77                            | $Z_{t+10}^{\text{OCF}}$ | 0.32                       | 0.20 | 0.47                        |
| Panel D: $t$ -statistics for Panel C |                         |      |                   |                         |                               |      |                                 |                         |                            |      |                             |
|                                      | $\Delta R^2$<br>(RD-FF) | Intc | $Z_t^{\text{RD}}$ |                         | $\Delta R^2$<br>(RD+.3 SG-FF) | Intc | $Z_t^{\text{RD}+.3 \text{ SG}}$ |                         | $\Delta R^2$<br>(RD+SG-FF) | Intc | $Z_t^{\text{RD}+\text{SG}}$ |
| 1                                    | -7.04                   | 3.77 | 23.23             | 1                       | -6.87                         | 2.60 | 22.61                           | 1                       | -10.93                     | 5.25 | 16.91                       |
| 2                                    | -1.71                   | 4.87 | 26.01             | 2                       | -2.74                         | 3.35 | 23.39                           | 2                       | -7.99                      | 5.67 | 16.99                       |
| 3                                    | 0.42                    | 6.09 | 24.38             | 3                       | -0.67                         | 4.48 | 23.52                           | 3                       | -6.56                      | 6.87 | 17.90                       |
| 4                                    | 2.03                    | 6.78 | 24.17             | 4                       | 0.73                          | 5.03 | 23.68                           | 4                       | -5.35                      | 7.31 | 17.85                       |
| 5                                    | 4.22                    | 9.12 | 21.57             | 5                       | 1.86                          | 7.40 | 22.57                           | 5                       | -3.68                      | 9.16 | 17.24                       |
| 6                                    | 5.38                    | 9.66 | 19.52             | 6                       | 3.18                          | 8.30 | 21.60                           | 6                       | -2.88                      | 9.68 | 16.93                       |
| 7                                    | 4.75                    | 8.75 | 16.27             | 7                       | 3.13                          | 6.94 | 16.63                           | 7                       | -2.02                      | 7.23 | 13.03                       |
| 8                                    | 6.01                    | 7.21 | 11.85             | 8                       | 4.60                          | 5.52 | 11.65                           | 8                       | -1.10                      | 5.80 | 9.42                        |
| 9                                    | 7.03                    | 9.78 | 15.70             | 9                       | 5.72                          | 8.26 | 15.87                           | 9                       | 0.00                       | 8.63 | 13.00                       |
| 10                                   | 6.52                    | 9.16 | 16.14             | 10                      | 5.39                          | 8.95 | 18.99                           | 10                      | 0.82                       | 9.55 | 15.67                       |

Table 2: Cross-section multiple regression of future profitability (operating cash flows) at the portfolio level (1978 - 2022).

This table reports the time-series averages of the slope coefficients, t-statistics, and adjusted r-squared for the cross-section regressions of future operating cash flows (divided by current book value of equity) at the portfolio level. The operating cash flow is the Compustat data item OANCF which becomes available on Compustat after 1988. The explanatory variables include Fama-French operating profits divided by book equity ( $Z_{t+\tau}^{FF}$ ), R&D costs divided by book equity ( $RD_t$ ), SG&A costs divided by book equity ( $SGA_t$ ), change of total assets divided by one-year lagged total assets ( $AG_t$ ), and book-to-market ( $BM_t$ ). We separate R&D costs and SG&A costs following Peters and Taylor (2017). Firms are assigned into 5-by-10 industry and ROE(FF) groups in year  $t$ . Industry groups are based on Fama-French five industry classification and ROE(FF) cut points are NYSE ROE(FF) decile within each industry group. The portfolio-level earnings, R&D costs, SG&A costs, total assets, and book equity are the value-weighted for a given portfolio in year  $t$ . The portfolio-level future cash flows is computed following equation (4). We remove firms with total assets less than \$12.5 million or book equity less than \$25 million in year  $t$ .

| Panel A: Average slopes |            |      |            |            |             |        |        |
|-------------------------|------------|------|------------|------------|-------------|--------|--------|
|                         | Adj. $R^2$ | Intc | $Z_t^{FF}$ | $RD_t/B_t$ | $SGA_t/B_t$ | $AG_t$ | $BM_t$ |
| $Z_{t+1}^{OCF}$         | 0.76       | 0.04 | 0.70       | 0.17       | 0.02        | -0.12  | 0.03   |
| $Z_{t+2}^{OCF}$         | 0.70       | 0.06 | 0.70       | 0.24       | 0.05        | -0.15  | 0.02   |
| $Z_{t+3}^{OCF}$         | 0.65       | 0.11 | 0.68       | 0.25       | 0.05        | -0.13  | -0.04  |
| $Z_{t+4}^{OCF}$         | 0.61       | 0.14 | 0.69       | 0.29       | 0.05        | -0.07  | -0.07  |
| $Z_{t+5}^{OCF}$         | 0.57       | 0.18 | 0.65       | 0.44       | 0.07        | -0.09  | -0.12  |
| $Z_{t+6}^{OCF}$         | 0.55       | 0.21 | 0.65       | 0.48       | 0.08        | -0.10  | -0.13  |
| $Z_{t+7}^{OCF}$         | 0.51       | 0.26 | 0.64       | 0.57       | 0.09        | -0.09  | -0.17  |
| $Z_{t+8}^{OCF}$         | 0.44       | 0.32 | 0.60       | 0.66       | 0.09        | -0.04  | -0.24  |
| $Z_{t+9}^{OCF}$         | 0.43       | 0.36 | 0.60       | 0.69       | 0.12        | 0.03   | -0.27  |
| $Z_{t+10}^{OCF}$        | 0.41       | 0.47 | 0.52       | 0.64       | 0.13        | 0.06   | -0.35  |
| Panel B: t-statistics   |            |      |            |            |             |        |        |
|                         |            | Intc | $Z_t^{FF}$ | $RD_t/B_t$ | $SGA_t/B_t$ | $AG_t$ | $BM_t$ |
| 1                       |            | 2.76 | 24.23      | 4.56       | 1.21        | -4.30  | 1.39   |
| 2                       |            | 2.47 | 22.77      | 4.58       | 2.04        | -5.78  | 0.51   |
| 3                       |            | 3.93 | 16.34      | 4.26       | 2.35        | -3.02  | -1.09  |
| 4                       |            | 5.56 | 17.29      | 5.57       | 2.40        | -1.19  | -3.11  |
| 5                       |            | 5.88 | 12.59      | 6.44       | 2.95        | -1.73  | -3.98  |
| 6                       |            | 5.45 | 12.14      | 4.97       | 3.06        | -1.96  | -4.21  |
| 7                       |            | 5.01 | 10.50      | 4.11       | 2.74        | -1.60  | -4.39  |
| 8                       |            | 5.42 | 7.49       | 5.30       | 3.23        | -0.70  | -5.38  |
| 9                       |            | 6.25 | 9.19       | 6.76       | 3.28        | 0.33   | -6.50  |
| 10                      |            | 6.28 | 5.34       | 4.96       | 3.02        | 0.69   | -7.47  |



Table 3: Summary statistics for 2-by-3 ME-ROE(X) portfolios and factor percent returns (July 1976 - June 2023).

In the June end of each year  $t$ , firms are assigned into two size (ME) groups based on median June end market equity of NYSE firms, and assigned independently into three profitability (Y/B) groups based on the 30 and 70 percentiles profitability of all NYSE firms reported annual earnings in calendar year  $t-1$ . We report the time-series mean of the portfolio return over the one-month Treasury bill rate and the associated t-statistics. H-L is a portfolio that is long the high-ROE group and short the low-ROE group within each size group, and Mean(H-L) is the equal-weighted average of the two H-L portfolios.  $a(RMW)$  is the intercept and the associated t-statistics in the regression of Mean(H-L) on the profitability factor RMW of the Fama-French 5-factor model. All other statistics reported in this table are the time-series average of the monthly statistics of each portfolio. N(firms) is the number of firms in each portfolio.  $RD/B$  is the total R&D expense as fractions of total book equity. Overlap(FF) is the fraction of firms that are in both  $Y(X)/B$  portfolio and the corresponding  $Y(FF)/B$  portfolio. Results for our three intangibles-adjusted factors are in bold.

|              |                          | Small       |             |             |             | Big         |             |             |             | Mean(H-L)   | $a(RMW)$    |
|--------------|--------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|              |                          | Low         | Med         | High        | H-L         | Low         | Med         | High        | H-L         |             |             |
| Y(IB)/B      | Mean                     | 0.67        | 0.89        | 0.98        | 0.31        | 0.59        | 0.64        | 0.69        | 0.10        | 0.20        | -0.13       |
|              | t-statistics             | 2.33        | 4.19        | 3.98        | 2.57        | 2.56        | 3.44        | 3.73        | 0.88        | 1.98        | -2.86       |
|              | N(firms)                 | 1732        | 1017        | 596         |             | 171         | 353         | 343         |             |             |             |
| Y(SPI)/B     | Mean                     | 0.63        | 0.92        | 1.00        | 0.36        | 0.53        | 0.64        | 0.70        | 0.17        | 0.26        | -0.06       |
|              | t-statistics             | 2.27        | 4.28        | 3.99        | 3.24        | 2.27        | 3.47        | 3.78        | 1.41        | 2.68        | -1.42       |
|              | N(firms)                 | 1818        | 957         | 570         |             | 170         | 358         | 338         |             |             |             |
| Y(EBT)/B     | Mean                     | 0.62        | 0.90        | 1.00        | 0.38        | 0.50        | 0.61        | 0.73        | 0.23        | 0.30        | -0.07       |
|              | t-statistics             | 2.16        | 4.08        | 4.25        | 2.75        | 2.14        | 3.28        | 3.95        | 1.97        | 2.72        | -1.44       |
|              | N(firms)                 | 1728        | 994         | 614         |             | 175         | 345         | 345         |             |             |             |
| Y(FF)/B      | Mean                     | 0.60        | 0.92        | 1.03        | 0.43        | 0.45        | 0.63        | 0.75        | 0.30        | 0.36        | -0.01       |
|              | t-statistics             | 2.18        | 4.24        | 4.15        | 3.35        | 1.99        | 3.30        | 4.16        | 2.64        | 3.53        | -0.47       |
|              | SD                       | 6.57        | 5.18        | 5.88        | 3.02        | 5.38        | 4.51        | 4.28        | 2.70        | 2.45        |             |
|              | N(firms)                 | 1808        | 969         | 560         |             | 188         | 356         | 321         |             |             |             |
|              | $RD/B$                   | 0.05        | 0.02        | 0.03        |             | 0.02        | 0.03        | 0.06        |             |             |             |
| Y(RD)/B      | <b>Mean</b>              | <b>0.52</b> | <b>0.94</b> | <b>1.09</b> | <b>0.58</b> | <b>0.40</b> | <b>0.63</b> | <b>0.74</b> | <b>0.35</b> | <b>0.46</b> | <b>0.24</b> |
|              | <b>t-statistics</b>      | <b>1.96</b> | <b>4.28</b> | <b>4.19</b> | <b>6.80</b> | <b>1.81</b> | <b>3.39</b> | <b>3.98</b> | <b>3.28</b> | <b>6.08</b> | <b>5.12</b> |
|              | <b>SD</b>                | <b>6.29</b> | <b>5.23</b> | <b>6.19</b> | <b>2.01</b> | <b>5.20</b> | <b>4.42</b> | <b>4.43</b> | <b>2.50</b> | <b>1.80</b> |             |
|              | <b>N(firms)</b>          | <b>1690</b> | <b>1022</b> | <b>624</b>  |             | <b>176</b>  | <b>355</b>  | <b>334</b>  |             |             |             |
|              | <b>Overlap(FF)</b>       | <b>0.90</b> | <b>0.83</b> | <b>0.88</b> |             | <b>0.87</b> | <b>0.84</b> | <b>0.90</b> |             |             |             |
| Y(RD+.3SG)/B | <b><math>RD/B</math></b> | <b>0.03</b> | <b>0.03</b> | <b>0.08</b> |             | <b>0.01</b> | <b>0.02</b> | <b>0.08</b> |             |             |             |
|              | <b>Mean</b>              | <b>0.50</b> | <b>0.93</b> | <b>1.10</b> | <b>0.60</b> | <b>0.43</b> | <b>0.62</b> | <b>0.77</b> | <b>0.34</b> | <b>0.47</b> | <b>0.28</b> |
|              | <b>t-statistics</b>      | <b>1.95</b> | <b>4.13</b> | <b>4.18</b> | <b>6.78</b> | <b>2.10</b> | <b>3.22</b> | <b>4.19</b> | <b>3.37</b> | <b>6.16</b> | <b>4.90</b> |
| Y(RD+SG)/B   | <b>N(firms)</b>          | <b>1430</b> | <b>1131</b> | <b>775</b>  |             | <b>216</b>  | <b>355</b>  | <b>295</b>  |             |             |             |
|              | <b>Mean</b>              | <b>0.54</b> | <b>0.91</b> | <b>1.04</b> | <b>0.50</b> | <b>0.51</b> | <b>0.62</b> | <b>0.79</b> | <b>0.28</b> | <b>0.39</b> | <b>0.23</b> |
|              | <b>t-statistics</b>      | <b>2.16</b> | <b>3.87</b> | <b>4.00</b> | <b>4.77</b> | <b>2.62</b> | <b>3.18</b> | <b>4.33</b> | <b>2.79</b> | <b>4.74</b> | <b>3.21</b> |
| Y(NM)/B      | <b>N(firms)</b>          | <b>1111</b> | <b>1273</b> | <b>953</b>  |             | <b>249</b>  | <b>365</b>  | <b>251</b>  |             |             |             |
|              | Mean                     | 0.55        | 0.93        | 1.03        | 0.49        | 0.50        | 0.64        | 0.78        | 0.27        | 0.38        | 0.21        |
|              | t-statistics             | 2.24        | 3.94        | 3.93        | 4.72        | 2.59        | 3.32        | 4.15        | 2.72        | 4.65        | 3.06        |
| Y(REVT)/B    | N(firms)                 | 1198        | 1199        | 940         |             | 257         | 363         | 244         |             |             |             |
|              | Mean                     | 0.64        | 0.93        | 1.02        | 0.38        | 0.57        | 0.70        | 0.76        | 0.19        | 0.28        | 0.09        |
|              | t-statistics             | 2.63        | 3.90        | 3.90        | 3.39        | 2.97        | 3.77        | 3.93        | 2.09        | 3.26        | 1.30        |
|              | N(firms)                 | 1434        | 1004        | 899         |             | 334         | 334         | 197         |             |             |             |

Table 4: Pricing tests on annual- and monthly-rebalanced profitability factors (July 1976 - June 2023).

To construct annual-rebalanced profitability factors,  $r_{ROE}^X$ , in the June end of each year  $t$ , firms are assigned into two size groups based on the median June end market equity of NYSE firms and assigned independently into three profitability groups based on the 30 and 70 percentiles profitability of all NYSE firms reported annual earnings in calendar year  $t-1$ . To construct monthly-rebalanced profitability factors,  $r_{ROEQ}^X$ , at the end of each month, firms are assigned into two size (ME) groups based on median June end market equity of NYSE firms, and assigned independently into three profitability groups based on the 30 and 70 percentiles (most recent available) profitability of all NYSE firms. The profitability factor return in any given month is the average return of the portfolios that are long the high profitability portfolios and short the low profitability portfolios within each size group. Firms' most recent available profitability is updated at the month end of the most recent quarterly earnings announcement date (quarterly Compustat data item RDQ, exclude if missing). To remove stale earnings, we exclude a firm if its most recent available quarter-end is 6 months away. The quarterly R&D expenses (quarterly Compustat data item XRDQ) are only recorded after 1989 and some firms may report only once a year, we simply set missing XRDQ to zero with no further interpolation. The considered profitability measures include  $Y(IB)/B$ ,  $Y(SPI)/B$ ,  $Y(EBT)/B$ ,  $Y(FF)/B$ ,  $Y(RD)/B$ ,  $Y(RD+.3SG)/B$ ,  $Y(RD+SG)/B$ ,  $Y(NM)/B$ , and  $Y(REVT)/B$ . Long- and short-side of the factors is the equal-weighted average excess return of the high- and low-profitability portfolios in each month. We run time-series regressions of the constructed profitability factors (Table A-1 for annual-rebalanced long-minus-short, A-2 for monthly-rebalanced long-minus-short, B-1 for annual-rebalanced long side only, B-2 for monthly-rebalanced long side only, C-1 for annual-rebalanced short side only, and C-2 for monthly-rebalanced short side only) on the existing factor models and report their intercept and associated t-statistics in italic. The factor models include Fama-French five factor model (FF5), Hou-Mo-Xue-Zhang five factor model (HMXZ5), and Daniel-Hirshleifer-Sun behavioral three factor model (DHS3). The data of the factor models is taken from the authors' websites. HMXZ5 and DHS3 data are available until December 2022, FF5 data are available until June 2023. GLPZ7 is the proposed factor models by Gulen et al. (2025) to replace FF5, data are from the author's website and the sample is from July 1977 to June 2022. Results for our three intangibles-adjusted factors are in bold.

| Moving from bottom-line to top-line profitability: left to right |                       |                       |                       |                       |                            |                            |                            |                       |                       |
|------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------------------|----------------------------|----------------------------|-----------------------|-----------------------|
| Panel A-1: Annual-rebalanced factors                             |                       |                       |                       |                       |                            |                            |                            |                       |                       |
|                                                                  | $r_{ROE}^{IB}$        | $r_{ROE}^{SPI}$       | $r_{ROE}^{EBT}$       | $r_{ROE}^{FF}$        | $r_{ROE}^{RD}$             | $r_{ROE}^{RD+.3SG}$        | $r_{ROE}^{RD+SG}$          | $r_{ROE}^{NM}$        | $r_{ROE}^{REVT}$      |
| Mean                                                             | 0.20<br><i>1.98</i>   | 0.26<br><i>2.68</i>   | 0.30<br><i>2.72</i>   | 0.36<br><i>3.53</i>   | <b>0.46</b><br><b>6.08</b> | <b>0.47</b><br><b>6.16</b> | <b>0.39</b><br><b>4.74</b> | 0.38<br><i>4.65</i>   | 0.28<br><i>3.26</i>   |
| $\alpha_{FF5}$                                                   | 0.01<br><i>0.20</i>   | 0.04<br><i>0.90</i>   | 0.06<br><i>1.27</i>   | 0.01<br><i>0.50</i>   | <b>0.24</b><br><b>6.18</b> | <b>0.24</b><br><b>5.35</b> | <b>0.17</b><br><b>2.69</b> | 0.11<br><i>1.94</i>   | -0.12<br><i>-1.78</i> |
| $\alpha_{HMXZ5}$                                                 | -0.06<br><i>-0.83</i> | -0.03<br><i>-0.34</i> | -0.02<br><i>-0.27</i> | 0.02<br><i>0.26</i>   | <b>0.14</b><br><b>2.24</b> | <b>0.16</b><br><b>2.22</b> | <b>0.13</b><br><b>1.50</b> | 0.07<br><i>0.87</i>   | -0.06<br><i>-0.64</i> |
| $\alpha_{DHS3}$                                                  | -0.03<br><i>-0.37</i> | 0.03<br><i>0.31</i>   | 0.01<br><i>0.16</i>   | 0.09<br><i>1.07</i>   | <b>0.28</b><br><b>3.63</b> | <b>0.30</b><br><b>3.69</b> | <b>0.23</b><br><b>2.62</b> | 0.18<br><i>2.11</i>   | -0.01<br><i>-0.09</i> |
| $\alpha_{GLPZ7}$                                                 | -0.11<br><i>-1.53</i> | -0.09<br><i>-1.40</i> | -0.12<br><i>-1.66</i> | -0.05<br><i>-1.26</i> | <b>0.14</b><br><b>3.21</b> | <b>0.09</b><br><b>1.98</b> | <b>0.00</b><br><b>0.00</b> | -0.04<br><i>-0.55</i> | -0.10<br><i>-0.97</i> |
| Panel A-2: Monthly-rebalanced factors                            |                       |                       |                       |                       |                            |                            |                            |                       |                       |
|                                                                  | $r_{ROEQ}^{IB}$       | $r_{ROEQ}^{SPI}$      | $r_{ROEQ}^{EBT}$      | $r_{ROEQ}^{FF}$       | $r_{ROEQ}^{RD}$            | $r_{ROEQ}^{RD+.3SG}$       | $r_{ROEQ}^{RD+SG}$         | $r_{ROEQ}^{NM}$       | $r_{ROEQ}^{REVT}$     |
| Mean                                                             | 0.61<br><i>5.08</i>   | 0.64<br><i>5.52</i>   | 0.62<br><i>5.08</i>   | 0.63<br><i>5.65</i>   | <b>0.76</b><br><b>8.31</b> | <b>0.70</b><br><b>8.01</b> | <b>0.53</b><br><b>6.12</b> | 0.53<br><i>6.17</i>   | 0.38<br><i>3.93</i>   |
| $\alpha_{FF5}$                                                   | 0.35<br><i>4.37</i>   | 0.36<br><i>4.81</i>   | 0.33<br><i>4.24</i>   | 0.29<br><i>4.54</i>   | <b>0.55</b><br><b>8.41</b> | <b>0.46</b><br><b>7.58</b> | <b>0.28</b><br><b>4.39</b> | 0.25<br><i>3.95</i>   | -0.05<br><i>-0.71</i> |
| $\alpha_{HMXZ5}$                                                 | 0.07<br><i>1.77</i>   | 0.12<br><i>2.36</i>   | 0.06<br><i>1.20</i>   | 0.11<br><i>1.77</i>   | <b>0.26</b><br><b>4.94</b> | <b>0.24</b><br><b>4.08</b> | <b>0.16</b><br><b>2.10</b> | 0.13<br><i>1.77</i>   | -0.05<br><i>-0.55</i> |
| $\alpha_{DHS3}$                                                  | 0.21<br><i>1.87</i>   | 0.28<br><i>2.52</i>   | 0.24<br><i>2.04</i>   | 0.28<br><i>2.63</i>   | <b>0.50</b><br><b>5.25</b> | <b>0.44</b><br><b>4.71</b> | <b>0.31</b><br><b>3.33</b> | 0.28<br><i>3.13</i>   | 0.05<br><i>0.57</i>   |
| $\alpha_{GLPZ7}$                                                 | 0.27<br><i>2.21</i>   | 0.27<br><i>2.40</i>   | 0.21<br><i>1.73</i>   | 0.21<br><i>2.06</i>   | <b>0.42</b><br><b>4.00</b> | <b>0.26</b><br><b>2.90</b> | <b>0.13</b><br><b>1.51</b> | 0.12<br><i>1.34</i>   | -0.04<br><i>-0.43</i> |

| Panel B-1: Long side of the annual-rebalanced factors   |                       |                       |                       |                       |                              |                              |                              |                       |                       |
|---------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|------------------------------|------------------------------|------------------------------|-----------------------|-----------------------|
|                                                         | $r_{ROE}^{IB}$        | $r_{ROE}^{SPI}$       | $r_{ROE}^{EBT}$       | $r_{ROE}^{FF}$        | $r_{ROE}^{RD}$               | $r_{ROE}^{RD+.3SG}$          | $r_{ROE}^{RD+SG}$            | $r_{ROE}^{NM}$        | $r_{ROE}^{REVT}$      |
| Mean                                                    | 0.83<br><i>4.05</i>   | 0.85<br><i>4.08</i>   | 0.86<br><i>4.31</i>   | 0.89<br><i>4.35</i>   | <b>0.92</b><br><i>4.30</i>   | <b>0.94</b><br><i>4.39</i>   | <b>0.92</b><br><i>4.34</i>   | 0.91<br><i>4.23</i>   | 0.89<br><i>4.07</i>   |
| $\alpha_{FF5}$                                          | -0.01<br><i>-0.45</i> | -0.01<br><i>-0.47</i> | 0.00<br><i>-0.20</i>  | -0.02<br><i>-0.97</i> | <b>0.09</b><br><i>4.07</i>   | <b>0.08</b><br><i>3.30</i>   | <b>0.05</b><br><i>1.41</i>   | 0.02<br><i>0.48</i>   | -0.07<br><i>-1.69</i> |
| $\alpha_{HMXZ5}$                                        | 0.03<br><i>0.85</i>   | 0.04<br><i>0.92</i>   | 0.03<br><i>0.83</i>   | 0.05<br><i>1.00</i>   | <b>0.11</b><br><i>3.29</i>   | <b>0.10</b><br><i>2.67</i>   | <b>0.10</b><br><i>2.04</i>   | 0.07<br><i>1.53</i>   | 0.02<br><i>0.32</i>   |
| $\alpha_{DHS3}$                                         | 0.14<br><i>2.61</i>   | 0.15<br><i>2.77</i>   | 0.14<br><i>2.68</i>   | 0.16<br><i>2.82</i>   | <b>0.26</b><br><i>4.43</i>   | <b>0.27</b><br><i>4.26</i>   | <b>0.24</b><br><i>3.50</i>   | 0.21<br><i>3.12</i>   | 0.14<br><i>1.79</i>   |
| $\alpha_{GLPZ7}$                                        | 0.00<br><i>0.12</i>   | -0.01<br><i>-0.24</i> | -0.03<br><i>-0.87</i> | -0.03<br><i>-0.65</i> | <b>0.06</b><br><i>1.79</i>   | <b>0.04</b><br><i>1.25</i>   | <b>0.01</b><br><i>0.10</i>   | -0.02<br><i>-0.32</i> | -0.05<br><i>-0.72</i> |
| Panel B-2: Long side of the monthly-rebalanced factors  |                       |                       |                       |                       |                              |                              |                              |                       |                       |
|                                                         | $r_{ROEQ}^{IB}$       | $r_{ROEQ}^{SPI}$      | $r_{ROEQ}^{EBT}$      | $r_{ROEQ}^{FF}$       | $r_{ROEQ}^{RD}$              | $r_{ROEQ}^{RD+.3SG}$         | $r_{ROEQ}^{RD+SG}$           | $r_{ROEQ}^{NM}$       | $r_{ROEQ}^{REVT}$     |
| Mean                                                    | 1.00<br><i>4.75</i>   | 1.02<br><i>4.80</i>   | 1.02<br><i>4.83</i>   | 1.02<br><i>4.78</i>   | <b>1.05</b><br><i>4.81</i>   | <b>1.06</b><br><i>4.85</i>   | <b>0.99</b><br><i>4.58</i>   | 1.00<br><i>4.57</i>   | 0.94<br><i>4.22</i>   |
| $\alpha_{FF5}$                                          | 0.14<br><i>4.53</i>   | 0.15<br><i>5.06</i>   | 0.14<br><i>4.32</i>   | 0.12<br><i>3.66</i>   | <b>0.23</b><br><i>7.19</i>   | <b>0.22</b><br><i>6.90</i>   | <b>0.14</b><br><i>3.52</i>   | 0.13<br><i>3.32</i>   | -0.01<br><i>-0.25</i> |
| $\alpha_{HMXZ5}$                                        | 0.09<br><i>3.24</i>   | 0.11<br><i>3.54</i>   | 0.11<br><i>3.20</i>   | 0.12<br><i>2.99</i>   | <b>0.18</b><br><i>5.08</i>   | <b>0.19</b><br><i>5.09</i>   | <b>0.15</b><br><i>3.22</i>   | 0.14<br><i>3.17</i>   | 0.06<br><i>1.03</i>   |
| $\alpha_{DHS3}$                                         | 0.25<br><i>4.14</i>   | 0.28<br><i>4.51</i>   | 0.27<br><i>4.29</i>   | 0.27<br><i>4.25</i>   | <b>0.36</b><br><i>5.87</i>   | <b>0.36</b><br><i>5.57</i>   | <b>0.32</b><br><i>4.53</i>   | 0.31<br><i>4.43</i>   | 0.20<br><i>2.47</i>   |
| $\alpha_{GLPZ7}$                                        | 0.11<br><i>2.23</i>   | 0.12<br><i>2.52</i>   | 0.13<br><i>2.48</i>   | 0.12<br><i>2.32</i>   | <b>0.19</b><br><i>4.06</i>   | <b>0.15</b><br><i>3.35</i>   | <b>0.08</b><br><i>1.57</i>   | 0.09<br><i>1.65</i>   | 0.01<br><i>0.09</i>   |
| Panel C-1: Short side of the annual-rebalanced factors  |                       |                       |                       |                       |                              |                              |                              |                       |                       |
|                                                         | $r_{ROE}^{IB}$        | $r_{ROE}^{SPI}$       | $r_{ROE}^{EBT}$       | $r_{ROE}^{FF}$        | $r_{ROE}^{RD}$               | $r_{ROE}^{RD+.3SG}$          | $r_{ROE}^{RD+SG}$            | $r_{ROE}^{NM}$        | $r_{ROE}^{REVT}$      |
| Mean                                                    | 0.63<br><i>2.51</i>   | 0.58<br><i>2.35</i>   | 0.56<br><i>2.23</i>   | 0.53<br><i>2.17</i>   | <b>0.46</b><br><i>1.97</i>   | <b>0.46</b><br><i>2.10</i>   | <b>0.53</b><br><i>2.48</i>   | 0.53<br><i>2.50</i>   | 0.61<br><i>2.91</i>   |
| $\alpha_{FF5}$                                          | -0.02<br><i>-0.46</i> | -0.05<br><i>-1.26</i> | -0.06<br><i>-1.60</i> | -0.03<br><i>-1.37</i> | <b>-0.15</b><br><i>-5.01</i> | <b>-0.16</b><br><i>-5.27</i> | <b>-0.12</b><br><i>-3.11</i> | -0.10<br><i>-2.83</i> | 0.05<br><i>1.46</i>   |
| $\alpha_{HMXZ5}$                                        | 0.10<br><i>1.79</i>   | 0.06<br><i>1.14</i>   | 0.06<br><i>1.03</i>   | 0.02<br><i>0.51</i>   | <b>-0.04</b><br><i>-0.80</i> | <b>-0.06</b><br><i>-1.33</i> | <b>-0.03</b><br><i>-0.65</i> | 0.00<br><i>0.01</i>   | 0.07<br><i>1.81</i>   |
| $\alpha_{DHS3}$                                         | 0.17<br><i>1.89</i>   | 0.13<br><i>1.42</i>   | 0.13<br><i>1.46</i>   | 0.07<br><i>0.89</i>   | <b>-0.03</b><br><i>-0.33</i> | <b>-0.04</b><br><i>-0.48</i> | <b>0.00</b><br><i>0.04</i>   | 0.03<br><i>0.37</i>   | 0.15<br><i>2.35</i>   |
| $\alpha_{GLPZ7}$                                        | 0.11<br><i>1.86</i>   | 0.08<br><i>1.41</i>   | 0.09<br><i>1.42</i>   | 0.02<br><i>0.56</i>   | <b>-0.08</b><br><i>-1.93</i> | <b>-0.05</b><br><i>-1.42</i> | <b>0.01</b><br><i>0.11</i>   | 0.03<br><i>0.57</i>   | 0.05<br><i>1.03</i>   |
| Panel C-2: Short side of the monthly-rebalanced factors |                       |                       |                       |                       |                              |                              |                              |                       |                       |
|                                                         | $r_{ROEQ}^{IB}$       | $r_{ROEQ}^{SPI}$      | $r_{ROEQ}^{EBT}$      | $r_{ROEQ}^{FF}$       | $r_{ROEQ}^{RD}$              | $r_{ROEQ}^{RD+.3SG}$         | $r_{ROEQ}^{RD+SG}$           | $r_{ROEQ}^{NM}$       | $r_{ROEQ}^{REVT}$     |
| Mean                                                    | 0.39<br><i>1.52</i>   | 0.38<br><i>1.50</i>   | 0.39<br><i>1.53</i>   | 0.38<br><i>1.55</i>   | <b>0.29</b><br><i>1.23</i>   | <b>0.36</b><br><i>1.58</i>   | <b>0.46</b><br><i>2.15</i>   | 0.47<br><i>2.21</i>   | 0.56<br><i>2.59</i>   |
| $\alpha_{FF5}$                                          | -0.21<br><i>-3.60</i> | -0.21<br><i>-3.76</i> | -0.18<br><i>-3.22</i> | -0.17<br><i>-3.79</i> | <b>-0.33</b><br><i>-7.05</i> | <b>-0.24</b><br><i>-5.79</i> | <b>-0.15</b><br><i>-3.65</i> | -0.13<br><i>-3.16</i> | 0.04<br><i>1.02</i>   |
| $\alpha_{HMXZ5}$                                        | 0.02<br><i>0.49</i>   | -0.01<br><i>-0.24</i> | 0.05<br><i>1.10</i>   | 0.01<br><i>0.24</i>   | <b>-0.09</b><br><i>-2.37</i> | <b>-0.05</b><br><i>-1.35</i> | <b>-0.01</b><br><i>-0.23</i> | 0.01<br><i>0.25</i>   | 0.11<br><i>2.31</i>   |
| $\alpha_{DHS3}$                                         | 0.04<br><i>0.35</i>   | -0.01<br><i>-0.06</i> | 0.03<br><i>0.31</i>   | -0.01<br><i>-0.06</i> | <b>-0.14</b><br><i>-1.46</i> | <b>-0.07</b><br><i>-0.83</i> | <b>0.01</b><br><i>0.09</i>   | 0.03<br><i>0.32</i>   | 0.15<br><i>2.12</i>   |
| $\alpha_{GLPZ7}$                                        | -0.16<br><i>-1.78</i> | -0.15<br><i>-1.77</i> | -0.08<br><i>-0.88</i> | -0.09<br><i>-1.15</i> | <b>-0.22</b><br><i>-2.84</i> | <b>-0.10</b><br><i>-1.54</i> | <b>-0.05</b><br><i>-0.85</i> | -0.03<br><i>-0.52</i> | 0.05<br><i>0.92</i>   |

Table 5: Maximum drawdown of profitability factors (July 1976 - June 2023).

To compute the drawdowns for a strategy, we keep tracking the cumulative returns of the factor portfolio and identify the episodes where the cumulative returns are below a historical peak. The drawdowns are the cumulative percentage returns from the historical peak to the trough within each episode. Panels A and B report the ten maximum drawdowns of the annual- and monthly-rebalancing profitability factors respectively. Results for our three intangibles-adjusted factors are in bold.

| Panel A: Annual-rebalanced profitability factors  |                 |                  |                  |                 |                 |                      |                    |                 |                   |
|---------------------------------------------------|-----------------|------------------|------------------|-----------------|-----------------|----------------------|--------------------|-----------------|-------------------|
| Rank                                              | $r_{ROE}^{IB}$  | $r_{ROE}^{SPI}$  | $r_{ROE}^{EBT}$  | $r_{ROE}^{FF}$  | $r_{ROE}^{RD}$  | $r_{ROE}^{RD+.3SG}$  | $r_{ROE}^{RD+SG}$  | $r_{ROE}^{NM}$  | $r_{ROE}^{REVT}$  |
| 1                                                 | -36.86          | -34.12           | -45.52           | -42.01          | <b>-14.94</b>   | <b>-12.22</b>        | <b>-20.40</b>      | -21.11          | -42.13            |
| 2                                                 | -25.84          | -23.90           | -27.08           | -24.43          | <b>-12.11</b>   | <b>-12.13</b>        | <b>-19.83</b>      | -19.40          | -14.19            |
| 3                                                 | -20.59          | -18.45           | -20.74           | -12.96          | <b>-10.54</b>   | <b>-10.57</b>        | <b>-15.74</b>      | -13.81          | -11.32            |
| 4                                                 | -18.70          | -17.96           | -18.32           | -12.51          | <b>-9.51</b>    | <b>-10.49</b>        | <b>-12.22</b>      | -11.22          | -9.71             |
| 5                                                 | -17.51          | -17.24           | -15.76           | -12.43          | <b>-8.29</b>    | <b>-9.02</b>         | <b>-10.60</b>      | -9.97           | -9.25             |
| 6                                                 | -17.43          | -11.36           | -13.82           | -9.97           | <b>-6.77</b>    | <b>-7.61</b>         | <b>-8.75</b>       | -9.70           | -8.18             |
| 7                                                 | -12.46          | -10.85           | -9.47            | -6.75           | <b>-6.35</b>    | <b>-7.46</b>         | <b>-7.84</b>       | -7.40           | -6.94             |
| 8                                                 | -8.64           | -8.58            | -9.27            | -6.11           | <b>-5.96</b>    | <b>-6.95</b>         | <b>-7.12</b>       | -7.18           | -5.64             |
| 9                                                 | -8.56           | -7.09            | -8.58            | -5.68           | <b>-5.80</b>    | <b>-6.94</b>         | <b>-6.42</b>       | -6.95           | -5.10             |
| 10                                                | -8.24           | -5.69            | -7.61            | -5.66           | <b>-5.64</b>    | <b>-6.91</b>         | <b>-6.41</b>       | -6.93           | -4.74             |
| Panel B: Monthly-rebalanced profitability factors |                 |                  |                  |                 |                 |                      |                    |                 |                   |
| Rank                                              | $r_{ROEQ}^{IB}$ | $r_{ROEQ}^{SPI}$ | $r_{ROEQ}^{EBT}$ | $r_{ROEQ}^{FF}$ | $r_{ROEQ}^{RD}$ | $r_{ROEQ}^{RD+.3SG}$ | $r_{ROEQ}^{RD+SG}$ | $r_{ROEQ}^{NM}$ | $r_{ROEQ}^{REVT}$ |
| 1                                                 | -33.28          | -29.07           | -38.86           | -36.47          | <b>-19.12</b>   | <b>-13.78</b>        | <b>-21.43</b>      | -21.36          | -42.50            |
| 2                                                 | -31.60          | -25.92           | -30.62           | -23.78          | <b>-17.13</b>   | <b>-13.38</b>        | <b>-13.52</b>      | -12.04          | -13.06            |
| 3                                                 | -28.06          | -24.80           | -27.11           | -21.43          | <b>-12.28</b>   | <b>-11.41</b>        | <b>-11.34</b>      | -9.28           | -12.76            |
| 4                                                 | -25.12          | -23.34           | -26.51           | -20.61          | <b>-11.39</b>   | <b>-11.31</b>        | <b>-10.17</b>      | -9.20           | -11.63            |
| 5                                                 | -14.20          | -13.67           | -13.69           | -11.39          | <b>-10.59</b>   | <b>-9.47</b>         | <b>-8.19</b>       | -8.51           | -10.66            |
| 6                                                 | -9.22           | -11.64           | -11.17           | -10.66          | <b>-7.80</b>    | <b>-7.57</b>         | <b>-7.28</b>       | -8.51           | -8.27             |
| 7                                                 | -9.12           | -8.68            | -10.76           | -10.25          | <b>-7.63</b>    | <b>-6.70</b>         | <b>-7.26</b>       | -8.07           | -8.12             |
| 8                                                 | -6.91           | -8.68            | -8.17            | -8.58           | <b>-6.97</b>    | <b>-5.06</b>         | <b>-6.87</b>       | -7.47           | -7.35             |
| 9                                                 | -6.62           | -8.15            | -7.73            | -5.54           | <b>-5.60</b>    | <b>-4.69</b>         | <b>-6.77</b>       | -7.17           | -6.45             |
| 10                                                | -6.32           | -7.48            | -7.31            | -5.21           | <b>-4.70</b>    | <b>-4.69</b>         | <b>-6.69</b>       | -7.12           | -6.04             |

Table 6: Co-tail risk of factors with the market (July 1976 - June 2023).

The table reports the time-series average of the monthly percentage return of each factor and the associated t-statistics conditional on the market performance respectively.  $r_{\text{MKT}}$  is the market excess return plus the risk-free rate in the Ken French's data library. N(months) is the total number of months that satisfy the given condition on market performance. The considered market conditions include months with market returns that are  $> 0$ ,  $< 0$ ,  $< -5$ ,  $< -7$ , and  $< -10$ . The considered factors include the annual- and monthly-rebalanced profitability factors formed based on  $Y(\text{IB})/B$ ,  $Y(\text{FF})/B$ ,  $Y(\text{RD})/B$ , and  $Y(\text{NM})/B$ . Results for our intangibles-adjusted factors are in bold.

|                        | N(months) | $r_{\text{ROE}}^{\text{IB}}$ | $r_{\text{ROE}}^{\text{FF}}$ | $r_{\text{ROE}}^{\text{RD}}$ | $r_{\text{ROE}}^{\text{NM}}$ | $r_{\text{ROEQ}}^{\text{IB}}$ | $r_{\text{ROEQ}}^{\text{FF}}$ | $r_{\text{ROEQ}}^{\text{RD}}$ | $r_{\text{ROEQ}}^{\text{NM}}$ |
|------------------------|-----------|------------------------------|------------------------------|------------------------------|------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
| Full sample            | 564       | 0.20<br><i>1.98</i>          | 0.36<br><i>3.53</i>          | <b>0.46</b><br><b>6.08</b>   | 0.38<br><i>4.65</i>          | 0.61<br><i>5.08</i>           | 0.63<br><i>5.65</i>           | <b>0.76</b><br><b>8.31</b>    | 0.53<br><i>6.17</i>           |
| $r_{\text{MKT}} > 0$   | 361       | -0.22<br><i>-1.82</i>        | -0.04<br><i>-0.35</i>        | <b>0.33</b><br><b>3.82</b>   | 0.46<br><i>4.68</i>          | 0.27<br><i>1.84</i>           | 0.35<br><i>2.61</i>           | <b>0.68</b><br><b>6.11</b>    | 0.65<br><i>6.23</i>           |
| $r_{\text{MKT}} < 0$   | 203       | 0.96<br><i>5.42</i>          | 1.08<br><i>6.07</i>          | <b>0.70</b><br><b>4.85</b>   | 0.23<br><i>1.61</i>          | 1.21<br><i>6.00</i>           | 1.13<br><i>5.85</i>           | <b>0.90</b><br><b>5.66</b>    | 0.31<br><i>2.10</i>           |
| $r_{\text{MKT}} < -5$  | 51        | 2.27<br><i>5.99</i>          | 2.34<br><i>5.53</i>          | <b>1.59</b><br><b>5.21</b>   | 0.41<br><i>1.27</i>          | 2.86<br><i>5.60</i>           | 2.30<br><i>4.56</i>           | <b>1.79</b><br><b>4.31</b>    | 0.47<br><i>1.30</i>           |
| $r_{\text{MKT}} < -7$  | 30        | 2.29<br><i>4.29</i>          | 2.35<br><i>3.77</i>          | <b>1.87</b><br><b>4.52</b>   | 0.65<br><i>1.46</i>          | 3.13<br><i>4.27</i>           | 2.26<br><i>3.20</i>           | <b>1.99</b><br><b>3.35</b>    | 0.43<br><i>0.84</i>           |
| $r_{\text{MKT}} < -10$ | 9         | 3.93<br><i>3.48</i>          | 3.34<br><i>2.21</i>          | <b>2.87</b><br><b>3.11</b>   | 0.82<br><i>0.67</i>          | 3.58<br><i>2.40</i>           | 2.60<br><i>1.75</i>           | <b>2.48</b><br><b>1.96</b>    | 0.29<br><i>0.22</i>           |

Table 7: Henriksson and Merton regression (July 1976 - June 2023).

$r_{\text{ROE}}^{i,t} = \alpha_i^s + \beta_i^U \text{RMRF} + \beta_i^{U-D} \max[-\text{RMRF}, 0] + \epsilon_{i,t}$ , where  $i$  is IB, FF, RD, NM, and RMW(GLPZ); RMRF is the market excess percentage return. The last term in the regression aims to capture the put option feature embedded in the factor returns. We report the regression results for the long-short portfolio (factor return), long side only, and short side only separately. t-statistics with heteroscedasticity-corrected standard errors is reported. RMW(GLPZ) is the intangible-adjusted factors proposed by Gulen et al. (2025) to replace RMW in FF5, data is from the author's website and the sample is from July 1998 to June 2022.

| $r_{\text{ROE}}^{i,t} = \alpha_i^s + \beta_i^U \text{RMRF} + \beta_i^{U-D} \max[-\text{RMRF}, 0] + \epsilon_{i,t}$ |              |              |                 |                     |              |              |                 |                     |              |              |                 |
|--------------------------------------------------------------------------------------------------------------------|--------------|--------------|-----------------|---------------------|--------------|--------------|-----------------|---------------------|--------------|--------------|-----------------|
|                                                                                                                    | Long - Short |              |                 |                     | Long Side    |              |                 |                     | Short Side   |              |                 |
|                                                                                                                    | $\alpha_i^s$ | $\beta_i^U$  | $\beta_i^{U-D}$ |                     | $\alpha_i^s$ | $\beta_i^U$  | $\beta_i^{U-D}$ |                     | $\alpha_i^s$ | $\beta_i^U$  | $\beta_i^{U-D}$ |
| IB                                                                                                                 | 0.12         | -0.10        | 0.11            | IB                  | 0.17         | 1.04         | -0.02           | IB                  | 0.05         | 1.15         | -0.13           |
| <i>t-stat</i>                                                                                                      | <i>0.79</i>  | <i>-2.49</i> | <i>1.54</i>     | <i>t-stat</i>       | <i>2.20</i>  | <i>49.85</i> | <i>-0.71</i>    | <i>t-stat</i>       | <i>0.31</i>  | <i>27.24</i> | <i>-1.88</i>    |
| <i>White t-stat</i>                                                                                                | <i>0.81</i>  | <i>-2.13</i> | <i>1.39</i>     | <i>White t-stat</i> | <i>2.27</i>  | <i>44.98</i> | <i>-0.63</i>    | <i>White t-stat</i> | <i>0.31</i>  | <i>22.15</i> | <i>-1.74</i>    |
| FF                                                                                                                 | 0.29         | -0.10        | 0.10            | FF                  | 0.29         | 1.01         | -0.06           | FF                  | 0.01         | 1.11         | -0.15           |
| <i>t-stat</i>                                                                                                      | <i>1.82</i>  | <i>-2.26</i> | <i>1.42</i>     | <i>t-stat</i>       | <i>3.46</i>  | <i>44.78</i> | <i>-1.51</i>    | <i>t-stat</i>       | <i>0.04</i>  | <i>28.86</i> | <i>-2.44</i>    |
| <i>White t-stat</i>                                                                                                | <i>2.04</i>  | <i>-2.04</i> | <i>1.37</i>     | <i>White t-stat</i> | <i>3.63</i>  | <i>36.33</i> | <i>-1.19</i>    | <i>White t-stat</i> | <i>0.05</i>  | <i>25.55</i> | <i>-2.48</i>    |
| RD                                                                                                                 | <b>0.24</b>  | <b>0.02</b>  | <b>0.14</b>     | RD                  | <b>0.24</b>  | <b>1.07</b>  | <b>-0.03</b>    | RD                  | <b>0.00</b>  | <b>1.05</b>  | <b>-0.17</b>    |
| <i>t-stat</i>                                                                                                      | <i>2.02</i>  | <i>0.67</i>  | <i>2.79</i>     | <i>t-stat</i>       | <i>2.84</i>  | <i>46.91</i> | <i>-0.81</i>    | <i>t-stat</i>       | <i>0.03</i>  | <i>29.88</i> | <i>-3.03</i>    |
| <i>White t-stat</i>                                                                                                | <i>2.07</i>  | <i>0.55</i>  | <i>2.55</i>     | <i>White t-stat</i> | <i>2.96</i>  | <i>40.51</i> | <i>-0.78</i>    | <i>White t-stat</i> | <i>0.03</i>  | <i>24.44</i> | <i>-2.80</i>    |
| NM                                                                                                                 | 0.17         | 0.10         | 0.10            | NM                  | 0.26         | 1.06         | -0.04           | NM                  | 0.09         | 0.96         | -0.14           |
| <i>t-stat</i>                                                                                                      | <i>1.33</i>  | <i>2.84</i>  | <i>1.78</i>     | <i>t-stat</i>       | <i>2.61</i>  | <i>40.57</i> | <i>-0.98</i>    | <i>t-stat</i>       | <i>0.73</i>  | <i>31.01</i> | <i>-2.79</i>    |
| <i>White t-stat</i>                                                                                                | <i>1.25</i>  | <i>2.55</i>  | <i>1.40</i>     | <i>White t-stat</i> | <i>2.34</i>  | <i>34.45</i> | <i>-0.79</i>    | <i>White t-stat</i> | <i>0.77</i>  | <i>27.99</i> | <i>-2.52</i>    |
| RMW(GLPZ)                                                                                                          | 0.39         | -0.12        | 0.08            |                     |              |              |                 |                     |              |              |                 |
| <i>t-stat</i>                                                                                                      | <i>1.80</i>  | <i>-2.05</i> | <i>0.89</i>     |                     |              |              |                 |                     |              |              |                 |
| <i>White t-stat</i>                                                                                                | <i>1.84</i>  | <i>-1.70</i> | <i>0.88</i>     |                     |              |              |                 |                     |              |              |                 |

Table 8: Mean and t-statistics of the monthly factor returns for the full sample and two subsamples (July 1976 - June 2023).

RMW, robust-minus-weak, is the annual-rebalanced operating profitability factor in Fama-French five factor model.  $r_{CROE}^{FF}$  is self-constructed cash based operating profitability factor based on Fama and French (2018).  $r_{ROA}^{NM}$  is self-constructed gross profitability factor based on Novy-Marx (2013).  $r_{ROA}^{BGLN}$  is self-constructed operating profitability factor based on Ball et al. (2015).  $r_{CROA}^{BGLN}$  is self-constructed cash based operating profitability factor based on Ball et al. (2016). R.ROE is the monthly-rebalancing profitability factor in Hou-Xue-Zhang four factor model. PERF is the performance factor from Stambaugh-Yuan four factor model (SY4). UMD is the momentum factor taken from Kenneth French's data library. QMJ is the monthly-rebalanced quality-minus-junk factor by Asness et al. (2019). PEAD is the post earnings announcement drift factor from Daniel-Hirshleifer-Sun behavioral three factor model. HML(EKP) is the intangible-adjusted value factor by Eisfeldt et al. (2022).  $r_{ROE}^{RD}$  and  $r_{ROEQ}^{RD}$  are self constructed annual- and monthly-rebalanced profitability factors based on the profitability measures  $Y(RD)/B$  where  $Y(RD) = REVT - COGS - XSGA - XINT + XRD$  for annual-rebalanced factor and  $Y(RD) = REVTQ - COGSQ - XSGAQ - XINTQ + XRDQ$  for monthly-rebalanced factor. RMW, R.ROE, PEAD, PERF, UMD, QMJ are taken from the authors' websites. PERF is available until December 2016, HML(EKP) is available until March 2022, R.ROE and PEAD are available until December 2022, RMW, UMD, and QMJ are available until June 2023. Results for our two intangibles-adjusted factors are in bold. RMW(GLPZ) and ROE(GLPZ) are the intangible-adjusted factors proposed by Gulen et al. (2025) to replace RMW in FF5 and R.ROE in HXZ4 respectively, data is from the author's website and the sample is from July 1998 to June 2022.

|                                     | RMW  | $r_{CROE}^{FF}$ | $r_{ROA}^{NM}$ | $r_{ROA}^{BGLN}$ | $r_{CROA}^{BGLN}$ | R.ROE | PERF | UMD  | QMJ  | PEAD | HML<br>(EKP) | $r_{ROE}^{RD}$ | $r_{ROEQ}^{RD}$ | RMW<br>(GLPZ) | ROE<br>(GLPZ) |
|-------------------------------------|------|-----------------|----------------|------------------|-------------------|-------|------|------|------|------|--------------|----------------|-----------------|---------------|---------------|
| Full Sample: July 1976 to June 2023 |      |                 |                |                  |                   |       |      |      |      |      |              |                |                 |               |               |
| Mean                                | 0.36 | 0.37            | 0.27           | 0.33             | 0.38              | 0.56  | 0.68 | 0.57 | 0.43 | 0.54 | 0.45         | <b>0.46</b>    | <b>0.76</b>     |               |               |
| <i>t-stat</i>                       | 3.69 | 4.21            | 2.77           | 3.56             | 4.60              | 5.05  | 3.79 | 3.10 | 4.28 | 6.71 | 5.33         | <b>6.08</b>    | <b>8.31</b>     |               |               |
| July 1976 to December 1999          |      |                 |                |                  |                   |       |      |      |      |      |              |                |                 |               |               |
| Mean                                | 0.27 | 0.35            | 0.18           | 0.22             | 0.35              | 0.77  | 0.64 | 1.00 | 0.41 | 0.79 | 0.45         | <b>0.34</b>    | <b>0.79</b>     |               |               |
| <i>t-stat</i>                       | 2.92 | 3.84            | 1.32           | 1.84             | 3.51              | 6.62  | 3.67 | 5.07 | 4.16 | 8.51 | 4.58         | <b>3.92</b>    | <b>7.21</b>     |               |               |
| January 2000 to June 2023           |      |                 |                |                  |                   |       |      |      |      |      |              |                |                 |               |               |
| Mean                                | 0.46 | 0.40            | 0.36           | 0.43             | 0.42              | 0.34  | 0.74 | 0.15 | 0.45 | 0.29 | 0.45         | <b>0.58</b>    | <b>0.72</b>     | 0.50          | 0.38          |
| <i>t-stat</i>                       | 2.65 | 2.62            | 2.57           | 3.11             | 3.12              | 1.83  | 2.08 | 0.47 | 2.57 | 2.20 | 3.23         | <b>4.69</b>    | <b>4.96</b>     | 3.42          | 2.10          |

Table 9: Pricing tests of  $r_{ROE}^{RD}$  on related factors and anomalies and the pricing tests of related factors and anomalies on  $r_{ROE}^{RD}$  (July 1976 - June 2023).

RMW, robust-minus-weak, is the annual-rebalanced operating profitability factor in Fama-French five factor model.  $r_{CROE}^{FF}$  is self-constructed cash based operating profitability factor based on Fama and French (2018).  $r_{ROA}^{NM}$  is self-constructed gross profitability factor based on Novy-Marx (2013).  $r_{ROA}^{BGLN}$  is self-constructed operating profitability factor based on Ball et al. (2015).  $r_{CROA}^{BGLN}$  is self-constructed cash based operating profitability factor based on Ball et al. (2016). R\_ROE is the monthly-rebalancing profitability factor in Hou-Xue-Zhang four factor model. PERF is the performance factor from Stambaugh-Yuan four factor model (SY4). UMD is the momentum factor taken from Kenneth French's data library. QMJ is the monthly-rebalanced quality-minus-junk factor by Asness et al. (2019). PEAD is the post earnings announcement drift factor from Daniel-Hirshleifer-Sun behavioral three factor model. HML(EKP) is the intangible-adjusted value factor by Eisfeldt et al. (2022).  $r_{ROE}^{RD}$  and  $r_{ROEQ}^{RD}$  are self constructed annual- and monthly-rebalanced profitability factors based on the profitability measures  $Y(RD)/B$  where  $Y(RD) = REV - COGS - XSGA - XINT + XRD$  for annual-rebalanced factor and  $Y(RD) = REVTQ - COGSQ - XSGAQ - XINTQ + XRDQ$  for monthly-rebalanced factor. RMW, R\_ROE, PEAD, PERF, UMD, QMJ are taken from the authors' websites. PERF is available until December 2016, HML(EKP) is available until March 2022, R\_ROE and PEAD are available until December 2022, RMW, UMD, and QMJ are available until June 2023. The table reports the intercept and slope coefficient with the associated t-statistics and (adjusted) r-squared of the univariate and multiple time-series regressions. % drop in Table B-2 reports the percentage drop of alpha in adjusted FF5 relative to FF5. RMW(GLPZ) and ROE(GLPZ) are the intangible-adjusted factors proposed by Gulen et al. (2025) to replace RMW in FF5 and R\_ROE in HXZ4 respectively, data is from the author's website and the sample is from July 1998 to June 2022.

|                                                                                                                          | RMW | $r_{CROE}^{FF}$ | $r_{ROA}^{NM}$  | $r_{ROA}^{BGLN}$ | $r_{CROA}^{BGLN}$ | R_ROE          | PERF            | UMD            | QMJ             | PEAD           | HML<br>(EKP)  | RMW<br>(GLPZ)  | ROE<br>(GLPZ)  |
|--------------------------------------------------------------------------------------------------------------------------|-----|-----------------|-----------------|------------------|-------------------|----------------|-----------------|----------------|-----------------|----------------|---------------|----------------|----------------|
| Panel A-1: $X = \alpha + b \text{RMW} + \epsilon$                                                                        |     |                 |                 |                  |                   |                |                 |                |                 |                |               |                |                |
| $\alpha$                                                                                                                 |     | 0.10<br>2.00    | 0.21<br>2.17    | 0.22<br>2.51     | 0.30<br>3.73      | 0.28<br>3.46   | 0.43<br>2.65    | 0.52<br>2.78   | 0.17<br>2.33    | 0.57<br>6.94   | 0.42<br>4.91  | 0.18<br>3.65   | 0.09<br>0.66   |
| $b$                                                                                                                      |     | 0.77<br>37.99   | 0.16<br>3.85    | 0.29<br>7.70     | 0.22<br>6.29      | 0.76<br>21.88  | 0.71<br>10.37   | 0.15<br>1.87   | 0.72<br>23.63   | -0.07<br>-1.89 | 0.09<br>2.62  | 0.76<br>45.92  | 0.69<br>15.80  |
| $R^2$                                                                                                                    |     | 0.72            | 0.03            | 0.10             | 0.07              | 0.46           | 0.18            | 0.01           | 0.50            | 0.01           | 0.01          | 0.88           | 0.47           |
| Panel A-2: $X = \alpha + b_1 \text{RMRF} + b_2 \text{SMB} + b_3 \text{HML} + b_4 \text{CMA} + b_5 \text{RMW} + \epsilon$ |     |                 |                 |                  |                   |                |                 |                |                 |                |               |                |                |
| $\alpha$                                                                                                                 |     | 0.18<br>4.30    | 0.19<br>2.79    | 0.23<br>3.68     | 0.33<br>5.34      | 0.37<br>4.56   | 0.60<br>4.26    | 0.55<br>3.03   | 0.37<br>6.12    | 0.63<br>7.64   | 0.20<br>4.09  | 0.15<br>3.53   | 0.28<br>2.34   |
| $b_1$                                                                                                                    |     | -0.08<br>-8.65  | 0.02<br>1.06    | -0.02<br>-1.44   | -0.05<br>-3.10    | -0.05<br>-2.61 | -0.20<br>-5.85  | -0.15<br>-3.44 | -0.20<br>-14.09 | -0.07<br>-3.49 | 0.04<br>3.36  | 0.02<br>2.05   | -0.16<br>-5.73 |
| $b_2$                                                                                                                    |     | -0.10<br>-6.96  | 0.11<br>4.31    | 0.07<br>3.11     | 0.03<br>1.34      | -0.09<br>-3.12 | 0.11<br>2.21    | 0.12<br>1.77   | -0.10<br>-4.56  | 0.01<br>0.30   | 0.19<br>11.05 | 0.07<br>4.46   | -0.17<br>-4.12 |
| $b_3$                                                                                                                    |     | 0.05<br>2.68    | -0.61<br>-20.27 | -0.58<br>-21.10  | -0.52<br>-19.64   | -0.20<br>-5.70 | -0.89<br>-13.70 | -0.60<br>-7.71 | -0.19<br>-7.31  | -0.18<br>-5.01 | 0.41<br>19.56 | -0.09<br>-5.36 | -0.13<br>-2.87 |
| $b_4$                                                                                                                    |     | 0.10<br>3.61    | 0.13<br>2.91    | 0.19<br>4.40     | 0.24<br>5.91      | 0.07<br>1.29   | 0.56<br>5.56    | 0.49<br>4.05   | 0.10<br>2.51    | 0.06<br>1.11   | 0.26<br>7.90  | -0.11<br>-4.23 | -0.04<br>-0.56 |
| $b_5$                                                                                                                    |     | 0.65<br>35.02   | 0.38<br>11.78   | 0.46<br>15.62    | 0.33<br>11.64     | 0.74<br>19.94  | 0.85<br>13.15   | 0.26<br>3.09   | 0.63<br>23.00   | -0.05<br>-1.24 | 0.08<br>3.45  | 0.87<br>47.15  | 0.57<br>11.15  |
| Adj. $R^2$                                                                                                               |     | 0.81            | 0.54            | 0.57             | 0.49              | 0.51           | 0.45            | 0.12           | 0.68            | 0.07           | 0.71          | 0.92           | 0.59           |
| Panel B-1: $X = \alpha + b r_{ROE}^{RD} + \epsilon$                                                                      |     |                 |                 |                  |                   |                |                 |                |                 |                |               |                |                |
| $\alpha$                                                                                                                 |     | -0.11<br>-1.81  | 0.02<br>0.25    | -0.03<br>-0.33   | -0.06<br>-0.92    | 0.04<br>0.69   | 0.16<br>1.72    | 0.08<br>0.57   | 0.32<br>1.72    | 0.01<br>0.11   | 0.51<br>6.12  | 0.51<br>5.95   | -0.14<br>-2.04 |
| $b$                                                                                                                      |     | 1.03<br>30.97   | 0.77<br>20.82   | 0.65<br>13.80    | 0.85<br>23.03     | 0.73<br>21.30  | 0.88<br>17.97   | 1.46<br>18.02  | 0.55<br>5.47    | 0.92<br>22.62  | 0.07<br>1.63  | -0.14<br>-3.03 | 0.93<br>32.51  |

|                                                                                                                                                     |                |                 |                 |                 |                 |                 |                |                 |                 |                 |                 |                 |                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| $R^2$                                                                                                                                               | 0.63           | 0.44            | 0.25            | 0.49            | 0.45            | 0.37            | 0.40           | 0.05            | 0.48            | 0.00            | 0.02            | 0.79            | 0.40            |
| Panel B-2: $X = \alpha + b_1\text{RMRF} + b_2\text{SMB} + b_3\text{HML} + b_4\text{CMA} + b_5r_{\text{ROE}}^{\text{RD}} + \epsilon$                 |                |                 |                 |                 |                 |                 |                |                 |                 |                 |                 |                 |                 |
| $\alpha$                                                                                                                                            | -0.13<br>-2.66 | 0.06<br>1.23    | 0.05<br>0.79    | 0.02<br>0.41    | 0.13<br>2.50    | 0.27<br>2.93    | 0.31<br>2.27   | 0.43<br>2.29    | 0.22<br>3.55    | 0.60<br>7.09    | 0.16<br>3.29    | -0.07<br>-1.29  | 0.16<br>1.23    |
| $b_1$                                                                                                                                               | -0.01<br>-0.89 | -0.09<br>-7.86  | 0.02<br>1.39    | -0.01<br>-1.07  | -0.03<br>-2.95  | -0.06<br>-2.80  | -0.18<br>-5.58 | -0.14<br>-3.34  | -0.20<br>-14.29 | -0.06<br>-3.30  | 0.04<br>3.54    | -0.01<br>-0.44  | -0.18<br>-6.38  |
| $b_2$                                                                                                                                               | -0.12<br>-7.21 | -0.17<br>-10.54 | 0.09<br>3.97    | 0.06<br>3.51    | 0.04<br>2.16    | -0.18<br>-5.85  | 0.05<br>1.17   | 0.11<br>1.81    | -0.16<br>-7.50  | 0.02<br>0.86    | 0.19<br>11.43   | -0.08<br>-4.82  | -0.27<br>-6.89  |
| $b_3$                                                                                                                                               | 0.31<br>15.27  | 0.26<br>12.88   | -0.47<br>-16.67 | -0.40<br>-18.20 | -0.38<br>-17.82 | 0.03<br>0.89    | -0.55<br>-8.82 | -0.50<br>-6.47  | 0.03<br>1.02    | -0.18<br>-5.22  | 0.44<br>21.28   | 0.26<br>14.37   | 0.09<br>2.20    |
| $b_4$                                                                                                                                               | -0.11<br>-3.54 | 0.02<br>0.78    | 0.09<br>2.14    | 0.14<br>4.00    | 0.21<br>6.22    | -0.01<br>-0.24  | 0.41<br>4.37   | 0.46<br>3.86    | 0.03<br>0.74    | 0.07<br>1.22    | 0.25<br>7.77    | -0.21<br>-7.00  | -0.11<br>-1.49  |
| $b_5$                                                                                                                                               | 1.06<br>38.83  | 0.75<br>28.00   | 0.56<br>15.18   | 0.76<br>26.15   | 0.64<br>22.42   | 0.79<br>15.73   | 1.26<br>15.82  | 0.45<br>4.33    | 0.79<br>23.55   | 0.01<br>0.30    | 0.13<br>4.59    | 1.03<br>38.52   | 0.64<br>10.02   |
| Adj. $R_2$                                                                                                                                          | 0.79           | 0.75            | 0.60            | 0.72            | 0.67            | 0.42            | 0.51           | 0.13            | 0.69            | 0.07            | 0.71            | 0.88            | 0.56            |
| % drop                                                                                                                                              |                | 0.67            | 0.74            | 0.91            | 0.61            | 0.27            | 0.48           | 0.22            | 0.41            | 0.05            | 0.20            | 0.53            | 0.43            |
| Panel C-1: $r_{\text{ROE}}^{\text{RD}} = \alpha + bX + \epsilon$                                                                                    |                |                 |                 |                 |                 |                 |                |                 |                 |                 |                 |                 |                 |
| $\alpha$                                                                                                                                            | 0.24<br>5.12   | 0.25<br>4.33    | 0.36<br>5.39    | 0.27<br>4.97    | 0.23<br>3.97    | 0.22<br>3.60    | 0.22<br>3.64   | 0.41<br>5.48    | 0.24<br>4.25    | 0.42<br>5.33    | 0.50<br>6.48    | 0.22<br>3.97    | 0.42<br>4.50    |
| $b$                                                                                                                                                 | 0.61<br>30.97  | 0.56<br>20.82   | 0.39<br>13.80   | 0.57<br>23.03   | 0.61<br>21.30   | 0.42<br>17.97   | 0.28<br>18.02  | 0.09<br>5.47    | 0.52<br>22.62   | 0.06<br>1.63    | -0.12<br>-3.03  | 75.02<br>32.51  | 43.04<br>13.80  |
| $R^2$                                                                                                                                               | 0.63           | 0.44            | 0.25            | 0.49            | 0.45            | 0.37            | 0.40           | 0.05            | 0.48            | 0.00            | 0.02            | 0.79            | 0.40            |
| Panel C-2: $r_{\text{ROE}}^{\text{RD}} = \alpha + b_1\text{RMRF} + b_2\text{SMB} + b_3\text{HML} + b_4\text{CMA} + b_5\text{RMW} + b_6X + \epsilon$ |                |                 |                 |                 |                 |                 |                |                 |                 |                 |                 |                 |                 |
| $\alpha$                                                                                                                                            |                | 0.21<br>5.36    | 0.20<br>5.52    | 0.15<br>4.91    | 0.10<br>3.42    | 0.24<br>6.10    | 0.18<br>4.70   | 0.22<br>5.77    | 0.17<br>4.42    | 0.21<br>5.16    | 0.23<br>5.70    | 0.16<br>3.36    | 0.28<br>4.65    |
| $b_1$                                                                                                                                               |                | 0.01<br>0.97    | -0.01<br>-1.08  | 0.00<br>0.31    | 0.01<br>1.87    | 0.00<br>-0.54   | 0.00<br>0.11   | 0.00<br>-0.19   | 0.03<br>3.15    | 0.00<br>-0.17   | -0.01<br>-1.33  | -0.02<br>-1.73  | 0.01<br>0.51    |
| $b_2$                                                                                                                                               |                | 0.06<br>3.99    | 0.02<br>1.33    | 0.01<br>1.02    | 0.03<br>2.48    | 0.04<br>2.83    | 0.03<br>2.04   | 0.04<br>2.52    | 0.06<br>4.21    | 0.04<br>2.82    | 0.02<br>1.23    | 0.03<br>1.75    | 0.10<br>4.53    |
| $b_3$                                                                                                                                               |                | -0.26<br>-15.78 | -0.14<br>-6.57  | -0.03<br>-1.81  | -0.03<br>-2.05  | -0.25<br>-14.50 | -0.18<br>-8.76 | -0.23<br>-13.46 | -0.21<br>-12.91 | -0.24<br>-14.17 | -0.29<br>-13.41 | -0.20<br>-11.12 | -0.28<br>-11.97 |
| $b_4$                                                                                                                                               |                | 0.06<br>2.32    | 0.05<br>2.11    | 0.01<br>0.31    | -0.02<br>-1.22  | 0.08<br>2.90    | 0.06<br>2.08   | 0.06<br>2.44    | 0.06<br>2.35    | 0.07<br>2.80    | 0.04<br>1.56    | 0.18<br>6.45    | 0.08<br>2.41    |
| $b_5$                                                                                                                                               |                | 0.57<br>18.26   | 0.62<br>33.11   | 0.52<br>31.03   | 0.55<br>37.36   | 0.69<br>29.47   | 0.57<br>27.57  | 0.68<br>38.40   | 0.57<br>24.00   | 0.69<br>39.11   | 0.67<br>37.08   | -0.08<br>-1.35  | 0.68<br>22.03   |
| $b_6$                                                                                                                                               |                | 0.18<br>4.64    | 0.19<br>8.61    | 0.38<br>18.90   | 0.41<br>21.11   | 0.00<br>0.16    | 0.09<br>7.61   | 0.03<br>3.22    | 0.19<br>7.30    | 0.05<br>2.64    | 0.11<br>3.09    | 0.90<br>14.21   | 0.05<br>1.58    |
| Adj. $R_2$                                                                                                                                          |                | 0.77            | 0.79            | 0.86            | 0.87            | 0.76            | 0.78           | 0.77            | 0.78            | 0.77            | 0.76            | 0.87            | 0.78            |



Table 10: Summary statistics and performance of within-industry profitability factors (July 1976 - June 2023).

Firms are assigned into industry groups following Fama-French five industry definition. In Panel A,  $N(\text{firms})$  is the time-series average of the number of firms,  $RD > 0$  is the time-series average fraction of firms that report positive R&D expense,  $RD/B$  is the time-series average total R&D expense divided by total book equity,  $RD/REVT$  is the time-series average total R&D expense divided by total revenue. In Panel B, we form within-industry profitability factors based on various profitability measures including  $Y(IB)/B$ ,  $Y(SPI)/B$ ,  $Y(EBT)/B$ ,  $Y(FF)/B$ ,  $Y(RD)/B$ ,  $Y(RD + .3SG)/B$ ,  $Y(RD + SG)/B$ ,  $Y(NM)/B$ , and  $Y(REVT)/B$  from the bottom-line to the top-line earnings. Specifically, in the June end of each year  $t$ , firms within each industry are assigned into two size (ME) groups based on median June end market equity of NYSE firms within the given industry, and assigned independently into three profitability groups based on the 30 and 70 percentiles profitability of all NYSE firms within the given industry reported annual earnings in calendar year  $t-1$ . The profitability factor return within each industry in any given month is the average return of the portfolios that are long the high profitability portfolios and short the low profitability portfolios within each size group. Results for our three intangibles-adjusted factors are in bold.

| Panel A: Summary statistics                                     |                                                                |                     |                     |                     |                            |                            |                            |                     |                     |
|-----------------------------------------------------------------|----------------------------------------------------------------|---------------------|---------------------|---------------------|----------------------------|----------------------------|----------------------------|---------------------|---------------------|
|                                                                 | N(firms)                                                       |                     | RD > 0              |                     | RD/B                       |                            | RD/REVT                    |                     |                     |
| Cnsmr                                                           | 778                                                            |                     | 0.21                |                     | 0.04                       |                            | 0.01                       |                     |                     |
| Manuf                                                           | 905                                                            |                     | 0.42                |                     | 0.02                       |                            | 0.01                       |                     |                     |
| HiTec                                                           | 719                                                            |                     | 0.69                |                     | 0.07                       |                            | 0.04                       |                     |                     |
| Hlth                                                            | 381                                                            |                     | 0.75                |                     | 0.12                       |                            | 0.08                       |                     |                     |
| Other                                                           | 1102                                                           |                     | 0.08                |                     | 0.00                       |                            | 0.00                       |                     |                     |
| Panel B: Profitability factors formed on within-industry Y(X)/B |                                                                |                     |                     |                     |                            |                            |                            |                     |                     |
|                                                                 | Monthly Percentage Returns and Associated <i>t</i> -statistics |                     |                     |                     |                            |                            |                            |                     |                     |
|                                                                 | IB                                                             | SPI                 | EBT                 | FF                  | RD                         | RD+.3SG                    | RD+SG                      | NM                  | REVT                |
| Cnsmr                                                           | 0.16<br><i>1.24</i>                                            | 0.10<br><i>0.74</i> | 0.26<br><i>1.95</i> | 0.39<br><i>3.70</i> | <b>0.46</b><br><b>4.80</b> | <b>0.46</b><br><b>5.18</b> | <b>0.40</b><br><b>3.85</b> | 0.37<br><i>3.58</i> | 0.30<br><i>3.22</i> |
| Manuf                                                           | 0.13<br><i>1.02</i>                                            | 0.19<br><i>1.53</i> | 0.29<br><i>2.32</i> | 0.31<br><i>2.91</i> | <b>0.36</b><br><b>3.42</b> | <b>0.38</b><br><b>3.36</b> | <b>0.35</b><br><b>2.69</b> | 0.34<br><i>2.69</i> | 0.36<br><i>3.01</i> |
| HiTec                                                           | 0.14<br><i>0.95</i>                                            | 0.14<br><i>0.97</i> | 0.07<br><i>0.45</i> | 0.23<br><i>1.44</i> | <b>0.41</b><br><b>3.32</b> | <b>0.30</b><br><b>2.66</b> | <b>0.28</b><br><b>2.59</b> | 0.23<br><i>2.22</i> | 0.25<br><i>2.11</i> |
| Hlth                                                            | -0.05<br><i>-0.25</i>                                          | 0.06<br><i>0.30</i> | 0.04<br><i>0.21</i> | 0.12<br><i>0.58</i> | <b>0.36</b><br><b>2.16</b> | <b>0.49</b><br><b>2.87</b> | <b>0.41</b><br><b>2.11</b> | 0.44<br><i>2.33</i> | 0.19<br><i>1.03</i> |
| Other                                                           | 0.33<br><i>3.34</i>                                            | 0.34<br><i>3.55</i> | 0.38<br><i>3.47</i> | 0.37<br><i>4.24</i> | <b>0.39</b><br><b>4.63</b> | <b>0.36</b><br><b>4.37</b> | <b>0.27</b><br><b>3.31</b> | 0.27<br><i>3.10</i> | 0.22<br><i>1.97</i> |

Table 11: Performance of profitability factors with various denominators (July 1976 - June 2023).

In the June end of each year  $t$ , firms are assigned into two size (ME) groups based on median June end market equity of NYSE firms, and assigned independently into three profitability groups based on the 30 and 70 percentiles profitability of all NYSE firms reported annual earnings in calendar year  $t-1$ . The profitability factor return in any given month is the average return of the portfolios that are long the high profitability portfolios and short the low profitability portfolios within each size group. The profitability is measured using various numerators and various denominators. The included numerators are  $Y(\text{IB})$ ,  $Y(\text{SPI})$ ,  $Y(\text{EBT})$ ,  $Y(\text{FF})$ ,  $Y(\text{RD})$ ,  $Y(\text{RD} + .3\text{SG})$ ,  $Y(\text{RD} + \text{SG})$ ,  $Y(\text{NM})$ , and  $Y(\text{REVT})$  from the bottom-line to the top-line earnings. The included denominators are total revenue (REVT), book equity ( $B$ ), total assets (AT), intangible-adjusted book value by Peters and Taylor (2017) ( $B^{\text{PT}}$ ), and market equity (ME). The intangible-adjusted book value is the physical capital plus the intangible capital where the physical capital is measured as the gross property, plant, and equipment and the intangible is measured by capitalizing 100% of R&D expenses and 30% of the SG&A expenses plus the balance sheet intangibles. The data on  $B^{\text{PT}}$  is taken from Peters and Taylor Total Q on Wharton Research Data Services which is available from June 1950 to May 2022 at the time we access the data. In each panel, we report the time-series average of the monthly percentage returns, Fama-French five factor alpha, and the associated t-statistics (in italic) of each factor. Results for our three intangibles-adjusted factors are in bold.

| Moving from bottom-line to top-line profitability: left to right |                       |                     |                       |                       |                            |                            |                            |                     |                       |
|------------------------------------------------------------------|-----------------------|---------------------|-----------------------|-----------------------|----------------------------|----------------------------|----------------------------|---------------------|-----------------------|
| Panel A: $Y(X)/\text{REVT}$                                      |                       |                     |                       |                       |                            |                            |                            |                     |                       |
|                                                                  | IB                    | SPI                 | EBT                   | FF                    | <b>RD</b>                  | <b>RD+.3 SG</b>            | <b>RD+SG</b>               | NM                  | REVT                  |
| Mean                                                             | 0.02<br><i>0.15</i>   | 0.05<br><i>0.50</i> | 0.09<br><i>0.84</i>   | 0.07<br><i>0.68</i>   | <b>0.18</b><br><b>2.28</b> | <b>0.17</b><br><b>2.42</b> | <b>0.08</b><br><b>1.18</b> | 0.02<br><i>0.34</i> |                       |
| $\alpha_{FF5}$                                                   | -0.02<br><i>-0.27</i> | 0.02<br><i>0.37</i> | 0.03<br><i>0.51</i>   | 0.01<br><i>0.18</i>   | <b>0.26</b><br><b>3.58</b> | <b>0.30</b><br><b>4.70</b> | <b>0.22</b><br><b>3.84</b> | 0.18<br><i>3.19</i> |                       |
| Panel B: $Y(X)/B$                                                |                       |                     |                       |                       |                            |                            |                            |                     |                       |
|                                                                  | IB                    | SPI                 | EBT                   | FF                    | <b>RD</b>                  | <b>RD+.3 SG</b>            | <b>RD+SG</b>               | NM                  | REVT                  |
| Mean                                                             | 0.22<br><i>2.12</i>   | 0.28<br><i>2.81</i> | 0.32<br><i>2.88</i>   | 0.38<br><i>3.66</i>   | <b>0.47</b><br><b>6.24</b> | <b>0.48</b><br><b>6.30</b> | <b>0.40</b><br><b>4.83</b> | 0.38<br><i>4.72</i> | 0.28<br><i>3.29</i>   |
| $\alpha_{FF5}$                                                   | 0.01<br><i>0.19</i>   | 0.04<br><i>0.89</i> | 0.06<br><i>1.34</i>   | 0.01<br><i>0.63</i>   | <b>0.25</b><br><b>6.41</b> | <b>0.25</b><br><b>5.58</b> | <b>0.17</b><br><b>2.79</b> | 0.12<br><i>2.02</i> | -0.11<br><i>-1.75</i> |
| Panel C: $Y(X)/\text{AT}$                                        |                       |                     |                       |                       |                            |                            |                            |                     |                       |
|                                                                  | IB                    | SPI                 | EBT                   | FF                    | <b>RD</b>                  | <b>RD+.3 SG</b>            | <b>RD+SG</b>               | NM                  | REVT                  |
| Mean                                                             | 0.14<br><i>1.43</i>   | 0.18<br><i>1.94</i> | 0.23<br><i>2.32</i>   | 0.24<br><i>2.52</i>   | <b>0.33</b><br><b>3.39</b> | <b>0.33</b><br><b>3.51</b> | <b>0.28</b><br><b>2.81</b> | 0.28<br><i>2.82</i> | 0.26<br><i>3.03</i>   |
| $\alpha_{FF5}$                                                   | 0.03<br><i>0.46</i>   | 0.07<br><i>1.20</i> | 0.10<br><i>1.76</i>   | 0.08<br><i>1.34</i>   | <b>0.26</b><br><b>3.95</b> | <b>0.26</b><br><b>4.03</b> | <b>0.22</b><br><b>3.24</b> | 0.20<br><i>2.95</i> | 0.02<br><i>0.22</i>   |
| Panel D: $Y(X)/B^{\text{PT}}$                                    |                       |                     |                       |                       |                            |                            |                            |                     |                       |
|                                                                  | IB                    | SPI                 | EBT                   | FF                    | <b>RD</b>                  | <b>RD+.3 SG</b>            | <b>RD+SG</b>               | NM                  | REVT                  |
| Mean                                                             | 0.15<br><i>1.47</i>   | 0.18<br><i>1.87</i> | 0.21<br><i>2.07</i>   | 0.22<br><i>2.28</i>   | <b>0.31</b><br><b>3.48</b> | <b>0.29</b><br><b>3.00</b> | <b>0.25</b><br><b>2.40</b> | 0.23<br><i>2.14</i> | 0.17<br><i>1.74</i>   |
| $\alpha_{FF5}$                                                   | -0.01<br><i>-0.15</i> | 0.01<br><i>0.12</i> | 0.04<br><i>0.69</i>   | 0.02<br><i>0.32</i>   | <b>0.21</b><br><b>3.04</b> | <b>0.22</b><br><b>2.74</b> | <b>0.24</b><br><b>2.61</b> | 0.20<br><i>2.16</i> | -0.07<br><i>-0.92</i> |
| Panel E: $Y(X)/\text{ME}$                                        |                       |                     |                       |                       |                            |                            |                            |                     |                       |
|                                                                  | IB                    | SPI                 | EBT                   | FF                    | <b>RD</b>                  | <b>RD+.3 SG</b>            | <b>RD+SG</b>               | NM                  | REVT                  |
| Mean                                                             | 0.27<br><i>1.89</i>   | 0.35<br><i>2.51</i> | 0.32<br><i>2.14</i>   | 0.36<br><i>2.52</i>   | <b>0.48</b><br><b>3.95</b> | <b>0.47</b><br><b>3.95</b> | <b>0.45</b><br><b>3.93</b> | 0.43<br><i>3.77</i> | 0.38<br><i>3.23</i>   |
| $\alpha_{FF5}$                                                   | 0.01<br><i>0.11</i>   | 0.05<br><i>0.94</i> | -0.01<br><i>-0.16</i> | -0.01<br><i>-0.14</i> | <b>0.16</b><br><b>3.42</b> | <b>0.08</b><br><b>1.64</b> | <b>0.04</b><br><b>0.68</b> | 0.01<br><i>0.23</i> | -0.09<br><i>-1.21</i> |

Table 12: Predicting analysts' forecast errors (1986 to 2022)

The table presents the average slopes and corresponding Fama-MacBeth t-statistics from annual cross-sectional regressions used to predict analysts' forecast errors (FEs) at the June end of each year. FE is calculated as the difference between the consensus EPS estimate (IBES meanest) and the actual EPS, multiplied by shares outstanding (from IBES) and scaled by the most recent available total assets. EQ1 to EQ4 are the FEs for the future four fiscal quarters. Explanatory variables include profitability measures and control variables. Considered profitability measures (ROE) include  $Y(\text{FF})/B$ ,  $Y(\text{RD})/B$ , and  $Y(\text{NM})/B$ . Control variables include log market equity (ME), market-to-book (M/B), prior 1-month return ( $R_{0\sim 1}$ ), prior 12-month return excluding the most recent month ( $R_{2\sim 12}$ ), and asset growth (AG). The regressions are estimated annually. The slope coefficients are multiplied by 100. All dependent and explanatory variables are winsorized at 1 and 99 percentiles. Firms with most recent available total assets less than \$25 million or book equity less than \$12.5 million are excluded. *Firms* is the average number of firms per regression, *Int* is the regression intercept, and  $R^2$  is the adjusted  $R^2$ .

| Panel A: ROE = $Y(\text{FF})/B$ |       |       |       |        |      |               |                |      |       |
|---------------------------------|-------|-------|-------|--------|------|---------------|----------------|------|-------|
|                                 | Firms | $R^2$ | Int   | ME     | M/B  | $R_{1\sim 2}$ | $R_{2\sim 12}$ | AG   | ROE   |
| <i>Average Slopes</i>           |       |       |       |        |      |               |                |      |       |
| EQ1                             | 2608  | 0.04  | 0.33  | -0.04  | 0.01 | -0.72         | -0.29          | 0.05 | -0.10 |
| EQ2                             | 2467  | 0.05  | 0.70  | -0.08  | 0.03 | -1.16         | -0.38          | 0.12 | -0.09 |
| EQ3                             | 2305  | 0.06  | 1.29  | -0.14  | 0.07 | -1.17         | -0.40          | 0.19 | -0.37 |
| EQ4                             | 1864  | 0.07  | 1.15  | -0.13  | 0.08 | -1.03         | -0.16          | 0.26 | -0.40 |
| <i>t-statistics</i>             |       |       |       |        |      |               |                |      |       |
| EQ1                             |       |       | 5.62  | -7.17  | 1.75 | -8.15         | -7.08          | 3.11 | -2.51 |
| EQ2                             |       |       | 8.92  | -11.29 | 4.03 | -10.39        | -7.73          | 7.98 | -1.31 |
| EQ3                             |       |       | 11.14 | -14.08 | 6.09 | -6.58         | -6.89          | 6.77 | -3.33 |
| EQ4                             |       |       | 12.14 | -13.56 | 8.19 | -4.80         | -2.63          | 6.62 | -3.79 |
| Panel B: ROE = $Y(\text{RD})/B$ |       |       |       |        |      |               |                |      |       |
|                                 | Firms | $R^2$ | Int   | ME     | M/B  | $R_{1\sim 2}$ | $R_{2\sim 12}$ | AG   | ROE   |
| <i>Average Slopes</i>           |       |       |       |        |      |               |                |      |       |
| EQ1                             | 2608  | 0.04  | 0.32  | -0.04  | 0.01 | -0.72         | -0.29          | 0.04 | -0.12 |
| EQ2                             | 2467  | 0.05  | 0.66  | -0.08  | 0.03 | -1.16         | -0.37          | 0.12 | -0.08 |
| EQ3                             | 2305  | 0.05  | 1.28  | -0.14  | 0.08 | -1.18         | -0.38          | 0.19 | -0.34 |
| EQ4                             | 1864  | 0.07  | 1.16  | -0.14  | 0.08 | -1.03         | -0.16          | 0.26 | -0.22 |
| <i>t-statistics</i>             |       |       |       |        |      |               |                |      |       |
| EQ1                             |       |       | 5.23  | -6.04  | 2.27 | -7.84         | -7.09          | 2.82 | -4.39 |
| EQ2                             |       |       | 7.92  | -8.93  | 4.18 | -10.28        | -7.46          | 7.96 | -1.51 |
| EQ3                             |       |       | 10.60 | -12.09 | 6.74 | -6.30         | -6.42          | 6.57 | -4.77 |
| EQ4                             |       |       | 11.92 | -12.71 | 8.26 | -4.65         | -2.42          | 6.43 | -2.19 |
| Panel C: ROE = $Y(\text{NM})/B$ |       |       |       |        |      |               |                |      |       |
|                                 | Firms | $R^2$ | Int   | ME     | M/B  | $R_{1\sim 2}$ | $R_{2\sim 12}$ | AG   | ROE   |
| <i>Average Slopes</i>           |       |       |       |        |      |               |                |      |       |
| EQ1                             | 2608  | 0.04  | 0.33  | -0.04  | 0.01 | -0.71         | -0.29          | 0.05 | -0.01 |
| EQ2                             | 2467  | 0.05  | 0.65  | -0.08  | 0.03 | -1.16         | -0.37          | 0.13 | 0.03  |
| EQ3                             | 2305  | 0.05  | 1.29  | -0.16  | 0.06 | -1.17         | -0.38          | 0.21 | -0.01 |
| EQ4                             | 1864  | 0.07  | 1.20  | -0.15  | 0.08 | -1.01         | -0.15          | 0.26 | -0.05 |
| <i>t-statistics</i>             |       |       |       |        |      |               |                |      |       |
| EQ1                             |       |       | 5.48  | -6.56  | 1.42 | -7.84         | -7.07          | 3.03 | -1.01 |
| EQ2                             |       |       | 7.33  | -8.31  | 3.17 | -10.36        | -7.49          | 8.44 | 1.51  |
| EQ3                             |       |       | 10.17 | -11.37 | 5.46 | -6.30         | -6.33          | 7.18 | -0.33 |
| EQ4                             |       |       | 11.85 | -12.76 | 7.73 | -4.52         | -2.26          | 6.36 | -1.80 |

Table 13: Predicting analysts' forecast errors (1986 to 2022, dummy variables of profitability)

The table presents the average slopes and corresponding Fama-MacBeth t-statistics from annual cross-sectional regressions used to predict analysts' forecast errors (FEs) at the June end of each year. FE is calculated as the difference between the consensus EPS estimate (IBES meanest) and the actual EPS, multiplied by shares outstanding (from IBES) and scaled by the most recent available total assets. EQ1 to EQ4 are the FEs for the future four fiscal quarters. Explanatory variables include dummy variables of profitability measures and control variables. Considered profitability measures (ROE) include  $Y(FF)/B$ ,  $Y(RD)/B$ , and  $Y(NM)/B$ . ROE-L and ROE-H are dummy variables based on profitability measures, equal to one for firms below the 30th percentile (ROE-L) and above the 70th percentile (ROE-H) of NYSE profitability, respectively, for a given measure. Control variables include log market equity (ME), market-to-book (M/B), prior 1-month return ( $R_{0\sim 1}$ ), prior 12-month return excluding the most recent month ( $R_{2\sim 12}$ ), and asset growth (AG). The regressions are estimated annually. The slope coefficients are multiplied by 100. All dependent and explanatory variables are winsorized at 1 and 99 percentiles. Firms with most recent available total assets less than \$25 million or book equity less than \$12.5 million are excluded.  $Firms$  is the average number of firms per regression,  $Int$  is the regression intercept, and  $R^2$  is the adjusted  $R^2$ .

| Panel A: ROE = Y(FF)/B |       |       |       |        |      |               |                |      |       |       |
|------------------------|-------|-------|-------|--------|------|---------------|----------------|------|-------|-------|
|                        | Firms | $R^2$ | Int   | ME     | M/B  | $R_{1\sim 2}$ | $R_{2\sim 12}$ | AG   | ROE-L | ROE-H |
| <i>Average Slopes</i>  |       |       |       |        |      |               |                |      |       |       |
| EQ1                    | 2608  | 0.04  | 0.28  | -0.04  | 0.01 | -0.71         | -0.28          | 0.05 | 0.05  | -0.01 |
| EQ2                    | 2467  | 0.05  | 0.61  | -0.08  | 0.03 | -1.16         | -0.37          | 0.12 | 0.07  | 0.01  |
| EQ3                    | 2305  | 0.05  | 1.11  | -0.14  | 0.07 | -1.18         | -0.38          | 0.19 | 0.18  | -0.01 |
| EQ4                    | 1864  | 0.07  | 1.02  | -0.14  | 0.08 | -1.05         | -0.15          | 0.26 | 0.17  | -0.01 |
| <i>t-statistics</i>    |       |       |       |        |      |               |                |      |       |       |
| EQ1                    |       |       | 5.08  | -6.43  | 1.48 | -8.14         | -7.06          | 3.00 | 4.08  | -0.66 |
| EQ2                    |       |       | 8.74  | -9.61  | 4.13 | -10.35        | -7.57          | 7.86 | 2.48  | 0.75  |
| EQ3                    |       |       | 11.60 | -12.49 | 6.20 | -6.56         | -6.45          | 6.81 | 4.36  | -0.44 |
| EQ4                    |       |       | 11.29 | -12.85 | 7.99 | -4.79         | -2.28          | 6.57 | 4.34  | -0.37 |
| Panel B: ROE = Y(RD)/B |       |       |       |        |      |               |                |      |       |       |
|                        | Firms | $R^2$ | Int   | ME     | M/B  | $R_{1\sim 2}$ | $R_{2\sim 12}$ | AG   | ROE-L | ROE-H |
| <i>Average Slopes</i>  |       |       |       |        |      |               |                |      |       |       |
| EQ1                    | 2608  | 0.04  | 0.26  | -0.04  | 0.01 | -0.72         | -0.28          | 0.04 | 0.06  | -0.02 |
| EQ2                    | 2467  | 0.05  | 0.62  | -0.08  | 0.03 | -1.16         | -0.37          | 0.12 | 0.06  | 0.02  |
| EQ3                    | 2305  | 0.05  | 1.16  | -0.15  | 0.07 | -1.19         | -0.38          | 0.20 | 0.14  | 0.02  |
| EQ4                    | 1864  | 0.07  | 1.13  | -0.15  | 0.07 | -1.02         | -0.14          | 0.27 | 0.05  | 0.03  |
| <i>t-statistics</i>    |       |       |       |        |      |               |                |      |       |       |
| EQ1                    |       |       | 4.21  | -5.45  | 1.68 | -7.85         | -7.05          | 2.88 | 6.08  | -1.89 |
| EQ2                    |       |       | 7.95  | -8.64  | 4.13 | -10.25        | -7.65          | 8.01 | 2.16  | 1.38  |
| EQ3                    |       |       | 11.00 | -11.83 | 6.29 | -6.31         | -6.42          | 6.99 | 3.67  | 1.32  |
| EQ4                    |       |       | 11.97 | -13.27 | 8.09 | -4.57         | -2.25          | 6.66 | 1.50  | 0.92  |
| Panel C: ROE = Y(NM)/B |       |       |       |        |      |               |                |      |       |       |
|                        | Firms | $R^2$ | Int   | ME     | M/B  | $R_{1\sim 2}$ | $R_{2\sim 12}$ | AG   | ROE-L | ROE-H |
| <i>Average Slopes</i>  |       |       |       |        |      |               |                |      |       |       |
| EQ1                    | 2608  | 0.04  | 0.31  | -0.04  | 0.01 | -0.71         | -0.29          | 0.04 | 0.04  | -0.02 |
| EQ2                    | 2467  | 0.05  | 0.69  | -0.08  | 0.03 | -1.15         | -0.37          | 0.13 | -0.06 | -0.01 |
| EQ3                    | 2305  | 0.05  | 1.32  | -0.16  | 0.06 | -1.16         | -0.38          | 0.21 | -0.07 | -0.04 |
| EQ4                    | 1864  | 0.07  | 1.22  | -0.15  | 0.08 | -0.99         | -0.15          | 0.28 | -0.07 | -0.08 |
| <i>t-statistics</i>    |       |       |       |        |      |               |                |      |       |       |
| EQ1                    |       |       | 5.06  | -6.35  | 1.57 | -7.71         | -6.99          | 2.81 | 3.54  | -1.46 |
| EQ2                    |       |       | 8.49  | -8.74  | 3.81 | -10.16        | -7.64          | 8.88 | -2.41 | -0.34 |
| EQ3                    |       |       | 11.72 | -11.99 | 5.94 | -6.19         | -6.44          | 7.53 | -2.04 | -1.43 |
| EQ4                    |       |       | 13.12 | -13.33 | 7.71 | -4.44         | -2.30          | 6.78 | -2.00 | -2.06 |

Table 14: Summary statistics for 2-by-3 ME-ROE(X) portfolios with restricted sample (July 1977 - June 2023).

In the June end of each year  $t$ , all firms are assigned into two size (ME) groups based on median June end market equity of NYSE firms, and assigned independently into three profitability (Y/B) groups based on the 30 and 70 percentiles profitability of all NYSE firms reported annual earnings in calendar year  $t-1$ .  $N(\text{Firms})$  is the time-series average of the number of firms assigned into each portfolio.  $N(\text{ICC}[0])/N(\text{Firms})$  is the fraction of firms that have available implied cost of capital (ICC) estimates.  $R_i - R_f$  is monthly percent excess return over the one-month Treasury bill rate. ICC is the simple average of the four (annualized) ICC estimates from Eskildsen et al. (2024). Data is available from June 1976 to December 2022.  $\text{ICC}[0]$  and  $\text{ICC}[12]$  are the most recent available ICC at the portfolio formation time and 12-month after the portfolio formation time. VW is the time-series average of value-weighted portfolio-level statistics for all firms with available ICC estimates. H-L is a portfolio that is long the high-ROE group and short the low-ROE group within each size group, and  $\text{Mean}(\text{H-L})$  is the equal-weighted average of the two H-L portfolios. Their corresponding t-statistics are also reported. To account for autocorrelation in the ICC estimates, we compute t-statistics with ICC using Newey-West standard errors. The considered profitability measures include  $Y(\text{FF})/B$ ,  $Y(\text{RD})/B$ , and  $Y(\text{NM})/B$ .

| Panel A: $Y(\text{FF})/B$            |       |       |       |       |       |       |      |       |           |
|--------------------------------------|-------|-------|-------|-------|-------|-------|------|-------|-----------|
|                                      | Small |       |       |       | Big   |       |      |       | Mean(H-L) |
|                                      | Low   | Med   | High  | H-L   | Low   | Med   | High | H-L   |           |
| $N(\text{Firms})$                    | 1926  | 996   | 572   |       | 189   | 358   | 324  |       |           |
| $N(\text{ICC}[0])/N(\text{Firms})$   | 0.40  | 0.66  | 0.65  |       | 0.89  | 0.95  | 0.96 |       |           |
| $VW(R_i - R_f)$                      | 0.72  | 0.91  | 1.04  | 0.32  | 0.42  | 0.63  | 0.74 | 0.33  | 0.32      |
| t-statistics                         | 2.61  | 4.04  | 4.11  | 2.75  | 1.82  | 3.23  | 4.05 | 2.88  | 3.30      |
| $VW(\text{ICC}[0])$                  | 10.48 | 10.74 | 11.08 | 0.60  | 10.55 | 10.07 | 9.85 | -0.70 | -0.05     |
| t-statistics                         |       |       |       | 1.29  |       |       |      | -2.98 | -0.16     |
| $VW(\text{ICC}[12] - \text{ICC}[0])$ | 0.07  | -0.06 | -0.19 | -0.26 | 0.32  | 0.09  | 0.02 | -0.30 | -0.28     |
| t-statistics                         |       |       |       | -3.71 |       |       |      | -3.16 | -4.15     |
| Panel B: $Y(\text{RD})/B$            |       |       |       |       |       |       |      |       |           |
|                                      | Small |       |       |       | Big   |       |      |       | Mean(H-L) |
|                                      | Low   | Med   | High  | H-L   | Low   | Med   | High | H-L   |           |
| $N(\text{Firms})$                    | 1799  | 1054  | 641   |       | 177   | 357   | 338  |       |           |
| $N(\text{ICC}[0])/N(\text{Firms})$   | 0.39  | 0.66  | 0.64  |       | 0.89  | 0.95  | 0.96 |       |           |
| $VW(R_i - R_f)$                      | 0.63  | 0.93  | 1.11  | 0.48  | 0.38  | 0.64  | 0.73 | 0.36  | 0.42      |
| t-statistics                         | 2.38  | 4.10  | 4.17  | 5.74  | 1.70  | 3.35  | 3.86 | 3.33  | 5.40      |
| $VW(\text{ICC}[0])$                  | 10.62 | 10.64 | 10.97 | 0.35  | 10.73 | 10.12 | 9.81 | -0.92 | -0.29     |
| t-statistics                         |       |       |       | 0.91  |       |       |      | -4.35 | -1.14     |
| $VW(\text{ICC}[12] - \text{ICC}[0])$ | 0.07  | -0.03 | -0.21 | -0.28 | 0.22  | 0.12  | 0.03 | -0.19 | -0.23     |
| t-statistics                         |       |       |       | -3.78 |       |       |      | -1.88 | -3.06     |
| Panel C: $Y(\text{NM})/B$            |       |       |       |       |       |       |      |       |           |
|                                      | Small |       |       |       | Big   |       |      |       | Mean(H-L) |
|                                      | Low   | Med   | High  | H-L   | Low   | Med   | High | H-L   |           |
| $N(\text{Firms})$                    | 1269  | 1246  | 981   |       | 259   | 367   | 246  |       |           |
| $N(\text{ICC}[0])/N(\text{Firms})$   | 0.42  | 0.60  | 0.52  |       | 0.91  | 0.95  | 0.95 |       |           |
| $VW(R_i - R_f)$                      | 0.63  | 0.93  | 1.09  | 0.46  | 0.51  | 0.63  | 0.78 | 0.28  | 0.37      |
| t-statistics                         | 2.63  | 3.84  | 4.04  | 4.45  | 2.56  | 3.20  | 4.10 | 2.71  | 4.36      |
| $VW(\text{ICC}[0])$                  | 10.48 | 10.70 | 10.96 | 0.48  | 10.48 | 10.07 | 9.64 | -0.84 | -0.18     |
| t-statistics                         |       |       |       | 2.24  |       |       |      | -2.96 | -0.74     |
| $VW(\text{ICC}[12] - \text{ICC}[0])$ | 0.02  | -0.05 | -0.09 | -0.11 | 0.16  | 0.07  | 0.07 | -0.09 | -0.10     |
| t-statistics                         |       |       |       | -1.49 |       |       |      | -1.14 | -1.65     |

Table 15: Buy-and-Hold EAD Return (July 1976 - June 2023).

This table reports the time-series averages of the 3-day buy-and-hold percentage returns around earnings announcement dates (EADs) for each profitability portfolio, along with the corresponding t-statistics. The 3-day EAD return, denoted as  $R_{i,s}^{\text{EAD}}$ , is the buy-and-hold return from day  $s - 1$  to  $s + 1$ , where  $s$  is the EAD. The annual EAD return for portfolio year  $t$  is the sum of  $R_{i,s}^{\text{EAD}}$  across all EADs occurring within that portfolio year, defined from the end of June in year  $t$  to the end of June in year  $t + 1$ . The excess EAD return is calculated as the 3-day buy-and-hold return minus the 3-day buy-and-hold market return over the same period. Each portfolio's EAD return is computed as the equal-weighted average across all firm-level EAD returns in that portfolio and year. Portfolios are formed at the June end of each year  $t$  based on the 30 and 70 percentiles profitability of all NYSE firms reported annual earnings in calendar year  $t - 1$ . The considered profitability measures include Y(FF)/B, Y(RD)/B, and Y(NM)/B. Y(RD)/B–Y(FF)/B is the difference in EAD return between Y(RD)/B and Y(FF)/B for a given portfolio, low, middle, high, and H-L.

| Buy-and-Hold EAD Return          |                     |                                                                          |       |      |      |                                                          |       |      |      |
|----------------------------------|---------------------|--------------------------------------------------------------------------|-------|------|------|----------------------------------------------------------|-------|------|------|
| Panel A: Raw Return              |                     |                                                                          |       |      |      |                                                          |       |      |      |
|                                  |                     | Annual, $\sum_{i,s \in t} R_{i,s}^{\text{EAD}}$                          |       |      |      | Quarterly, $R_{i,s}^{\text{EAD}}$                        |       |      |      |
|                                  |                     | Low                                                                      | Mid   | High | H-L  | Low                                                      | Mid   | High | H-L  |
| Y(FF)/B                          | Mean                | 0.87                                                                     | 1.56  | 1.04 | 0.16 | 0.25                                                     | 0.41  | 0.27 | 0.02 |
|                                  | <i>t-statistics</i> | 3.18                                                                     | 9.02  | 4.65 | 0.66 | 3.29                                                     | 8.99  | 4.68 | 0.36 |
| Y(RD)/B                          | Mean                | 0.83                                                                     | 1.48  | 1.15 | 0.32 | 0.24                                                     | 0.39  | 0.30 | 0.07 |
|                                  | <i>t-statistics</i> | 3.08                                                                     | 8.13  | 5.27 | 1.38 | 3.22                                                     | 8.10  | 5.26 | 1.05 |
| Y(NM)/B                          | Mean                | 0.36                                                                     | 1.32  | 1.71 | 1.35 | 0.10                                                     | 0.35  | 0.46 | 0.36 |
|                                  | <i>t-statistics</i> | 1.73                                                                     | 6.68  | 6.20 | 5.65 | 1.87                                                     | 6.67  | 6.24 | 5.70 |
| Y(RD)/B–Y(FF)/B                  | Mean                | -0.05                                                                    | -0.08 | 0.11 | 0.16 | -0.01                                                    | -0.02 | 0.03 | 0.04 |
|                                  | <i>t-statistics</i> | -1.11                                                                    | -1.20 | 1.53 | 1.70 | -0.98                                                    | -1.18 | 1.59 | 1.68 |
| Panel B: Excess Return to Market |                     |                                                                          |       |      |      |                                                          |       |      |      |
|                                  |                     | Annual, $\sum_{i,s \in t} (R_{i,s}^{\text{EAD}} - R_{m,s}^{\text{EAD}})$ |       |      |      | Quarterly, $R_{i,s}^{\text{EAD}} - R_{m,s}^{\text{EAD}}$ |       |      |      |
|                                  |                     | Low                                                                      | Mid   | High | H-L  | Low                                                      | Mid   | High | H-L  |
| Y(FF)/B                          | Mean                | 0.29                                                                     | 1.03  | 0.51 | 0.22 | 0.09                                                     | 0.27  | 0.13 | 0.05 |
|                                  | <i>t-statistics</i> | 1.26                                                                     | 6.89  | 2.82 | 0.98 | 1.38                                                     | 6.94  | 2.85 | 0.75 |
| Y(RD)/B                          | Mean                | 0.23                                                                     | 0.96  | 0.62 | 0.38 | 0.07                                                     | 0.25  | 0.16 | 0.09 |
|                                  | <i>t-statistics</i> | 1.02                                                                     | 6.51  | 3.60 | 1.87 | 1.18                                                     | 6.55  | 3.61 | 1.60 |
| Y(NM)/B                          | Mean                | -0.22                                                                    | 0.81  | 1.14 | 1.36 | -0.05                                                    | 0.21  | 0.31 | 0.36 |
|                                  | <i>t-statistics</i> | -1.41                                                                    | 5.20  | 4.69 | 6.04 | -1.30                                                    | 5.23  | 4.76 | 6.10 |
| Y(RD)/B–Y(FF)/B                  | Mean                | -0.06                                                                    | -0.06 | 0.10 | 0.16 | -0.02                                                    | -0.02 | 0.03 | 0.04 |
|                                  | <i>t-statistics</i> | -1.48                                                                    | -0.95 | 1.42 | 1.77 | -1.36                                                    | -0.94 | 1.48 | 1.77 |

Table 16: Cross-section regression of annual stock returns

The table reports the average slopes and corresponding Fama-MacBeth t-statistics from cross-section regressions of annual stock returns at the June end of each year. Annual stock returns are the compounded stock returns between June end in year  $t$  to year  $t + 1$ . The missing returns are filled using the market return (value-weighted return of all firms) for the given month. Explanatory variables include profitability measures and control variables. Considered profitability measures (ROE) include  $Y(FF)/B$ ,  $Y(RD)/B$ , and  $Y(NM)/B$ . Control variables include log market equity (ME), market-to-book (M/B), prior 1-month return ( $R_{0\sim 1}$ ), prior 12-month return excluding the most recent month ( $R_{2\sim 12}$ ), asset growth (AG), and analysts' earnings forecast errors for the next one to four quarters (EQ1 to EQ4). Forecast errors is calculated as the difference between the consensus EPS estimate (IBES meanest) made that the June end of year  $t$  and the actual EPS, multiplied by shares outstanding (from IBES) and scaled by the most recent available total assets. The regressions are estimated annually. All explanatory variables are winsorized at 1 and 99 percentiles. Firms with most recent available total assets less than \$25 million or book equity less than \$12.5 million are excluded.  $Firms$  is the average number of firms per regression,  $Int$  is the regression intercept, and  $R^2$  is the adjusted  $R^2$ .

| Panel A: ROE = Y(FF)/B                             |       |       |      |       |       |               |                |       |       |      |       |       |        |
|----------------------------------------------------|-------|-------|------|-------|-------|---------------|----------------|-------|-------|------|-------|-------|--------|
|                                                    | Firms | $R^2$ | Int  | ME    | M/B   | $R_{1\sim 2}$ | $R_{2\sim 12}$ | AG    | ROE   | EQ1  | EQ2   | EQ3   | EQ4    |
| Full sample (1976-2023)                            |       |       |      |       |       |               |                |       |       |      |       |       |        |
| Average Slopes                                     | 3132  | 0.04  | 0.17 | -0.01 | 0.00  | 0.03          | 0.03           | -0.02 | 0.05  |      |       |       |        |
| <i>t-statistics</i>                                |       |       | 3.13 | -2.27 | -1.06 | 0.72          | 1.49           | -3.10 | 1.91  |      |       |       |        |
| Restricted sample with forecast errors (1986-2023) |       |       |      |       |       |               |                |       |       |      |       |       |        |
| Average Slopes                                     | 1731  | 0.17  | 0.24 | -0.02 | 0.01  | -0.20         | -0.02          | 0.01  | -0.03 | 0.07 | -1.50 | -2.07 | -5.82  |
| <i>t-statistics</i>                                |       |       | 3.75 | -3.10 | 1.60  | -3.34         | -1.26          | 0.54  | -0.81 | 0.16 | -5.31 | -7.73 | -11.29 |
| Restricted sample with forecast errors (1986-2023) |       |       |      |       |       |               |                |       |       |      |       |       |        |
| Average Slopes                                     | 1731  | 0.05  | 0.13 | -0.01 | 0.00  | -0.02         | 0.01           | -0.02 | 0.02  |      |       |       |        |
| <i>t-statistics</i>                                |       |       | 2.07 | -1.08 | -0.74 | -0.30         | 0.41           | -1.66 | 0.65  |      |       |       |        |
| Panel B: ROE = Y(RD)/B                             |       |       |      |       |       |               |                |       |       |      |       |       |        |
|                                                    | Firms | $R^2$ | Int  | ME    | M/B   | $R_{1\sim 2}$ | $R_{2\sim 12}$ | AG    | ROE   | EQ1  | EQ2   | EQ3   | EQ4    |
| Full sample (1976-2023)                            |       |       |      |       |       |               |                |       |       |      |       |       |        |
| Average Slopes                                     | 3132  | 0.04  | 0.16 | -0.01 | -0.01 | 0.03          | 0.03           | -0.02 | 0.10  |      |       |       |        |
| <i>t-statistics</i>                                |       |       | 3.09 | -2.39 | -1.78 | 0.76          | 1.45           | -2.61 | 5.56  |      |       |       |        |
| Restricted sample with forecast errors (1986-2023) |       |       |      |       |       |               |                |       |       |      |       |       |        |
| Average Slopes                                     | 1731  | 0.17  | 0.23 | -0.02 | 0.00  | -0.20         | -0.02          | 0.01  | 0.04  | 0.09 | -1.54 | -2.05 | -5.79  |
| <i>t-statistics</i>                                |       |       | 3.70 | -3.21 | 0.97  | -3.17         | -1.15          | 0.89  | 1.57  | 0.19 | -5.38 | -7.68 | -11.33 |
| Restricted sample with forecast errors (1986-2023) |       |       |      |       |       |               |                |       |       |      |       |       |        |
| Average Slopes                                     | 1731  | 0.04  | 0.13 | -0.01 | 0.00  | -0.01         | 0.01           | -0.01 | 0.06  |      |       |       |        |
| <i>t-statistics</i>                                |       |       | 2.00 | -1.14 | -1.28 | -0.22         | 0.37           | -1.18 | 2.53  |      |       |       |        |
| Panel C: ROE = Y(NM)/B                             |       |       |      |       |       |               |                |       |       |      |       |       |        |
|                                                    | Firms | $R^2$ | Int  | ME    | M/B   | $R_{1\sim 2}$ | $R_{2\sim 12}$ | AG    | ROE   | EQ1  | EQ2   | EQ3   | EQ4    |
| Full sample (1976-2023)                            |       |       |      |       |       |               |                |       |       |      |       |       |        |
| Average Slopes                                     | 3132  | 0.04  | 0.15 | -0.01 | 0.00  | 0.02          | 0.03           | -0.02 | 0.02  |      |       |       |        |
| <i>t-statistics</i>                                |       |       | 3.00 | -1.78 | -1.21 | 0.56          | 1.43           | -2.55 | 2.26  |      |       |       |        |
| Restricted sample with forecast errors (1986-2023) |       |       |      |       |       |               |                |       |       |      |       |       |        |
| Average Slopes                                     | 1731  | 0.17  | 0.23 | -0.02 | 0.00  | -0.21         | -0.02          | 0.01  | 0.01  | 0.08 | -1.50 | -2.07 | -5.80  |
| <i>t-statistics</i>                                |       |       | 3.73 | -3.05 | 1.09  | -3.27         | -1.07          | 0.81  | 0.66  | 0.19 | -5.25 | -7.79 | -11.30 |
| Restricted sample with forecast errors (1986-2023) |       |       |      |       |       |               |                |       |       |      |       |       |        |
| Average Slopes                                     | 1731  | 0.04  | 0.12 | -0.01 | 0.00  | -0.02         | 0.01           | -0.01 | 0.01  |      |       |       |        |
| <i>t-statistics</i>                                |       |       | 1.99 | -0.92 | -0.87 | -0.32         | 0.42           | -1.22 | 0.98  |      |       |       |        |

Table 17: Cross-section regression of annual stock returns (dummy variables of profitability)

The table reports the average slopes and corresponding Fama-MacBeth  $t$ -statistics from cross-section regressions of annual stock returns at the June end of each year. Annual stock returns are the compounded stock returns between June end in year  $t$  to year  $t + 1$ . The missing returns are filled using the market return (value-weighted return of all firms) for the given month. Explanatory variables include dummy variables of profitability and control variables. Considered profitability measures (ROE) include  $Y(FF)/B$ ,  $Y(RD)/B$ , and  $Y(NM)/B$ . ROE-L and ROE-H are dummy variables based on profitability measures, equal to one for firms below the 30th percentile (ROE-L) and above the 70th percentile (ROE-H) of NYSE profitability, respectively, for a given measure. Control variables include log market equity (ME), market-to-book (M/B), prior 1-month return ( $R_{0\sim 1}$ ), prior 12-month return excluding the most recent month ( $R_{2\sim 12}$ ), asset growth (AG), and analysts' earnings forecast errors for the next one to four quarters (EQ1 to EQ4). Forecast errors is calculated as the difference between the consensus EPS estimate (IBES meanest) made that the June end of year  $t$  and the actual EPS, multiplied by shares outstanding (from IBES) and scaled by the most recent available total assets. The regressions are estimated annually. All explanatory variables are winsorized at 1 and 99 percentiles. Firms with most recent available total assets less than \$25 million or book equity less than \$12.5 million are excluded.  $Firms$  is the average number of firms per regression,  $Int$  is the regression intercept, and  $R^2$  is the adjusted  $R^2$ .

| Panel A: ROE = $Y(FF)/B$         |       |       |      |       |       |               |                |       |       |       |      |       |       |        |
|----------------------------------|-------|-------|------|-------|-------|---------------|----------------|-------|-------|-------|------|-------|-------|--------|
|                                  | Firms | $R^2$ | Int  | ME    | M/B   | $R_{1\sim 2}$ | $R_{2\sim 12}$ | AG    | ROE-L | ROE-H | EQ1  | EQ2   | EQ3   | EQ4    |
| Full sample (1976-2023)          |       |       |      |       |       |               |                |       |       |       |      |       |       |        |
| Average Slopes                   | 3131  | 0.04  | 0.18 | -0.01 | 0.00  | 0.03          | 0.03           | -0.02 | -0.01 | 0.02  |      |       |       |        |
| <i>t</i> -statistics             |       |       | 3.41 | -2.23 | -1.08 | 0.76          | 1.56           | -2.89 | -1.09 | 2.23  |      |       |       |        |
| Restricted sample (1986-2023)    |       |       |      |       |       |               |                |       |       |       |      |       |       |        |
| Average Slopes                   | 1731  | 0.17  | 0.23 | -0.02 | 0.01  | -0.20         | -0.02          | 0.01  | 0.01  | 0.01  | 0.08 | -1.51 | -2.10 | -5.76  |
| <i>t</i> -statistics             |       |       | 3.68 | -3.18 | 1.39  | -3.22         | -1.20          | 0.65  | 0.66  | 1.22  | 0.18 | -5.28 | -7.83 | -11.31 |
| Restricted sample (1986-2023)    |       |       |      |       |       |               |                |       |       |       |      |       |       |        |
| Average Slopes                   | 1731  | 0.05  | 0.14 | -0.01 | 0.00  | -0.01         | 0.01           | -0.02 | -0.01 | 0.01  |      |       |       |        |
| <i>t</i> -statistics             |       |       | 2.20 | -1.17 | -0.82 | -0.22         | 0.43           | -1.57 | -0.67 | 1.39  |      |       |       |        |
| Panel B: ROE = $Y(RD)/B$         |       |       |      |       |       |               |                |       |       |       |      |       |       |        |
|                                  | Firms | $R^2$ | Int  | ME    | M/B   | $R_{1\sim 2}$ | $R_{2\sim 12}$ | AG    | ROE-L | ROE-H | EQ1  | EQ2   | EQ3   | EQ4    |
| Full sample (1976-2023)          |       |       |      |       |       |               |                |       |       |       |      |       |       |        |
| Average Slopes                   | 3131  | 0.04  | 0.19 | -0.01 | -0.01 | 0.03          | 0.03           | -0.02 | -0.03 | 0.03  |      |       |       |        |
| <i>t</i> -statistics             |       |       | 3.66 | -2.43 | -1.61 | 0.74          | 1.53           | -2.59 | -3.66 | 3.83  |      |       |       |        |
| Full sample with forecast errors |       |       |      |       |       |               |                |       |       |       |      |       |       |        |
| Average Slopes                   | 1731  | 0.17  | 0.25 | -0.02 | 0.00  | -0.20         | -0.02          | 0.01  | -0.01 | 0.03  | 0.11 | -1.53 | -2.07 | -5.76  |
| <i>t</i> -statistics             |       |       | 3.96 | -3.36 | 1.03  | -3.18         | -1.07          | 0.92  | -1.03 | 3.12  | 0.26 | -5.41 | -7.71 | -11.34 |
| Restricted sample (1986-2023)    |       |       |      |       |       |               |                |       |       |       |      |       |       |        |
| Average Slopes                   | 1731  | 0.04  | 0.15 | -0.01 | 0.00  | -0.02         | 0.01           | -0.01 | -0.02 | 0.02  |      |       |       |        |
| <i>t</i> -statistics             |       |       | 2.35 | -1.28 | -1.16 | -0.25         | 0.45           | -1.30 | -2.15 | 2.32  |      |       |       |        |
| Panel C: ROE = $Y(NM)/B$         |       |       |      |       |       |               |                |       |       |       |      |       |       |        |
|                                  | Firms | $R^2$ | Int  | ME    | M/B   | $R_{1\sim 2}$ | $R_{2\sim 12}$ | AG    | ROE-L | ROE-H | EQ1  | EQ2   | EQ3   | EQ4    |
| Full sample (1976-2023)          |       |       |      |       |       |               |                |       |       |       |      |       |       |        |
| Average Slopes                   | 3131  | 0.04  | 0.17 | -0.01 | 0.00  | 0.03          | 0.03           | -0.02 | -0.03 | 0.02  |      |       |       |        |
| <i>t</i> -statistics             |       |       | 3.28 | -1.88 | -1.31 | 0.60          | 1.46           | -2.35 | -3.13 | 1.85  |      |       |       |        |
| Restricted sample (1986-2023)    |       |       |      |       |       |               |                |       |       |       |      |       |       |        |
| Average Slopes                   | 1731  | 0.17  | 0.24 | -0.02 | 0.00  | -0.21         | -0.02          | 0.01  | -0.02 | 0.02  | 0.10 | -1.49 | -2.08 | -5.78  |
| <i>t</i> -statistics             |       |       | 3.82 | -3.08 | 1.04  | -3.28         | -1.09          | 1.12  | -1.85 | 1.08  | 0.23 | -5.19 | -7.84 | -11.33 |
| Restricted sample (1986-2023)    |       |       |      |       |       |               |                |       |       |       |      |       |       |        |
| Average Slopes                   | 1731  | 0.05  | 0.13 | -0.01 | 0.00  | -0.02         | 0.01           | -0.01 | -0.02 | 0.02  |      |       |       |        |
| <i>t</i> -statistics             |       |       | 2.11 | -0.95 | -1.05 | -0.36         | 0.39           | -0.95 | -1.77 | 1.26  |      |       |       |        |



Table 18: Variable definitions.

| Variable         | Definition                                                                                                                                                                                                                                                           |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REVT             | Revenue (quarterly equivalent: REVTQ), Data source: Compustat                                                                                                                                                                                                        |
| COGS             | Cost of good sold (quarterly equivalent: COGSQ), Data source: Compustat                                                                                                                                                                                              |
| XSGA             | Selling, general and administrative expense (quarterly equivalent: XSGAQ), Data source: Compustat                                                                                                                                                                    |
| XRD              | Research and development expense (quarterly equivalent: XRDQ), Data source: Compustat                                                                                                                                                                                |
| DP               | Depreciation (quarterly equivalent: DPQ), Data source: Compustat                                                                                                                                                                                                     |
| XINT             | Interest and related expense (quarterly equivalent: XINTQ), Data source: Compustat                                                                                                                                                                                   |
| SPI              | Special items (quarterly equivalent: SPIQ), Data source: Compustat                                                                                                                                                                                                   |
| IB               | Income before extraordinary items (quarterly equivalent: IBQ), Data source: Compustat                                                                                                                                                                                |
| CEQ              | Common/ordinary equity, Data source: Compustat                                                                                                                                                                                                                       |
| AT               | Total assets, Data source: Compustat                                                                                                                                                                                                                                 |
| DVC              | Common/ordinary dividends, Data source: Compustat                                                                                                                                                                                                                    |
| RDIP             | The portion of R&D considered to be “purchased” and written off immediately upon acquisition if the R&D items are deemed not to have an alternative use, Data source: Compustat                                                                                      |
| RDQ              | Report date of quarterly earnings, Data source: Compustat                                                                                                                                                                                                            |
| PRIMEXCH         | Primary exchange, Data source: CRSP                                                                                                                                                                                                                                  |
| SHRCD            | Share code, Data source: CRSP                                                                                                                                                                                                                                        |
| SGA              | XSGA excluding research and development related expense following Peters and Taylor (2017)                                                                                                                                                                           |
| ME               | Market equity, price times number of shares outstanding. For factor construction and cross-section return regressions, we use June-end market equity; while for cross-section earnings forecasting regressions, we use December-end market equity                    |
| $B$              | Book equity, equal to CEQ                                                                                                                                                                                                                                            |
| BQ               | Quarterly book equity computed following Hou et al. (2015)                                                                                                                                                                                                           |
| $B^{PT}$         | Intangible-adjusted book equity following Peters and Taylor (2017), data source: <a href="https://wrds-www.wharton.upenn.edu/">https://wrds-www.wharton.upenn.edu/</a> , date visited: December 2, 2023                                                              |
| BM               | Book-to-market ratio, $B/ME$                                                                                                                                                                                                                                         |
| AG               | Asset growth, $dAT(t)/AT(t-1)$                                                                                                                                                                                                                                       |
| $Y(REVT)$        | REVT                                                                                                                                                                                                                                                                 |
| $Y(NM)$          | REVT-COGS, following Novy-Marx (2013)                                                                                                                                                                                                                                |
| $Y(RD + SG)$     | REVT-COGS-XINT, following Eisfeldt et al. (2022) definition of investment in intangibles                                                                                                                                                                             |
| $Y(RD + .3SG)$   | REVT-COGS-XSGA-XINT+0.3*SGA+XRD, following Peters and Taylor (2017) definition of investment in intangibles                                                                                                                                                          |
| $Y(RD)$          | REVT-COGS-XSGA-XINT+XRD                                                                                                                                                                                                                                              |
| $Y(FF)$          | REVT-COGS-XSGA-XINT, following Fama and French (2015) instruction of RMW factor                                                                                                                                                                                      |
| $Y(EBT)$         | REVT-COGS-XSGA-XINT-DP                                                                                                                                                                                                                                               |
| $Y(SPI)$         | IB-SPI                                                                                                                                                                                                                                                               |
| $Y(IB)$          | IB, following Fama and French (2006) using annual figure and following Hou et al. (2015) using quarterly figure                                                                                                                                                      |
| FF5              | Fama-French five factor model, Fama and French (2015), data source: <a href="http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html">http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html</a> , date visited: December 2, 2023 |
| FF5(C)           | Fama-French five factor model with RMW replaced by the cash-based operating profitability factor, Fama and French (2018)                                                                                                                                             |
| FF6              | Fama-French five factor model plus momentum factor, Fama and French (2016)                                                                                                                                                                                           |
| FF6 <sup>1</sup> | FF6 with HML replaced by HML(Devil)                                                                                                                                                                                                                                  |
| FF6 <sup>2</sup> | FF6 with HML replaced by HML(EKP)                                                                                                                                                                                                                                    |
| HXZ4             | Hou-Xue-Zhang four factor model, Hou et al. (2015), data source: <a href="https://global-q.org/factors.html">https://global-q.org/factors.html</a> , date visited: December 2, 2023                                                                                  |
| HMXZ5            | Hou-Mo-Xue-Zhang five factor model, HXZ4 plus expected growth factor, Hou et al. (2021), data source: <a href="https://global-q.org/factors.html">https://global-q.org/factors.html</a> , date visited: December 2, 2023                                             |
| DHS3             | Daniel-Hirshleifer-Sun three factor model, Daniel et al. (2020), data source: <a href="https://sites.google.com/view/1insunhome">https://sites.google.com/view/1insunhome</a> , date visited: December 2, 2023                                                       |
| SY4              | Stambaugh-Yuan four factor model Stambaugh and Yuan (2017), available up to the end of 2016, data source: <a href="https://finance.wharton.upenn.edu/~stambaug/">https://finance.wharton.upenn.edu/~stambaug/</a> , date visited: December 2, 2023                   |
| UMD              | Momentum factor, data source: <a href="http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html">http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html</a> , date visited: December 2, 2023                                       |
| RMW              | Robust-minus-weak factor in FF5                                                                                                                                                                                                                                      |

|            |                                                                                                                                                                                                                                                                                                                                              |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HML(EKP)   | Intangible-adjusted value factor, Eisfeldt et al. (2022), data source: <a href="https://github.com/edwardtkim/intangiblevalue">https://github.com/edwardtkim/intangiblevalue</a> , date visited: December 2, 2023                                                                                                                            |
| HML(Devil) | Monthly-rebalanced HML factor with most recent market equity, Asness and Frazzini (2013), data source: <a href="https://www.aqr.com/Insights/Datasets/The-Devil-in-HMLs-Details-Factors-Monthly">https://www.aqr.com/Insights/Datasets/The-Devil-in-HMLs-Details-Factors-Monthly</a> , date visited: December 2, 2023                        |
| RMW(GLPZ)  | intangibles-adjusted profitability factor by Asness and Frazzini (2013) to replace RMW in FF5, data source: <a href="https://sites.google.com/site/rheathpeters">https://sites.google.com/site/rheathpeters</a> , date visited: August 30, 2024                                                                                              |
| ROE(GLPZ)  | intangibles-adjusted profitability factor by Asness and Frazzini (2013) to replace R.ROE in HXZ4, data source: <a href="https://sites.google.com/site/rheathpeters">https://sites.google.com/site/rheathpeters</a> , date visited: August 30, 2024                                                                                           |
| ICC        | Implied cost of capital. It is the simple average of the four ICC estimates from Eskildsen et al. (2024). Data is available from June 1976 to December 2022.                                                                                                                                                                                 |
| EAD        | Earnings announcement date. Compustat data item RDQ.                                                                                                                                                                                                                                                                                         |
| EQ1-EQ4    | Analysts' earnings forecast errors for the future one to four quarters. Earnings forecast errors is calculated as the difference between the consensus EPS estimate (IBES meanest) made that the June end of year $t$ and the actual EPS, multiplied by shares outstanding (from IBES) and scaled by the most recent available total assets. |

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# Internet Appendix for “Intangibles Investment and Asset Quality”

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September 23, 2025

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## A Additional Figures and Tables

### A.1 Figure A1: $R^2$ Difference in Forecasting Future Cash Flows

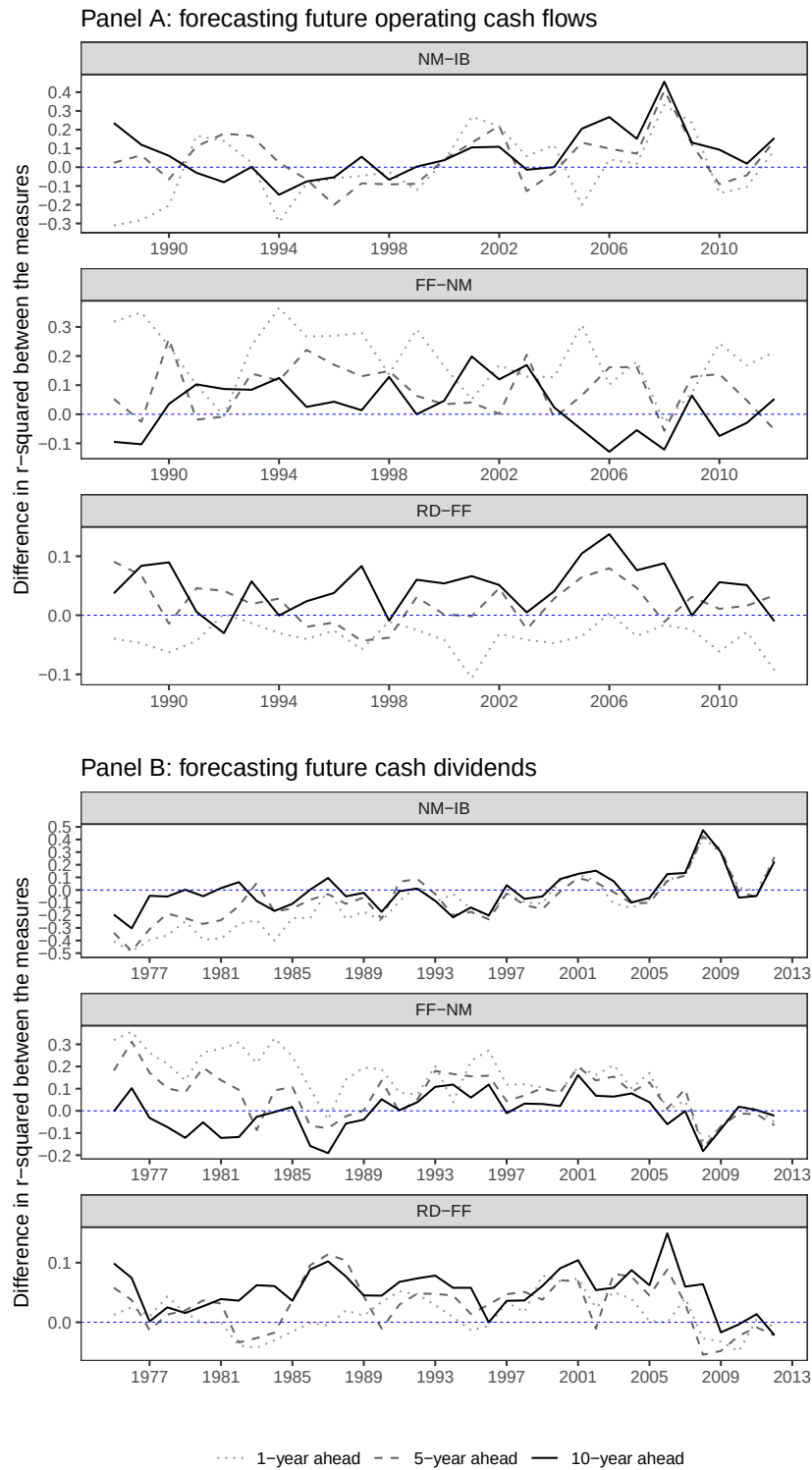


Figure A1: The differences in  $R^2$  of the cross section regressions when forecasting future cash flows using different profitability measures at 1-, 5-, 10-year horizons.

## A.2 Figure A2: Timeline for Portfolio Formation

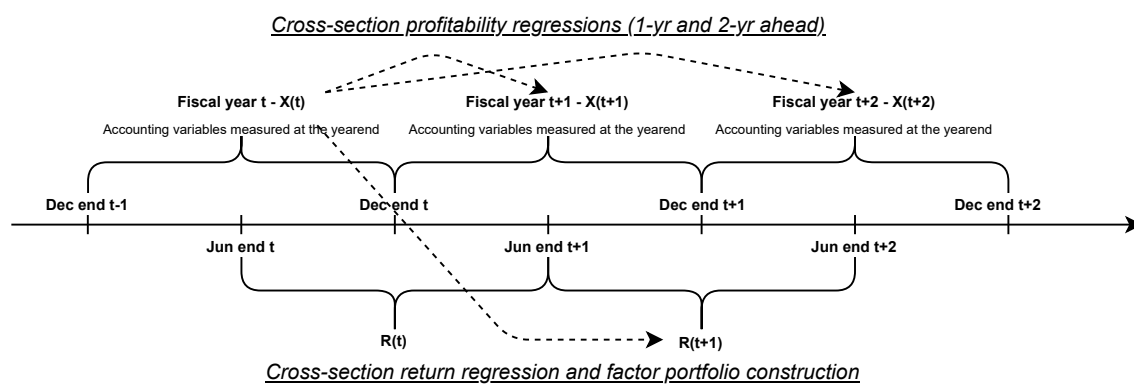


Figure A2: Timeline for portfolio formation and cross-section profitability regressions.

### A.3 Figure A3: Profitability Portfolio versus Quality ETFs

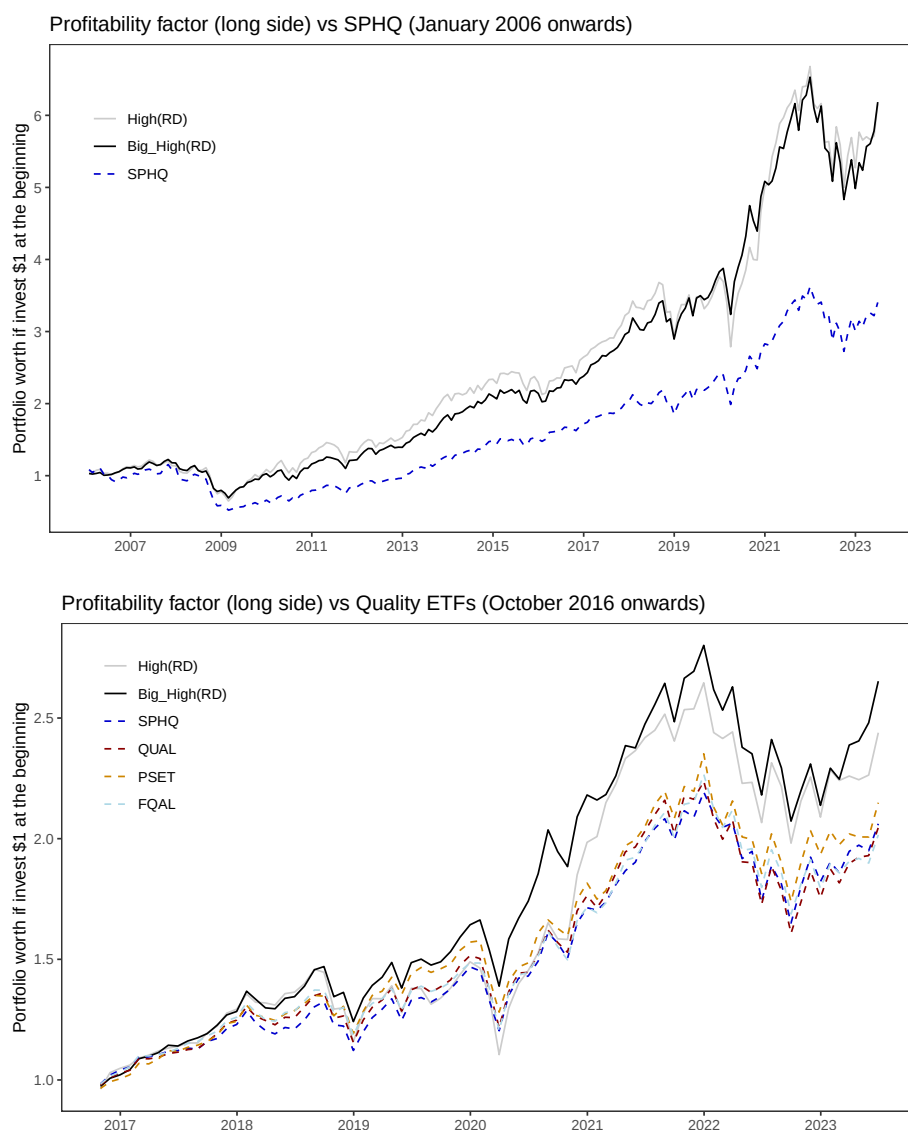


Figure A3: Cumulative return of annual-rebalanced high Y(RD)/B portfolio and quality ETFs. The considered quality ETFs include Invesco S&P 500 Quality ETF (SPHQ, available from January 2006), iShares MSCI USA Quality Factor ETF (QUAL, available from August 2013), Principal Quality ETF (PSET, available from April 2016), and Fidelity Quality Factor ETF (FQAL, available from October 2016). Since some of the quality ETFs are concentrated on large firms, we show the cumulative return of the big-high Y(RD)/B portfolio separately.

## A.4 Figure A4: R&D/Sales and R&D/BE Across Industries

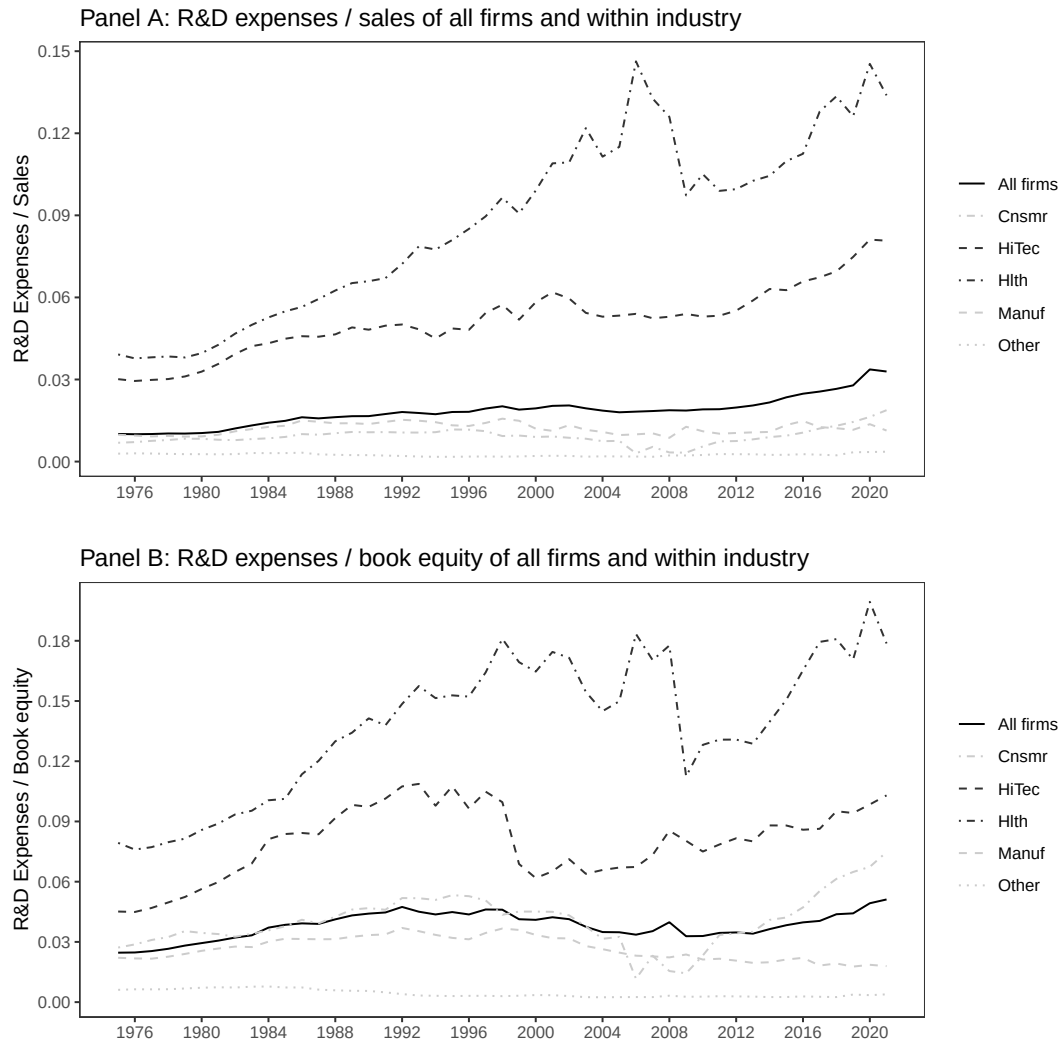


Figure A4: Aggregate R&D expenses scaled by sales and book equity for all firms and across Fama-French five industries, 1975 to 2022.

ALT TEXT: The figure shows R&D expenses scaled by sales and book equity for all firms and across industries. High Tech and Health are the most R&D-intensive industries and display the fastest growth in R&D intensity over the sample period, highlighting substantial cross-industry heterogeneity.



## A.5 Figure A5: Number of EADs on Calendar Date

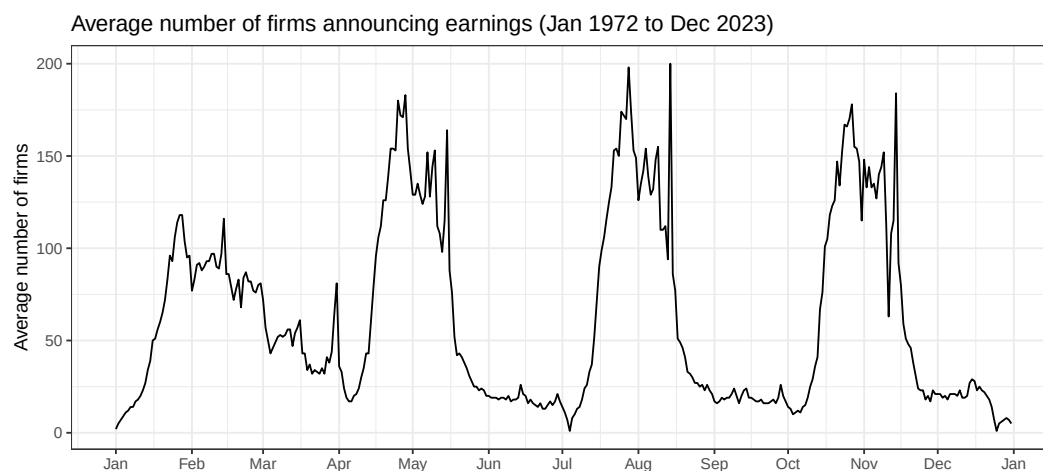


Figure A5: Average number of EADs on each calendar date, 1977 to 2023.

EAD is the quarterly earnings announcement dates (Compustat item RDQ). The figure plots the average EADs occurred on a given calendar day.

## A.6 Figure A6: Cumulative Returns Spreads Over Trading Dates

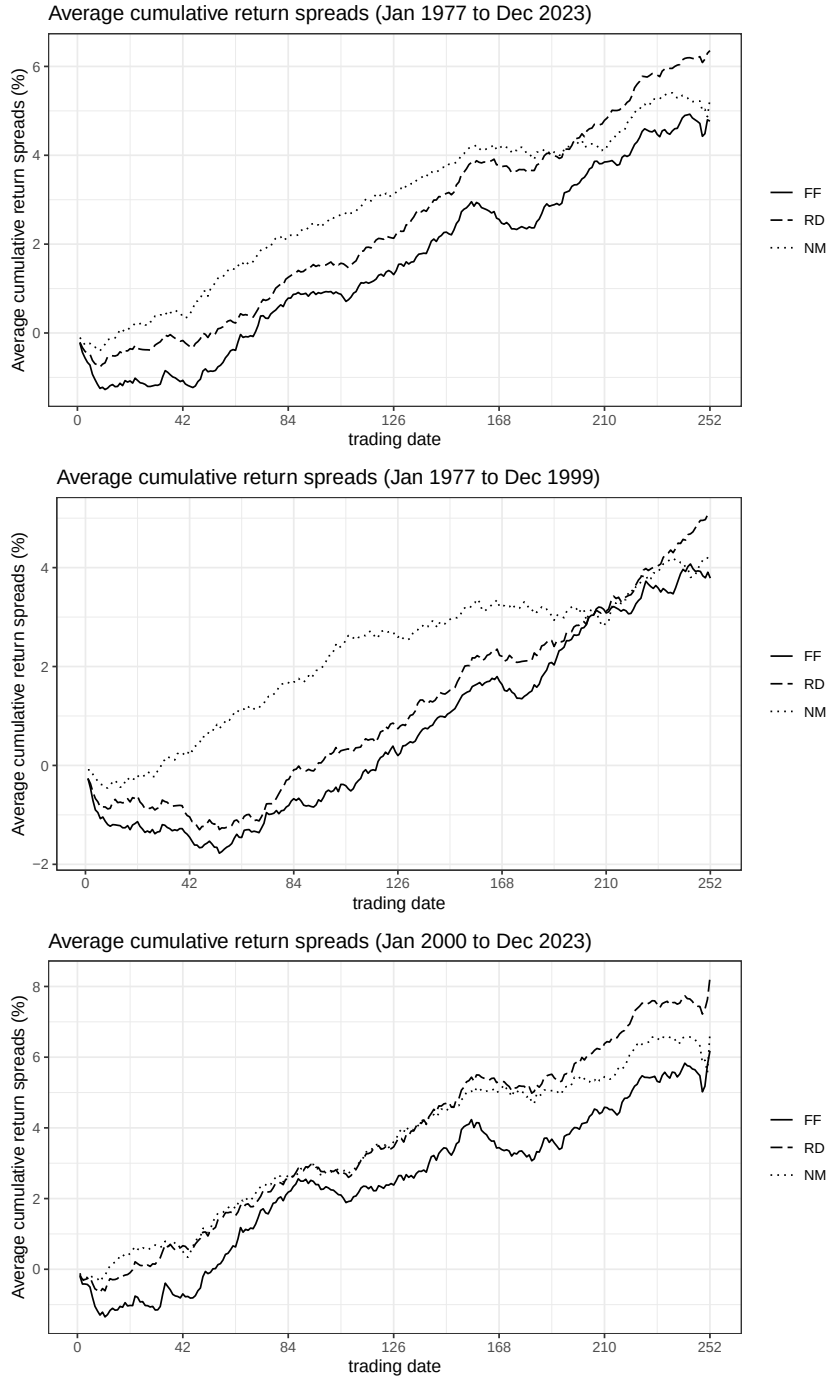


Figure A6: Average cumulative returns spread between high and low profitability portfolios over trading dates, 1977 to 2023.

In the June end of each year  $t$ , all firms are assigned into three profitability ( $Y/B$ ) groups based on the 30 and 70 percentiles profitability of all NYSE firms reported annual earnings in calendar year  $t-1$ . The consider profitability measures include  $Y(FF)/B$ ,  $Y(RD)/B$ , and  $Y(NM)/B$ . The cumulative returns spread between high and low profitability portfolios is the difference in cumulative value-weighted portfolio return between the high and low profitability portfolios.

## A.7 Table A1: Fractions of Firms Delisted After $\tau$ Years

Table A1: Fractions of firms delisted after  $\tau$  years (1975 - 2022).

This table reports the time-series average of the fractions of firms that are delisted after  $\tau = 1, 2, \dots, 10$  years,  $(N_0 - N_\tau)/N_0$ . The reasons of the delisting correspond to the first digit of of delisting code (CRSP data item DLSTCD). To be included, a firm must be listed in one of the three major exchanges and have a share code of 10 or 11. Merger, Dropped, Exchanges, and Liquidation are the fractions of firms that are delisted for the given reason divided by the total number of firms delisted. We remove firms with total assets less than \$12.5 million or book equity less than \$25 million in year  $t$ .

| $\tau$                                                     | 0    | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   |
|------------------------------------------------------------|------|------|------|------|------|------|------|------|------|------|------|
| Panel A: All firms                                         |      |      |      |      |      |      |      |      |      |      |      |
| $N_\tau$                                                   | 3515 | 3323 | 3092 | 2876 | 2679 | 2495 | 2330 | 2181 | 2045 | 1921 | 1808 |
| $(N_0 - N_\tau)/N_0$                                       |      | 0.05 | 0.11 | 0.17 | 0.23 | 0.28 | 0.32 | 0.37 | 0.40 | 0.44 | 0.47 |
| Panel B: Fraction of firms delisted due to various reasons |      |      |      |      |      |      |      |      |      |      |      |
| Merger                                                     |      | 0.04 | 0.09 | 0.13 | 0.17 | 0.21 | 0.24 | 0.27 | 0.30 | 0.33 | 0.35 |
| Dropped                                                    |      | 0.01 | 0.02 | 0.03 | 0.05 | 0.06 | 0.07 | 0.08 | 0.08 | 0.09 | 0.10 |
| Exchanges                                                  |      | 0.00 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.02 | 0.02 | 0.02 |
| Liquidation                                                |      | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.01 | 0.01 |

## A.8 Table A2: Cross-Section Univariate Regression of Future Cash Dividends

Table A2: Cross-section univariate regression of future profitability (cash dividends) at the portfolio level (1975 - 2022).

This table reports the time-series averages of the slope coefficients, t-statistics, and r-squared for the cross-section regressions of future cash dividends (divided by current book value of equity) at the portfolio level. Cash dividends is Compustat data item DVC. Firms are assigned into 5-by-10 industry and ROE(FF) groups in year  $t$ . Industry groups are based on Fama-French five industry classification and ROE(FF) cut points are NYSE ROE(FF) decile within each industry group. For notation convenience, we denote  $Z_{t+\tau}^X$  as  $Y_{t+\tau}^X/B_t$  where  $X$  includes IB, NM, FF, RD, RD+.3SG, and RD+SG. The portfolio-level earnings and book equity are the value-weighted earnings and book equity for a given portfolio in year  $t$ . The portfolio-level future cash flows is computed following equation (4). We also test the null hypothesis that the difference of the r-squared time series between  $Z_{t+\tau}^{\text{NM}}$ ,  $Z_{t+\tau}^{\text{FF}}$ ,  $Z_{t+\tau}^{\text{RD}}$ ,  $Z_{t+\tau}^{\text{RD+.3SG}}$ ,  $Z_{t+\tau}^{\text{RD+SG}}$  and  $Z_{t+\tau}^{\text{IB}}$ ,  $Z_{t+\tau}^{\text{NM}}$ ,  $Z_{t+\tau}^{\text{FF}}$ ,  $Z_{t+\tau}^{\text{RD}}$ ,  $Z_{t+\tau}^{\text{RD+.3SG}}$  is zero and report the t-statistics in the panel B and D. We remove firms with total assets less than \$12.5 million or book equity less than \$25 million in year  $t$ .

| Panel A: Average slopes for $Z_t^{\text{IB}}$ , $Z_t^{\text{NM}}$ , and $Z_t^{\text{FF}}$         |       |      |                   |                         |                         |       |                        |                         |                         |      |                      |
|---------------------------------------------------------------------------------------------------|-------|------|-------------------|-------------------------|-------------------------|-------|------------------------|-------------------------|-------------------------|------|----------------------|
|                                                                                                   | $R^2$ | Intc | $Z_t^{\text{IB}}$ |                         | $R^2$                   | Intc  | $Z_t^{\text{NM}}$      |                         | $R^2$                   | Intc | $Z_t^{\text{FF}}$    |
| $Z_{t+1}^{\text{DVC}}$                                                                            | 0.50  | 0.03 | 0.26              | $Z_{t+1}^{\text{DVC}}$  | 0.45                    | 0.00  | 0.07                   | $Z_{t+1}^{\text{DVC}}$  | 0.57                    | 0.01 | 0.15                 |
| $Z_{t+2}^{\text{DVC}}$                                                                            | 0.51  | 0.03 | 0.29              | $Z_{t+2}^{\text{DVC}}$  | 0.46                    | 0.00  | 0.08                   | $Z_{t+2}^{\text{DVC}}$  | 0.57                    | 0.01 | 0.17                 |
| $Z_{t+3}^{\text{DVC}}$                                                                            | 0.51  | 0.03 | 0.32              | $Z_{t+3}^{\text{DVC}}$  | 0.45                    | 0.00  | 0.08                   | $Z_{t+3}^{\text{DVC}}$  | 0.55                    | 0.01 | 0.18                 |
| $Z_{t+4}^{\text{DVC}}$                                                                            | 0.50  | 0.03 | 0.36              | $Z_{t+4}^{\text{DVC}}$  | 0.45                    | 0.00  | 0.09                   | $Z_{t+4}^{\text{DVC}}$  | 0.53                    | 0.01 | 0.20                 |
| $Z_{t+5}^{\text{DVC}}$                                                                            | 0.49  | 0.03 | 0.39              | $Z_{t+5}^{\text{DVC}}$  | 0.45                    | 0.00  | 0.10                   | $Z_{t+5}^{\text{DVC}}$  | 0.52                    | 0.01 | 0.22                 |
| $Z_{t+6}^{\text{DVC}}$                                                                            | 0.49  | 0.03 | 0.43              | $Z_{t+6}^{\text{DVC}}$  | 0.45                    | -0.01 | 0.11                   | $Z_{t+6}^{\text{DVC}}$  | 0.50                    | 0.01 | 0.23                 |
| $Z_{t+7}^{\text{DVC}}$                                                                            | 0.47  | 0.03 | 0.48              | $Z_{t+7}^{\text{DVC}}$  | 0.44                    | -0.01 | 0.12                   | $Z_{t+7}^{\text{DVC}}$  | 0.48                    | 0.01 | 0.25                 |
| $Z_{t+8}^{\text{DVC}}$                                                                            | 0.46  | 0.03 | 0.53              | $Z_{t+8}^{\text{DVC}}$  | 0.43                    | -0.01 | 0.13                   | $Z_{t+8}^{\text{DVC}}$  | 0.46                    | 0.01 | 0.27                 |
| $Z_{t+9}^{\text{DVC}}$                                                                            | 0.44  | 0.03 | 0.57              | $Z_{t+9}^{\text{DVC}}$  | 0.43                    | -0.01 | 0.14                   | $Z_{t+9}^{\text{DVC}}$  | 0.43                    | 0.02 | 0.29                 |
| $Z_{t+10}^{\text{DVC}}$                                                                           | 0.42  | 0.04 | 0.60              | $Z_{t+10}^{\text{DVC}}$ | 0.41                    | -0.01 | 0.15                   | $Z_{t+10}^{\text{DVC}}$ | 0.41                    | 0.02 | 0.30                 |
| Panel B: $t$ -statistics for Panel A                                                              |       |      |                   |                         |                         |       |                        |                         |                         |      |                      |
|                                                                                                   |       | Intc | $Z_t^{\text{IB}}$ |                         | $\Delta R^2$<br>(NM-IB) | Intc  | $Z_t^{\text{NM}}$      |                         | $\Delta R^2$<br>(FF-NM) | Intc | $Z_t^{\text{FF}}$    |
| 1                                                                                                 |       | 6.00 | 16.47             | 1                       | -1.37                   | 0.50  | 6.66                   | 1                       | 6.22                    | 2.44 | 13.71                |
| 2                                                                                                 |       | 3.28 | 15.02             | 2                       | -1.66                   | 0.10  | 5.20                   | 2                       | 6.28                    | 1.18 | 11.32                |
| 3                                                                                                 |       | 5.26 | 13.47             | 3                       | -1.81                   | 0.03  | 6.24                   | 3                       | 5.44                    | 1.97 | 14.98                |
| 4                                                                                                 |       | 5.04 | 14.13             | 4                       | -1.75                   | -0.25 | 7.66                   | 4                       | 5.20                    | 1.47 | 15.44                |
| 5                                                                                                 |       | 4.24 | 11.79             | 5                       | -1.38                   | -0.72 | 10.67                  | 5                       | 4.39                    | 1.53 | 16.17                |
| 6                                                                                                 |       | 4.70 | 12.70             | 6                       | -1.42                   | -0.94 | 10.63                  | 6                       | 3.58                    | 1.57 | 19.65                |
| 7                                                                                                 |       | 4.35 | 9.99              | 7                       | -1.21                   | -1.52 | 12.33                  | 7                       | 2.71                    | 1.39 | 18.78                |
| 8                                                                                                 |       | 4.76 | 10.97             | 8                       | -1.24                   | -1.66 | 11.57                  | 8                       | 1.79                    | 1.55 | 15.15                |
| 9                                                                                                 |       | 5.55 | 10.02             | 9                       | -0.64                   | -2.13 | 12.39                  | 9                       | 0.43                    | 1.71 | 16.24                |
| 10                                                                                                |       | 6.63 | 10.43             | 10                      | -0.40                   | -1.45 | 13.83                  | 10                      | -0.43                   | 2.38 | 17.64                |
| Panel C: Average slopes for $Z_t^{\text{RD}}$ , $Z_t^{\text{RD+.3SG}}$ , and $Z_t^{\text{RD+SG}}$ |       |      |                   |                         |                         |       |                        |                         |                         |      |                      |
|                                                                                                   | $R^2$ | Intc | $Z_t^{\text{RD}}$ |                         | $R^2$                   | Intc  | $Z_t^{\text{RD+.3SG}}$ |                         | $R^2$                   | Intc | $Z_t^{\text{RD+SG}}$ |
| $Z_{t+1}^{\text{DVC}}$                                                                            | 0.58  | 0.00 | 0.15              | $Z_{t+1}^{\text{DVC}}$  | 0.57                    | 0.00  | 0.12                   | $Z_{t+1}^{\text{DVC}}$  | 0.45                    | 0.01 | 0.07                 |
| $Z_{t+2}^{\text{DVC}}$                                                                            | 0.59  | 0.00 | 0.16              | $Z_{t+2}^{\text{DVC}}$  | 0.58                    | 0.00  | 0.13                   | $Z_{t+2}^{\text{DVC}}$  | 0.46                    | 0.00 | 0.08                 |
| $Z_{t+3}^{\text{DVC}}$                                                                            | 0.58  | 0.00 | 0.17              | $Z_{t+3}^{\text{DVC}}$  | 0.57                    | 0.00  | 0.14                   | $Z_{t+3}^{\text{DVC}}$  | 0.45                    | 0.00 | 0.08                 |
| $Z_{t+4}^{\text{DVC}}$                                                                            | 0.56  | 0.00 | 0.19              | $Z_{t+4}^{\text{DVC}}$  | 0.56                    | 0.00  | 0.16                   | $Z_{t+4}^{\text{DVC}}$  | 0.46                    | 0.00 | 0.09                 |
| $Z_{t+5}^{\text{DVC}}$                                                                            | 0.55  | 0.00 | 0.21              | $Z_{t+5}^{\text{DVC}}$  | 0.55                    | -0.01 | 0.17                   | $Z_{t+5}^{\text{DVC}}$  | 0.46                    | 0.00 | 0.10                 |
| $Z_{t+6}^{\text{DVC}}$                                                                            | 0.54  | 0.00 | 0.22              | $Z_{t+6}^{\text{DVC}}$  | 0.54                    | -0.01 | 0.19                   | $Z_{t+6}^{\text{DVC}}$  | 0.46                    | 0.00 | 0.11                 |
| $Z_{t+7}^{\text{DVC}}$                                                                            | 0.52  | 0.00 | 0.24              | $Z_{t+7}^{\text{DVC}}$  | 0.53                    | -0.01 | 0.21                   | $Z_{t+7}^{\text{DVC}}$  | 0.45                    | 0.00 | 0.12                 |
| $Z_{t+8}^{\text{DVC}}$                                                                            | 0.50  | 0.00 | 0.26              | $Z_{t+8}^{\text{DVC}}$  | 0.52                    | -0.01 | 0.23                   | $Z_{t+8}^{\text{DVC}}$  | 0.44                    | 0.00 | 0.14                 |
| $Z_{t+9}^{\text{DVC}}$                                                                            | 0.48  | 0.00 | 0.28              | $Z_{t+9}^{\text{DVC}}$  | 0.51                    | -0.01 | 0.24                   | $Z_{t+9}^{\text{DVC}}$  | 0.44                    | 0.00 | 0.15                 |

|                                      |                         |      |            |                  |                              |       |                 |                  |                            |       |               |
|--------------------------------------|-------------------------|------|------------|------------------|------------------------------|-------|-----------------|------------------|----------------------------|-------|---------------|
| $Z_{t+10}^{DVC}$                     | 0.46                    | 0.01 | 0.30       | $Z_{t+10}^{DVC}$ | 0.49                         | -0.01 | 0.26            | $Z_{t+10}^{DVC}$ | 0.43                       | 0.00  | 0.16          |
| Panel D: $t$ -statistics for Panel C |                         |      |            |                  |                              |       |                 |                  |                            |       |               |
|                                      | $\Delta R^2$<br>(RD-FF) | Intc | $Z_t^{RD}$ |                  | $\Delta R^2$<br>(RD+.3SG-FF) | Intc  | $Z_t^{RD+.3SG}$ |                  | $\Delta R^2$<br>(RD+SG-FF) | Intc  | $Z_t^{RD+SG}$ |
| 1                                    | 3.38                    | 0.81 | 9.20       | 1                | 0.21                         | 0.01  | 9.09            | 1                | -6.81                      | 0.93  | 6.27          |
| 2                                    | 3.45                    | 0.39 | 10.65      | 2                | 0.60                         | -0.34 | 7.84            | 2                | -6.96                      | 0.53  | 4.23          |
| 3                                    | 4.10                    | 0.53 | 11.31      | 3                | 1.90                         | -0.47 | 9.06            | 3                | -5.58                      | 0.61  | 5.69          |
| 4                                    | 4.30                    | 0.36 | 16.00      | 4                | 3.48                         | -0.74 | 14.16           | 4                | -5.14                      | 0.32  | 7.47          |
| 5                                    | 4.94                    | 0.29 | 16.22      | 5                | 4.79                         | -1.20 | 16.17           | 5                | -4.12                      | 0.18  | 10.39         |
| 6                                    | 6.15                    | 0.37 | 21.48      | 6                | 6.38                         | -1.41 | 19.79           | 6                | -3.10                      | 0.07  | 11.02         |
| 7                                    | 7.94                    | 0.23 | 21.10      | 7                | 7.61                         | -1.90 | 20.14           | 7                | -2.17                      | -0.25 | 12.15         |
| 8                                    | 8.56                    | 0.22 | 16.57      | 8                | 8.58                         | -1.89 | 17.79           | 8                | -1.05                      | -0.38 | 11.91         |
| 9                                    | 8.17                    | 0.41 | 18.14      | 9                | 8.68                         | -2.09 | 18.56           | 9                | 0.26                       | -0.72 | 13.33         |
| 10                                   | 9.01                    | 0.76 | 21.32      | 10               | 9.78                         | -1.69 | 20.71           | 10               | 1.41                       | -0.29 | 15.71         |

## A.9 Table A3: Cross-Section Multiple Regression of Future Cash Dividends

Table A3: Cross-section multiple regression of future profitability (cash dividends) at the portfolio level (1975 - 2022).

This table reports the time-series averages of the slope coefficients, t-statistics, and adjusted r-squared for the cross-section regressions of future operating cash flows (divided by current book value of equity) at the portfolio level. Cash dividends is Compustat data item DVC. The explanatory variables include Fama-French operating profits divided by book equity ( $Z_{t+\tau}^{FF}$ ), R&D costs divided by book equity ( $RD_t$ ), SG&A costs divided by book equity ( $SGA_t$ ), change of total assets divided by one-year lagged total assets ( $AG_t$ ), and book-to-market ( $BM_t$ ). We separate R&D costs and SG&A costs following Peters and Taylor (2017). Firms are assigned into 5-by-10 industry and ROE(FF) groups in year t. Industry groups are based on Fama-French five industry classification and ROE(FF) cut points are NYSE ROE(FF) decile within each industry group. The portfolio-level earnings, R&D costs, SG&A costs, total assets, and book equity are the value-weighted for a given portfolio in year t. The portfolio-level future cash flows is computed following equation (4). We remove firms with total assets less than \$12.5 million or book equity less than \$25 million in year t.

| Panel A: Average slopes |            |      |            |            |             |        |        |
|-------------------------|------------|------|------------|------------|-------------|--------|--------|
|                         | Adj. $R^2$ | Intc | $Z_t^{FF}$ | $RD_t/B_t$ | $SGA_t/B_t$ | $AG_t$ | $BM_t$ |
| $Z_{t+1}^{DVC}$         | 0.62       | 0.00 | 0.15       | 0.11       | 0.02        | -0.06  | 0.01   |
| $Z_{t+2}^{DVC}$         | 0.63       | 0.01 | 0.16       | 0.11       | 0.02        | -0.06  | 0.01   |
| $Z_{t+3}^{DVC}$         | 0.61       | 0.01 | 0.17       | 0.11       | 0.02        | -0.06  | 0.01   |
| $Z_{t+4}^{DVC}$         | 0.60       | 0.01 | 0.18       | 0.12       | 0.02        | -0.06  | 0.00   |
| $Z_{t+5}^{DVC}$         | 0.59       | 0.01 | 0.19       | 0.13       | 0.03        | -0.07  | 0.00   |
| $Z_{t+6}^{DVC}$         | 0.58       | 0.02 | 0.19       | 0.14       | 0.04        | -0.07  | -0.01  |
| $Z_{t+7}^{DVC}$         | 0.56       | 0.03 | 0.20       | 0.16       | 0.04        | -0.08  | -0.02  |
| $Z_{t+8}^{DVC}$         | 0.54       | 0.03 | 0.21       | 0.18       | 0.04        | -0.08  | -0.03  |
| $Z_{t+9}^{DVC}$         | 0.53       | 0.04 | 0.21       | 0.17       | 0.05        | -0.06  | -0.05  |
| $Z_{t+10}^{DVC}$        | 0.51       | 0.05 | 0.22       | 0.21       | 0.06        | -0.06  | -0.05  |
| Panel B: t-statistics   |            |      |            |            |             |        |        |
|                         |            | Intc | $Z_t^{FF}$ | $RD_t/B_t$ | $SGA_t/B_t$ | $AG_t$ | $BM_t$ |
| 1                       |            | 0.17 | 15.32      | 3.68       | 1.51        | -4.74  | 1.34   |
| 2                       |            | 0.35 | 13.61      | 2.94       | 1.22        | -4.95  | 0.80   |
| 3                       |            | 0.32 | 12.48      | 2.48       | 1.69        | -4.57  | 0.82   |
| 4                       |            | 0.53 | 10.94      | 2.39       | 2.31        | -3.74  | 0.24   |
| 5                       |            | 0.53 | 10.13      | 2.36       | 2.54        | -3.48  | -0.04  |
| 6                       |            | 0.86 | 10.92      | 2.61       | 2.62        | -3.63  | -0.56  |
| 7                       |            | 0.92 | 8.92       | 2.49       | 2.44        | -3.20  | -0.96  |
| 8                       |            | 1.03 | 7.82       | 2.48       | 2.78        | -3.01  | -1.11  |
| 9                       |            | 1.47 | 7.50       | 1.51       | 2.75        | -2.76  | -2.19  |
| 10                      |            | 1.31 | 7.07       | 2.48       | 3.63        | -2.00  | -1.87  |

## A.10 Table A4: Pricing Tests on Profitability Factors

Table A4: Pricing tests on annual- and monthly-rebalanced profitability factors (July 1976 - June 2023).

To construct annual-rebalanced profitability factors,  $r_{ROE}^X$ , in the June end of each year  $t$ , firms are assigned into two size groups based on the median June end market equity of NYSE firms and assigned independently into three profitability groups based on the 30 and 70 percentiles profitability of all NYSE firms reported annual earnings in calendar year  $t-1$ . To construct monthly-rebalanced profitability factors,  $r_{ROEQ}^X$ , at the end of each month, firms are assigned into two size (ME) groups based on median June end market equity of NYSE firms, and assigned independently into three profitability groups based on the 30 and 70 percentiles (most recent available) profitability of all NYSE firms. The profitability factor return in any given month is the average return of the portfolios that are long the high profitability portfolios and short the low profitability portfolios within each size group. Firms' most recent available profitability is updated at the month end of the most recent quarterly earnings announcement date (quarterly Compustat data item RDQ, exclude if missing). To remove stale earnings, we exclude a firm if its most recent available quarter-end is 6 months away. The quarterly R&D expenses (quarterly Compustat data item XRDQ) are only recorded after 1989 and some firms may report only once a year, we simply set missing XRDQ to zero with no further interpolation. The considered profitability measures include  $Y(IB)/B$ ,  $Y(SPI)/B$ ,  $Y(EBT)/B$ ,  $Y(FF)/B$ ,  $Y(RD)/B$ ,  $Y(RD + .3SG)/B$ ,  $Y(RD + SG)/B$ ,  $Y(NM)/B$ , and  $Y(REVT)/B$ . Long- and short-side of the factors is the equal-weighted average excess return of the high- and low-profitability portfolios in each month. We run time-series regressions of the constructed profitability factors (Table A-1 for annual-rebalanced long-minus-short, A-2 for monthly-rebalanced long-minus-short, B-1 for annual-rebalancing long side only, B-2 for monthly-rebalancing long side only, C-1 for annual-rebalancing short side only, and C-2 for monthly-rebalancing short side only) on the existing factor models and report their intercept and associated t-statistics in italic. The factor models include Fama-French five factor model with RMW replaced by the cash-based operating profitability factor (FF5(C)), FF5 plus momentum factor (FF6), FF6 with the value factor, HML, replaced by the monthly-rebalanced value factor with most recent market equity, HML(Devil), from Asness and Frazzini (2013), FF6 with the value factor, HML, replaced by the intangible-adjusted value factor, HML(EKP), from Eisfeldt et al. (2022), Hou-Xue-Zhang four factor model (HXZ4), and Stambaugh-Yuan four factor model (SY4). The data of the factor models is taken from the authors' websites. SY4 is available until December 2016, HML(EKP) are available until March 2022, HXZ4 data are available until December 2022, FF6, and HML(Devil) are available until June 2023. Cash-based operating profitability factor is self constructed following the method in Fama and French (2018). GLPZ5 is the proposed factor models by Gulen et al. (2025) to replace HXZ4, data are from the author's website and the sample is from July 1977 to June 2022. Results for our three intangibles-adjusted factors are in bold.

| Moving from bottom-line to top-line profitability: left to right |                       |                      |                     |                       |                            |                            |                            |                     |                       |
|------------------------------------------------------------------|-----------------------|----------------------|---------------------|-----------------------|----------------------------|----------------------------|----------------------------|---------------------|-----------------------|
| Panel A-1: Annual-rebalanced factors                             |                       |                      |                     |                       |                            |                            |                            |                     |                       |
|                                                                  | $r_{ROE}^{IB}$        | $r_{ROE}^{SPI}$      | $r_{ROE}^{EBT}$     | $r_{ROE}^{FF}$        | $r_{ROE}^{RD}$             | $r_{ROE}^{RD+.3SG}$        | $r_{ROE}^{RD+SG}$          | $r_{ROE}^{NM}$      | $r_{ROE}^{REVT}$      |
| $\alpha_{FF5(C)}$                                                | -0.03<br><i>-0.45</i> | 0.00<br><i>-0.02</i> | 0.02<br><i>0.34</i> | -0.06<br><i>-1.25</i> | <b>0.18</b><br><b>3.74</b> | <b>0.21</b><br><b>3.68</b> | <b>0.15</b><br><b>2.12</b> | 0.11<br><i>1.58</i> | -0.13<br><i>-1.73</i> |
| $\alpha_{FF6}$                                                   | 0.01<br><i>0.19</i>   | 0.03<br><i>0.79</i>  | 0.05<br><i>1.20</i> | 0.01<br><i>0.42</i>   | <b>0.22</b><br><b>5.77</b> | <b>0.25</b><br><b>5.43</b> | <b>0.19</b><br><b>3.17</b> | 0.14<br><i>2.42</i> | -0.09<br><i>-1.35</i> |
| $\alpha_{FF6/Devil}$                                             | 0.01<br><i>0.25</i>   | 0.04<br><i>1.04</i>  | 0.04<br><i>0.91</i> | -0.01<br><i>-0.29</i> | <b>0.28</b><br><b>7.05</b> | <b>0.34</b><br><b>7.09</b> | <b>0.28</b><br><b>4.42</b> | 0.22<br><i>3.58</i> | -0.08<br><i>-1.28</i> |
| $\alpha_{FF6/EKP}$                                               | 0.03<br><i>0.56</i>   | 0.05<br><i>1.05</i>  | 0.03<br><i>0.65</i> | 0.01<br><i>0.45</i>   | <b>0.27</b><br><b>5.99</b> | <b>0.27</b><br><b>4.92</b> | <b>0.20</b><br><b>2.82</b> | 0.14<br><i>2.10</i> | -0.16<br><i>-2.55</i> |
| $\alpha_{HXZ4}$                                                  | -0.02<br><i>-0.27</i> | 0.03<br><i>0.38</i>  | 0.03<br><i>0.43</i> | 0.07<br><i>0.90</i>   | <b>0.31</b><br><b>4.84</b> | <b>0.31</b><br><b>4.50</b> | <b>0.23</b><br><b>2.89</b> | 0.17<br><i>2.22</i> | -0.09<br><i>-1.15</i> |
| $\alpha_{SY4}$                                                   | 0.12<br><i>1.26</i>   | 0.14<br><i>1.54</i>  | 0.15<br><i>1.42</i> | 0.14<br><i>1.40</i>   | <b>0.26</b><br><b>3.93</b> | <b>0.24</b><br><b>3.26</b> | <b>0.15</b><br><b>1.66</b> | 0.10<br><i>1.15</i> | -0.10<br><i>-0.97</i> |
| $\alpha_{GLPZ5}$                                                 | 0.06<br><i>0.55</i>   | 0.12<br><i>1.15</i>  | 0.10<br><i>0.82</i> | 0.17<br><i>1.49</i>   | <b>0.44</b><br><b>4.60</b> | <b>0.48</b><br><b>4.67</b> | <b>0.40</b><br><b>3.49</b> | 0.35<br><i>3.11</i> | 0.09<br><i>0.76</i>   |
| Panel A-2: Monthly-rebalanced factors                            |                       |                      |                     |                       |                            |                            |                            |                     |                       |

|                      | $r_{\text{ROEQ}}^{\text{IB}}$ | $r_{\text{ROEQ}}^{\text{SPI}}$ | $r_{\text{ROEQ}}^{\text{EBT}}$ | $r_{\text{ROEQ}}^{\text{FF}}$ | $r_{\text{ROEQ}}^{\text{RD}}$ | $r_{\text{ROEQ}}^{\text{RD}+.3\text{SG}}$ | $r_{\text{ROEQ}}^{\text{RD}+\text{SG}}$ | $r_{\text{ROEQ}}^{\text{NM}}$ | $r_{\text{ROEQ}}^{\text{REVT}}$ |
|----------------------|-------------------------------|--------------------------------|--------------------------------|-------------------------------|-------------------------------|-------------------------------------------|-----------------------------------------|-------------------------------|---------------------------------|
| $\alpha_{FF5(C)}$    | 0.30<br><i>3.21</i>           | 0.31<br><i>3.51</i>            | 0.28<br><i>3.00</i>            | 0.22<br><i>2.82</i>           | <b>0.49</b><br><b>6.88</b>    | <b>0.42</b><br><b>5.97</b>                | <b>0.27</b><br><b>3.57</b>              | 0.25<br><i>3.33</i>           | -0.06<br><i>-0.69</i>           |
| $\alpha_{FF6}$       | 0.21<br><i>3.20</i>           | 0.25<br><i>3.76</i>            | 0.20<br><i>3.09</i>            | 0.19<br><i>3.45</i>           | <b>0.45</b><br><b>7.95</b>    | <b>0.38</b><br><b>6.88</b>                | <b>0.26</b><br><b>3.99</b>              | 0.23<br><i>3.58</i>           | -0.05<br><i>-0.66</i>           |
| $\alpha_{FF6/Devil}$ | 0.26<br><i>4.09</i>           | 0.29<br><i>4.56</i>            | 0.24<br><i>3.70</i>            | 0.21<br><i>3.78</i>           | <b>0.51</b><br><b>9.34</b>    | <b>0.46</b><br><b>8.72</b>                | <b>0.34</b><br><b>5.30</b>              | 0.30<br><i>4.67</i>           | -0.07<br><i>-0.92</i>           |
| $\alpha_{FF6/EKP}$   | 0.26<br><i>3.83</i>           | 0.30<br><i>4.52</i>            | 0.24<br><i>3.68</i>            | 0.23<br><i>4.05</i>           | <b>0.49</b><br><b>8.36</b>    | <b>0.41</b><br><b>6.88</b>                | <b>0.28</b><br><b>4.02</b>              | 0.24<br><i>3.47</i>           | -0.13<br><i>-1.82</i>           |
| $\alpha_{HXZ4}$      | 0.05<br><i>1.34</i>           | 0.11<br><i>2.39</i>            | 0.06<br><i>1.26</i>            | 0.12<br><i>2.06</i>           | <b>0.37</b><br><b>7.28</b>    | <b>0.33</b><br><b>5.94</b>                | <b>0.23</b><br><b>3.25</b>              | 0.20<br><i>2.83</i>           | -0.10<br><i>-1.21</i>           |
| $\alpha_{SY4}$       | 0.31<br><i>3.10</i>           | 0.37<br><i>3.79</i>            | 0.33<br><i>3.11</i>            | 0.36<br><i>3.65</i>           | <b>0.50</b><br><b>7.12</b>    | <b>0.40</b><br><b>5.36</b>                | <b>0.23</b><br><b>2.66</b>              | 0.21<br><i>2.37</i>           | -0.04<br><i>-0.34</i>           |
| $\alpha_{GLPZ5}$     | 0.11<br><i>1.67</i>           | 0.17<br><i>2.22</i>            | 0.07<br><i>0.95</i>            | 0.14<br><i>1.69</i>           | <b>0.42</b><br><b>5.46</b>    | <b>0.39</b><br><b>4.81</b>                | <b>0.38</b><br><b>3.71</b>              | 0.36<br><i>3.59</i>           | 0.11<br><i>0.92</i>             |

Panel B-1: Long side of the annual-rebalanced factors

|                      | $r_{\text{ROE}}^{\text{IB}}$ | $r_{\text{ROE}}^{\text{SPI}}$ | $r_{\text{ROE}}^{\text{EBT}}$ | $r_{\text{ROE}}^{\text{FF}}$ | $r_{\text{ROE}}^{\text{RD}}$ | $r_{\text{ROE}}^{\text{RD}+.3\text{SG}}$ | $r_{\text{ROE}}^{\text{RD}+\text{SG}}$ | $r_{\text{ROE}}^{\text{NM}}$ | $r_{\text{ROE}}^{\text{REVT}}$ |
|----------------------|------------------------------|-------------------------------|-------------------------------|------------------------------|------------------------------|------------------------------------------|----------------------------------------|------------------------------|--------------------------------|
| $\alpha_{FF5(C)}$    | -0.02<br><i>-0.58</i>        | -0.02<br><i>-0.87</i>         | -0.01<br><i>-0.45</i>         | -0.04<br><i>-1.63</i>        | <b>0.08</b><br><b>3.19</b>   | <b>0.08</b><br><b>2.71</b>               | <b>0.05</b><br><b>1.36</b>             | 0.02<br><i>0.61</i>          | -0.08<br><i>-1.68</i>          |
| $\alpha_{FF6}$       | 0.01<br><i>0.36</i>          | 0.01<br><i>0.28</i>           | 0.01<br><i>0.59</i>           | -0.01<br><i>-0.43</i>        | <b>0.09</b><br><b>4.08</b>   | <b>0.09</b><br><b>3.52</b>               | <b>0.06</b><br><b>1.78</b>             | 0.03<br><i>0.90</i>          | -0.05<br><i>-1.18</i>          |
| $\alpha_{FF6/Devil}$ | 0.01<br><i>0.38</i>          | 0.01<br><i>0.41</i>           | 0.00<br><i>-0.23</i>          | -0.03<br><i>-1.68</i>        | <b>0.11</b><br><b>4.66</b>   | <b>0.12</b><br><b>4.53</b>               | <b>0.09</b><br><b>2.42</b>             | 0.05<br><i>1.44</i>          | -0.07<br><i>-1.65</i>          |
| $\alpha_{FF6/EKP}$   | 0.00<br><i>0.13</i>          | 0.00<br><i>0.21</i>           | -0.01<br><i>-0.35</i>         | -0.03<br><i>-1.31</i>        | <b>0.10</b><br><b>3.99</b>   | <b>0.09</b><br><b>3.26</b>               | <b>0.06</b><br><b>1.49</b>             | 0.02<br><i>0.60</i>          | -0.11<br><i>-2.74</i>          |
| $\alpha_{HXZ4}$      | 0.02<br><i>0.54</i>          | 0.02<br><i>0.64</i>           | 0.02<br><i>0.55</i>           | 0.03<br><i>0.63</i>          | <b>0.14</b><br><b>4.51</b>   | <b>0.13</b><br><b>3.78</b>               | <b>0.09</b><br><b>2.15</b>             | 0.06<br><i>1.40</i>          | -0.04<br><i>-0.75</i>          |
| $\alpha_{SY4}$       | 0.01<br><i>0.33</i>          | 0.02<br><i>0.48</i>           | 0.02<br><i>0.51</i>           | 0.02<br><i>0.46</i>          | <b>0.07</b><br><b>2.29</b>   | <b>0.05</b><br><b>1.36</b>               | <b>0.01</b><br><b>0.21</b>             | -0.02<br><i>-0.32</i>        | -0.09<br><i>-1.50</i>          |
| $\alpha_{GLPZ5}$     | 0.07<br><i>1.32</i>          | 0.06<br><i>1.15</i>           | 0.04<br><i>0.74</i>           | 0.04<br><i>0.67</i>          | <b>0.16</b><br><b>3.27</b>   | <b>0.19</b><br><b>3.72</b>               | <b>0.16</b><br><b>2.56</b>             | 0.14<br><i>2.23</i>          | 0.04<br><i>0.57</i>            |

Panel B-2: Long side of the monthly-rebalanced factors

|                      | $r_{\text{ROEQ}}^{\text{IB}}$ | $r_{\text{ROEQ}}^{\text{SPI}}$ | $r_{\text{ROEQ}}^{\text{EBT}}$ | $r_{\text{ROEQ}}^{\text{FF}}$ | $r_{\text{ROEQ}}^{\text{RD}}$ | $r_{\text{ROEQ}}^{\text{RD}+.3\text{SG}}$ | $r_{\text{ROEQ}}^{\text{RD}+\text{SG}}$ | $r_{\text{ROEQ}}^{\text{NM}}$ | $r_{\text{ROEQ}}^{\text{REVT}}$ |
|----------------------|-------------------------------|--------------------------------|--------------------------------|-------------------------------|-------------------------------|-------------------------------------------|-----------------------------------------|-------------------------------|---------------------------------|
| $\alpha_{FF5(C)}$    | 0.13<br><i>3.63</i>           | 0.14<br><i>4.11</i>            | 0.12<br><i>3.32</i>            | 0.09<br><i>2.49</i>           | <b>0.20</b><br><b>6.33</b>    | <b>0.21</b><br><b>6.18</b>                | <b>0.14</b><br><b>3.41</b>              | 0.14<br><i>3.31</i>           | -0.02<br><i>-0.47</i>           |
| $\alpha_{FF6}$       | 0.10<br><i>3.60</i>           | 0.12<br><i>4.27</i>            | 0.11<br><i>3.51</i>            | 0.09<br><i>2.90</i>           | <b>0.19</b><br><b>6.50</b>    | <b>0.20</b><br><b>6.30</b>                | <b>0.13</b><br><b>3.42</b>              | 0.13<br><i>3.24</i>           | 0.00<br><i>0.04</i>             |
| $\alpha_{FF6/Devil}$ | 0.12<br><i>4.27</i>           | 0.14<br><i>4.75</i>            | 0.13<br><i>4.12</i>            | 0.10<br><i>2.97</i>           | <b>0.22</b><br><b>7.18</b>    | <b>0.23</b><br><b>7.14</b>                | <b>0.16</b><br><b>4.04</b>              | 0.15<br><i>3.75</i>           | -0.01<br><i>-0.28</i>           |
| $\alpha_{FF6/EKP}$   | 0.11<br><i>3.73</i>           | 0.13<br><i>4.34</i>            | 0.13<br><i>3.98</i>            | 0.10<br><i>3.14</i>           | <b>0.21</b><br><b>6.62</b>    | <b>0.21</b><br><b>6.20</b>                | <b>0.14</b><br><b>3.38</b>              | 0.13<br><i>3.09</i>           | -0.05<br><i>-1.11</i>           |
| $\alpha_{HXZ4}$      | 0.07<br><i>2.53</i>           | 0.09<br><i>3.32</i>            | 0.09<br><i>2.65</i>            | 0.09<br><i>2.33</i>           | <b>0.19</b><br><b>5.92</b>    | <b>0.20</b><br><b>5.82</b>                | <b>0.15</b><br><b>3.48</b>              | 0.14<br><i>3.26</i>           | 0.00<br><i>0.03</i>             |
| $\alpha_{SY4}$       | 0.11<br><i>2.81</i>           | 0.13<br><i>3.39</i>            | 0.12<br><i>2.99</i>            | 0.13<br><i>2.83</i>           | <b>0.18</b><br><b>4.99</b>    | <b>0.16</b><br><b>4.35</b>                | <b>0.08</b><br><b>1.60</b>              | 0.07<br><i>1.56</i>           | -0.03<br><i>-0.49</i>           |
| $\alpha_{GLPZ5}$     | 0.06<br><i>1.49</i>           | 0.10<br><i>2.16</i>            | 0.08<br><i>1.59</i>            | 0.08<br><i>1.45</i>           | <b>0.18</b><br><b>3.95</b>    | <b>0.21</b><br><b>4.28</b>                | <b>0.20</b><br><b>3.31</b>              | 0.20<br><i>3.31</i>           | 0.09<br><i>1.20</i>             |

Panel C-1: Short side of the annual-rebalanced factors

|                   | $r_{\text{ROE}}^{\text{IB}}$ | $r_{\text{ROE}}^{\text{SPI}}$ | $r_{\text{ROE}}^{\text{EBT}}$ | $r_{\text{ROE}}^{\text{FF}}$ | $r_{\text{ROE}}^{\text{RD}}$ | $r_{\text{ROE}}^{\text{RD}+.3\text{SG}}$ | $r_{\text{ROE}}^{\text{RD}+\text{SG}}$ | $r_{\text{ROE}}^{\text{NM}}$ | $r_{\text{ROE}}^{\text{REVT}}$ |
|-------------------|------------------------------|-------------------------------|-------------------------------|------------------------------|------------------------------|------------------------------------------|----------------------------------------|------------------------------|--------------------------------|
| $\alpha_{FF5(C)}$ | 0.01<br><i>0.26</i>          | -0.02<br><i>-0.46</i>         | -0.04<br><i>-0.73</i>         | 0.02<br><i>0.47</i>          | <b>-0.10</b><br><b>-2.89</b> | <b>-0.13</b><br><b>-3.58</b>             | <b>-0.10</b><br><b>-2.34</b>           | -0.09<br><i>-2.18</i>        | 0.06<br><i>1.47</i>            |
| $\alpha_{FF6}$    | 0.00<br><i>-0.02</i>         | -0.03<br><i>-0.75</i>         | -0.04<br><i>-1.13</i>         | -0.02<br><i>-0.79</i>        | <b>-0.13</b><br><b>-4.47</b> | <b>-0.16</b><br><b>-5.20</b>             | <b>-0.13</b><br><b>-3.52</b>           | -0.11<br><i>-3.20</i>        | 0.04<br><i>1.22</i>            |



|                      |                |                |                |                |                              |                              |                              |                |                |
|----------------------|----------------|----------------|----------------|----------------|------------------------------|------------------------------|------------------------------|----------------|----------------|
| $\alpha_{FF6/Devil}$ | 0.00<br>-0.09  | -0.04<br>-0.96 | -0.05<br>-1.23 | -0.03<br>-1.29 | <b>-0.17</b><br><b>-5.87</b> | <b>-0.22</b><br><b>-6.95</b> | <b>-0.19</b><br><b>-5.01</b> | -0.17<br>-4.63 | 0.02<br>0.49   |
| $\alpha_{FF6/EKP}$   | -0.02<br>-0.58 | -0.04<br>-1.09 | -0.04<br>-0.98 | -0.04<br>-1.76 | <b>-0.17</b><br><b>-5.25</b> | <b>-0.18</b><br><b>-5.02</b> | <b>-0.14</b><br><b>-3.26</b> | -0.12<br>-2.92 | 0.05<br>1.67   |
| $\alpha_{HXZ4}$      | 0.04<br>0.77   | 0.00<br>-0.08  | -0.01<br>-0.24 | -0.04<br>-0.93 | <b>-0.16</b><br><b>-3.78</b> | <b>-0.18</b><br><b>-4.10</b> | <b>-0.14</b><br><b>-2.92</b> | -0.11<br>-2.51 | 0.06<br>1.44   |
| $\alpha_{SY4}$       | -0.11<br>-1.53 | -0.12<br>-1.81 | -0.12<br>-1.72 | -0.11<br>-1.81 | <b>-0.18</b><br><b>-3.53</b> | <b>-0.19</b><br><b>-3.65</b> | <b>-0.14</b><br><b>-2.34</b> | -0.12<br>-2.09 | 0.00<br>0.10   |
| $\alpha_{GLPZ5}$     | 0.01<br>0.13   | -0.06<br>-0.82 | -0.05<br>-0.67 | -0.13<br>-1.88 | <b>-0.29</b><br><b>-4.23</b> | <b>-0.29</b><br><b>-4.39</b> | <b>-0.24</b><br><b>-3.51</b> | -0.22<br>-3.24 | -0.05<br>-0.83 |

Panel C-2: Short side of the monthly-rebalanced factors

|                      | $r_{ROEQ}^{IB}$ | $r_{ROEQ}^{SPI}$ | $r_{ROEQ}^{EBT}$ | $r_{ROEQ}^{FF}$ | $r_{ROEQ}^{RD}$              | $r_{ROEQ}^{RD+.3SG}$         | $r_{ROEQ}^{RD+SG}$           | $r_{ROEQ}^{NM}$ | $r_{ROEQ}^{REVT}$ |
|----------------------|-----------------|------------------|------------------|-----------------|------------------------------|------------------------------|------------------------------|-----------------|-------------------|
| $\alpha_{FF5(C)}$    | -0.17<br>-2.58  | -0.17<br>-2.66   | -0.15<br>-2.30   | -0.13<br>-2.38  | <b>-0.29</b><br><b>-5.60</b> | <b>-0.21</b><br><b>-4.37</b> | <b>-0.12</b><br><b>-2.70</b> | -0.11<br>-2.45  | 0.03<br>0.73      |
| $\alpha_{FF6}$       | -0.11<br>-2.26  | -0.12<br>-2.55   | -0.10<br>-1.92   | -0.10<br>-2.58  | <b>-0.25</b><br><b>-6.36</b> | <b>-0.18</b><br><b>-4.92</b> | <b>-0.12</b><br><b>-3.11</b> | -0.10<br>-2.64  | 0.05<br>1.29      |
| $\alpha_{FF6/Devil}$ | -0.14<br>-3.02  | -0.16<br>-3.29   | -0.11<br>-2.27   | -0.12<br>-2.99  | <b>-0.29</b><br><b>-7.67</b> | <b>-0.24</b><br><b>-6.75</b> | <b>-0.18</b><br><b>-4.66</b> | -0.15<br>-3.95  | 0.05<br>1.38      |
| $\alpha_{FF6/EKP}$   | -0.15<br>-3.03  | -0.17<br>-3.52   | -0.12<br>-2.36   | -0.13<br>-3.19  | <b>-0.28</b><br><b>-6.88</b> | <b>-0.21</b><br><b>-5.19</b> | <b>-0.14</b><br><b>-3.34</b> | -0.11<br>-2.71  | 0.08<br>2.02      |
| $\alpha_{HXZ4}$      | 0.01<br>0.45    | -0.02<br>-0.46   | 0.02<br>0.58     | -0.03<br>-0.82  | <b>-0.18</b><br><b>-5.06</b> | <b>-0.13</b><br><b>-3.57</b> | <b>-0.08</b><br><b>-2.00</b> | -0.06<br>-1.52  | 0.11<br>2.38      |
| $\alpha_{SY4}$       | -0.20<br>-2.62  | -0.24<br>-3.16   | -0.20<br>-2.49   | -0.23<br>-3.19  | <b>-0.32</b><br><b>-5.31</b> | <b>-0.23</b><br><b>-3.91</b> | <b>-0.16</b><br><b>-2.53</b> | -0.14<br>-2.18  | 0.01<br>0.11      |
| $\alpha_{GLPZ5}$     | -0.05<br>-0.98  | -0.07<br>-1.32   | 0.01<br>0.09     | -0.06<br>-1.06  | <b>-0.24</b><br><b>-4.28</b> | <b>-0.18</b><br><b>-3.27</b> | <b>-0.18</b><br><b>-2.95</b> | -0.17<br>-2.75  | -0.01<br>-0.23    |

## A.11 Table A5: Pricing Tests on Profitability Factors (Varying Fraction of SGA)

Table A5: Pricing tests on annual-rebalanced profitability factors varying the fraction of SGA (July 1976 - June 2023).

To construct annual-rebalanced profitability factors, in the June end of each year  $t$ , firms are assigned into two size groups based on median June end market equity of NYSE firms, and assigned independently into three profitability groups based on the 30 and 70 percentiles profitability of all NYSE firms reported annual earnings in calendar year  $t-1$ . The profitability factor return in any given month is the average return of the portfolios that are long the high profitability portfolios and short the low profitability portfolios within each size group. We consider the profitability measures of  $Y(\gamma)/B$  and  $Y(\gamma) = (\text{REVT} - \text{COGS} - \text{XSGA} - \text{XINT} + \frac{\gamma}{10}\text{SGA} + \text{XRD})$  where  $\gamma = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$ . We run time-series regressions of the constructed profitability factors on the existing factor models and report their intercept and associated  $t$ -statistics in italic. The factor models include Fama-French five factor model (FF5), FF5 with RMW replaced by the cash-based operating profitability factor (FF5(C)), FF5 plus momentum factor (FF6), FF6 with the value factor, HML, replaced by the monthly-rebalanced value factor with most recent market equity, HML(Devil), from Asness and Frazzini (2013), FF6 with the value factor, HML, replaced by the intangible-adjusted value factor, HML(EKP), from Eisfeldt et al. (2022), Hou-Xue-Zhang four factor model (HXZ4) and Hou-Mo-Xue-Zhang five factor model (HMXZ5), Daniel-Hirshleifer-Sun behavioral three factor model (DHS3), and Stambaugh-Yuan four factor model (SY4). The data of the factor models is taken from the authors' websites. Except for SY4 which is available only until December 2016, HML(EKP) are available until March 2022, HXZ4, HMXZ5, and DHS3 data are available until December 2022, FF5, FF6, and HML(Devil) are available until June 2023. Cash-based operating profitability factor is self constructed following the method in Fama and French (2018). GLPZ7 and GLPZ5 are the proposed factor models by Gulen et al. (2025) to replace FF5 and HXZ4 respectively, data is from the author's website and the sample is from July 1977 to June 2022.

|                                    | 0%                               | 10%                 | 20%                 | 30%                                          | 40%                 | 50%                 | 60%                 | 70%                 | 80%                 | 90%                 | 100%                                               |
|------------------------------------|----------------------------------|---------------------|---------------------|----------------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|----------------------------------------------------|
|                                    | ( $r_{\text{ROE}}^{\text{RD}}$ ) |                     |                     | ( $r_{\text{ROE}}^{\text{RD}+.3\text{SG}}$ ) |                     |                     |                     |                     |                     |                     | ( $\approx r_{\text{ROE}}^{\text{RD}+\text{SG}}$ ) |
| Mean                               | 0.46<br><i>6.08</i>              | 0.47<br><i>6.28</i> | 0.47<br><i>6.32</i> | 0.47<br><i>6.16</i>                          | 0.46<br><i>5.90</i> | 0.46<br><i>5.76</i> | 0.43<br><i>5.41</i> | 0.44<br><i>5.39</i> | 0.43<br><i>5.22</i> | 0.42<br><i>5.06</i> | 0.40<br><i>4.87</i>                                |
| $\alpha_{\text{FF5}}$              | 0.24<br><i>6.18</i>              | 0.24<br><i>6.09</i> | 0.25<br><i>5.96</i> | 0.24<br><i>5.35</i>                          | 0.22<br><i>4.67</i> | 0.22<br><i>4.36</i> | 0.20<br><i>3.74</i> | 0.20<br><i>3.67</i> | 0.20<br><i>3.44</i> | 0.19<br><i>3.16</i> | 0.18<br><i>2.93</i>                                |
| $\alpha_{\text{FF5}(C)}$           | 0.18<br><i>3.74</i>              | 0.19<br><i>3.79</i> | 0.21<br><i>3.92</i> | 0.21<br><i>3.68</i>                          | 0.19<br><i>3.21</i> | 0.20<br><i>3.12</i> | 0.18<br><i>2.72</i> | 0.19<br><i>2.78</i> | 0.18<br><i>2.65</i> | 0.17<br><i>2.50</i> | 0.17<br><i>2.35</i>                                |
| $\alpha_{\text{FF6}}$              | 0.22<br><i>5.77</i>              | 0.23<br><i>5.87</i> | 0.25<br><i>5.93</i> | 0.25<br><i>5.43</i>                          | 0.23<br><i>4.83</i> | 0.24<br><i>4.61</i> | 0.21<br><i>4.01</i> | 0.22<br><i>4.01</i> | 0.22<br><i>3.78</i> | 0.21<br><i>3.53</i> | 0.21<br><i>3.38</i>                                |
| $\alpha_{\text{FF6}/\text{Devil}}$ | 0.28<br><i>7.05</i>              | 0.30<br><i>7.42</i> | 0.33<br><i>7.54</i> | 0.34<br><i>7.09</i>                          | 0.32<br><i>6.43</i> | 0.33<br><i>6.18</i> | 0.30<br><i>5.55</i> | 0.31<br><i>5.43</i> | 0.31<br><i>5.16</i> | 0.30<br><i>4.92</i> | 0.30<br><i>4.69</i>                                |
| $\alpha_{\text{FF6}/\text{EKP}}$   | 0.27<br><i>5.99</i>              | 0.27<br><i>5.60</i> | 0.28<br><i>5.46</i> | 0.27<br><i>4.92</i>                          | 0.26<br><i>4.40</i> | 0.25<br><i>4.17</i> | 0.23<br><i>3.65</i> | 0.24<br><i>3.65</i> | 0.23<br><i>3.44</i> | 0.22<br><i>3.21</i> | 0.21<br><i>3.03</i>                                |
| $\alpha_{\text{HXZ4}}$             | 0.31<br><i>4.84</i>              | 0.31<br><i>4.77</i> | 0.32<br><i>4.83</i> | 0.31<br><i>4.50</i>                          | 0.29<br><i>4.04</i> | 0.28<br><i>3.89</i> | 0.26<br><i>3.48</i> | 0.26<br><i>3.50</i> | 0.26<br><i>3.37</i> | 0.25<br><i>3.17</i> | 0.24<br><i>3.09</i>                                |
| $\alpha_{\text{HMXZ5}}$            | 0.14<br><i>2.24</i>              | 0.15<br><i>2.28</i> | 0.17<br><i>2.48</i> | 0.16<br><i>2.22</i>                          | 0.15<br><i>2.01</i> | 0.16<br><i>2.04</i> | 0.13<br><i>1.73</i> | 0.15<br><i>1.83</i> | 0.15<br><i>1.81</i> | 0.13<br><i>1.64</i> | 0.13<br><i>1.61</i>                                |
| $\alpha_{\text{DHS3}}$             | 0.28<br><i>3.63</i>              | 0.30<br><i>3.76</i> | 0.31<br><i>3.84</i> | 0.30<br><i>3.69</i>                          | 0.29<br><i>3.39</i> | 0.28<br><i>3.31</i> | 0.26<br><i>3.05</i> | 0.27<br><i>3.11</i> | 0.26<br><i>2.96</i> | 0.25<br><i>2.85</i> | 0.25<br><i>2.77</i>                                |
| $\alpha_{\text{SY4}}$              | 0.26<br><i>3.93</i>              | 0.26<br><i>3.79</i> | 0.26<br><i>3.75</i> | 0.24<br><i>3.26</i>                          | 0.21<br><i>2.74</i> | 0.21<br><i>2.63</i> | 0.18<br><i>2.28</i> | 0.19<br><i>2.23</i> | 0.18<br><i>2.07</i> | 0.16<br><i>1.88</i> | 0.16<br><i>1.79</i>                                |
| $\alpha_{\text{GLPZ7}}$            | 0.14<br><i>3.21</i>              | 0.11<br><i>2.66</i> | 0.12<br><i>2.68</i> | 0.09<br><i>1.98</i>                          | 0.09<br><i>1.62</i> | 0.07<br><i>1.27</i> | 0.04<br><i>0.59</i> | 0.04<br><i>0.62</i> | 0.02<br><i>0.35</i> | 0.02<br><i>0.31</i> | 0.01<br><i>0.09</i>                                |
| $\alpha_{\text{GLPZ5}}$            | 0.44<br><i>4.60</i>              | 0.45<br><i>4.62</i> | 0.48<br><i>4.83</i> | 0.48<br><i>4.67</i>                          | 0.48<br><i>4.51</i> | 0.47<br><i>4.35</i> | 0.44<br><i>4.02</i> | 0.45<br><i>4.04</i> | 0.44<br><i>3.86</i> | 0.44<br><i>3.83</i> | 0.42<br><i>3.67</i>                                |

## A.12 Table A6: Model Comparison Tests

Table A6: Model comparison tests.

The R&D-adjusted profitability factors (annual-rebalanced  $r_{ROE}^{RD}$  and monthly-rebalanced  $r_{ROEQ}^{RD}$ ) are incorporated into the existing models in two ways. First, for models with annual- or monthly-rebalanced profitability related factors, the original profitability factor in the model is replaced with the annual- or monthly-rebalanced R&D-adjusted profitability factors respectively. Second, all models are augmented with the R&D-adjusted profitability factors. The squared Sharpe ratio test by Barillas et al. (2020) is used to compare the original models and the models with the R&D-adjusted profitability factors. The first row of each panel is the difference in the squared Sharpe ratios between the column model and the row model, and the second row report the p-value of the test with the null hypothesis that the difference in the squared Sharpe ratio is zero. The factor models include Fama-French five factor model (FF5), FF5 with RMW replaced by the cash-based operating profitability factor (FF5(C)), FF5 plus momentum factor (FF6), FF6 with the value factor, HML, replaced by the monthly-rebalanced value factor with most recent market equity, HML(Devil), from Asness and Frazzini (2013) (FF6<sup>1</sup>), FF6 with the value factor, HML, replaced by the intangible-adjusted value factor, HML(EKP), from Eisfeldt et al. (2022) (FF6<sup>2</sup>), Hou-Xue-Zhang four factor model (HXZ4), Hou-Mo-Xue-Zhang five factor model (HMXZ5), Daniel-Hirshleifer-Sun behavioral three factor model (DHS3), and Stambaugh-Yuan four factor model (SY4). The data of the factor models is taken from the authors' websites. Cash-based operating profitability factor is self constructed following the method in Fama and French (2018). SY4 is available until December 2016, HML(EKP) are available until March 2022, HXZ4, HMXZ5, and DHS3 data are available until December 2022, FF5, FF6, and HML(Devil) are available until June 2023. Cash-based operating profitability factor is self constructed following the method in Fama and French (2018).

| July 1976 - June 2023     |                                   |                                   | July 1976 - December 1999 |                                   |                                   | January 2000 - June 2023     |                                   |                                   |
|---------------------------|-----------------------------------|-----------------------------------|---------------------------|-----------------------------------|-----------------------------------|------------------------------|-----------------------------------|-----------------------------------|
|                           | FF5/ $r_{ROE}^{RD}$               | FF5+ $r_{ROE}^{RD}$               |                           | FF5/ $r_{ROE}^{RD}$               | FF5+ $r_{ROE}^{RD}$               |                              | FF5/ $r_{ROE}^{RD}$               | FF5+ $r_{ROE}^{RD}$               |
| FF5                       | 0.060                             | 0.073                             | FF5                       | 0.070                             | 0.069                             | FF5                          | 0.059                             | 0.062                             |
|                           | 0.000                             | 0.000                             |                           | 0.023                             | 0.000                             |                              | 0.010                             | 0.000                             |
|                           | FF5(C)/ $r_{ROE}^{RD}$            | FF5(C)+ $r_{ROE}^{RD}$            |                           | FF5(C)/ $r_{ROE}^{RD}$            | FF5(C)+ $r_{ROE}^{RD}$            |                              | FF5(C)/ $r_{ROE}^{RD}$            | FF5(C)+ $r_{ROE}^{RD}$            |
| FF5(C)                    | 0.023                             | 0.063                             | FF5(C)                    | 0.017                             | 0.042                             | FF5(C)                       | 0.058                             | 0.069                             |
|                           | 0.212                             | 0.000                             |                           | 0.473                             | 0.007                             |                              | 0.036                             | 0.000                             |
|                           | FF6/ $r_{ROE}^{RD}$               | FF6+ $r_{ROE}^{RD}$               |                           | FF6/ $r_{ROE}^{RD}$               | FF6+ $r_{ROE}^{RD}$               |                              | FF6/ $r_{ROE}^{RD}$               | FF6+ $r_{ROE}^{RD}$               |
| FF6                       | 0.053                             | 0.064                             | FF6                       | 0.060                             | 0.058                             | FF6                          | 0.058                             | 0.061                             |
|                           | 0.000                             | 0.000                             |                           | 0.039                             | 0.001                             |                              | 0.008                             | 0.000                             |
|                           | FF6 <sup>1</sup> / $r_{ROE}^{RD}$ | FF6 <sup>1</sup> + $r_{ROE}^{RD}$ |                           | FF6 <sup>1</sup> / $r_{ROE}^{RD}$ | FF6 <sup>1</sup> + $r_{ROE}^{RD}$ |                              | FF6 <sup>1</sup> / $r_{ROE}^{RD}$ | FF6 <sup>1</sup> + $r_{ROE}^{RD}$ |
| FF6 <sup>1</sup>          | 0.072                             | 0.098                             | FF6 <sup>1</sup>          | 0.078                             | 0.075                             | FF6 <sup>1</sup>             | 0.075                             | 0.092                             |
|                           | 0.000                             | 0.000                             |                           | 0.034                             | 0.000                             |                              | 0.001                             | 0.000                             |
| July 1976 - March 2022    |                                   |                                   | July 1976 - December 1999 |                                   |                                   | January 2000 - March 2022    |                                   |                                   |
|                           | FF6 <sup>2</sup> / $r_{ROE}^{RD}$ | FF6 <sup>2</sup> + $r_{ROE}^{RD}$ |                           | FF6 <sup>2</sup> / $r_{ROE}^{RD}$ | FF6 <sup>2</sup> + $r_{ROE}^{RD}$ |                              | FF6 <sup>2</sup> / $r_{ROE}^{RD}$ | FF6 <sup>2</sup> + $r_{ROE}^{RD}$ |
| FF6 <sup>2</sup>          | 0.059                             | 0.073                             | FF6 <sup>2</sup>          | 0.036                             | 0.032                             | FF6 <sup>2</sup>             | 0.086                             | 0.103                             |
|                           | 0.000                             | 0.000                             |                           | 0.112                             | 0.007                             |                              | 0.001                             | 0.000                             |
| July 1976 - December 2022 |                                   |                                   | July 1976 - December 1999 |                                   |                                   | January 2000 - December 2022 |                                   |                                   |
|                           | HXZ4/ $r_{ROEQ}^{RD}$             | HXZ4+ $r_{ROEQ}^{RD}$             |                           | HXZ4/ $r_{ROEQ}^{RD}$             | HXZ4+ $r_{ROEQ}^{RD}$             |                              | HXZ4/ $r_{ROEQ}^{RD}$             | HXZ4+ $r_{ROEQ}^{RD}$             |
| HXZ4                      | 0.091                             | 0.107                             | HXZ4                      | 0.114                             | 0.110                             | HXZ4                         | 0.106                             | 0.145                             |
|                           | 0.000                             | 0.000                             |                           | 0.015                             | 0.000                             |                              | 0.001                             | 0.000                             |
|                           | HMXZ5/ $r_{ROEQ}^{RD}$            | HMXZ5+ $r_{ROEQ}^{RD}$            |                           | HMXZ5/ $r_{ROEQ}^{RD}$            | HMXZ5+ $r_{ROEQ}^{RD}$            |                              | HMXZ5/ $r_{ROEQ}^{RD}$            | HMXZ5+ $r_{ROEQ}^{RD}$            |
| HMXZ5                     | 0.030                             | 0.054                             | HMXZ5                     | 0.040                             | 0.037                             | HMXZ5                        | 0.053                             | 0.094                             |
|                           | 0.044                             | 0.000                             |                           | 0.151                             | 0.012                             |                              | 0.053                             | 0.000                             |
|                           | DHS3+ $r_{ROE}^{RD}$              | DHS3+ $r_{ROEQ}^{RD}$             |                           | DHS3+ $r_{ROE}^{RD}$              | DHS3+ $r_{ROEQ}^{RD}$             |                              | DHS3+ $r_{ROE}^{RD}$              | DHS3+ $r_{ROEQ}^{RD}$             |
|                           |                                   |                                   |                           |                                   |                                   |                              |                                   |                                   |

|                           |                                         |                                         |                           |                                         |                                         |                              |                                         |                                         |
|---------------------------|-----------------------------------------|-----------------------------------------|---------------------------|-----------------------------------------|-----------------------------------------|------------------------------|-----------------------------------------|-----------------------------------------|
| DHS3                      | 0.026<br><i>0.001</i>                   | 0.057<br><i>0.000</i>                   | DHS3                      | 0.046<br><i>0.003</i>                   | 0.134<br><i>0.000</i>                   | DHS3                         | 0.048<br><i>0.000</i>                   | 0.068<br><i>0.000</i>                   |
| July 1976 - December 2016 |                                         |                                         | July 1976 - December 1999 |                                         |                                         | January 2000 - December 2016 |                                         |                                         |
|                           | $\text{SY4}/r_{\text{ROE}}^{\text{RD}}$ | $\text{SY4}+r_{\text{ROE}}^{\text{RD}}$ |                           | $\text{SY4}/r_{\text{ROE}}^{\text{RD}}$ | $\text{SY4}+r_{\text{ROE}}^{\text{RD}}$ |                              | $\text{SY4}/r_{\text{ROE}}^{\text{RD}}$ | $\text{SY4}+r_{\text{ROE}}^{\text{RD}}$ |
| SY4                       | 0.121<br><i>0.000</i>                   | 0.126<br><i>0.000</i>                   | SY4                       | 0.279<br><i>0.001</i>                   | 0.274<br><i>0.000</i>                   | SY4                          | 0.095<br><i>0.016</i>                   | 0.096<br><i>0.000</i>                   |

## B Dividend Discount Model, Present Value Relation, and Characteristics-based Factors

Fama and French (2006, 2015) use the dividend discount model (DDM) to motivate the characteristics-based five-factor model. DDM states that the market price of one share of stock is the sum of the discounted value of expected dividends per share,

$$P_{i,t} = \sum_{\tau=1}^{\infty} \mathbb{E}_t(D_{i,t+\tau}) / (1 + r_{i,t})^{\tau}, \quad (1)$$

where  $P_{i,t}$  is the market value of one share of firm  $i$ 's equity at time  $t$ ,  $\mathbb{E}_t(D_{i,t+\tau})$  is the expectation at time  $t$  for the dividend per share in period  $t + \tau$ , and  $r_{i,t}$  is the discount rate for firm  $i$ 's all future dividends at time  $t$ . Note that DDM is a valuation model that takes  $\mathbb{E}_t(D_{i,t+\tau})$  and  $r_{i,t}$  as inputs, and  $P_{i,t}$  as output, and it is an assumption to apply the same discount rate to all future dividends. DDM implies that if two firms have the same stream of expected dividends per share, the firm with riskier expected dividends should have higher discount rate and thereby lower price per share. Therefore, if pricing is rational, a strategy that long low price stocks and short high price stocks, controlling for the expected dividends, should have positive expected return (though zero risk-adjusted return).<sup>1</sup>

Equation (1) has another interpretation as a present value relation (PVR), which is a mathematical identity that is used to compute the internal rate of return, implied cost of capital, or stock yield (analogous to bond yield). It takes the observed stock price and estimates of expected dividends as inputs, and yields  $r$  as output. The PVR implies that a higher yield is obtained from buying the same expected stream of dividends at a lower price. This should hold both ex post and ex ante. Therefore, the same long-short strategy has positive expected return irrelevant of rational or irrational pricing (as noted in Fama and French (2015)). Campbell and Shiller (1988) use PVR to motivate dividend-to-price as a natural candidate to predict stock returns, and Gebhardt et al. (2001) compute the implied cost of capital based on PVR and find it strongly correlated with the future stock returns.

By applying the fact that dividend is the after-tax earnings minus investment, and the clearing surplus accounting, Fama and French (2015) rewrite equation (1) as follows:

$$\frac{P_{i,t}}{B_{i,t}} = \sum_{\tau=1}^{\infty} \mathbb{E} \left( \frac{Y_{i,t+\tau}}{B_{i,t}} - \frac{dB_{i,t+\tau}}{B_{i,t}} \right) / (1 + r_{i,t})^{\tau}. \quad (2)$$

where  $Y_{i,t+\tau}$  is the after-tax earnings of firm  $i$  in year  $t + \tau$ ,  $B_{i,t+\tau}$  is the book value of equity,

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<sup>1</sup>The dividend discount model, however, is silent about the source of the systematic risk, i.e., it does not endogenously specify why some firms have higher expected return than others.

$dB_{i,t+\tau}$  is the change of the book equity from  $t+\tau-1$  to  $t+\tau$ ,  $P_{i,t}$  is the market value of the equity, and  $r_{i,t}$  is internal rate of return. This motivates the four characteristics included in the Fama-French five factor model: size, book-to-market equity ratio, profitability, and investment rate.

In this section, we use the Gordon growth model (GGM), a DDM with perpetual profitability, investment rate, and discount rate, to relate the DDM/PVR to the characteristics-based linear factor model.<sup>2</sup> The GGM (by suppressing the time subscript) is given as follows:

$$\frac{P_i}{B_i} = \frac{\text{ROE}_i \times (1 - \text{PBR}_i)}{r_i - g_i} \quad (3)$$

where  $r_i$  is the discount rate,  $\text{ROE}_i$  is the return on equity,  $\text{PBR}_i$  is the equity plowback rate, and  $g_i = \text{ROE}_i \times \text{PBR}_i$  is the growth rate. Our simulated model economy assumes three exogenous variables, ROE, expected return, and PBR. The price-to-book ratio is determined by the present value relation in equation (3).<sup>3</sup> Note that the Fama-French investment factor is constructed based on the asset growth, measured by the annual change of total assets divided by the lagged total assets. In the GGM, this measure of investment is equivalent to the growth rate rather than PBR. Therefore, we consider four characteristics that are potentially related to the cross section of expected return: ROE, P/B, PBR, and growth rate.

In the simulation, we first focus on the cross section relation between characteristics and expected returns. We find significant relations between expected return and single characteristics except for ROE while ROE becomes significant when controlling for other characteristics. The growth rate is significant when holding PBR fixed but not vice versa. We therefore use growth rate rather than PBR when constructing factors. In our simulation, the PBR and growth rate are positively related to the expected return whereas it is negative in real data. We will argue that this contradiction is due to the interaction between future ROE and PBR which leaves the relation between the expected returns and PBR (and thereby growth rate) undetermined.

We then construct a model economy where returns are generated according to a linear latent factor structure in which the expected return of firms and their factor loadings are matched by construction. We examine whether in this example economy characteristics based factors are able to explain the cross section of returns, and whether they do as well as the Connor-Korajczyk (CK) factors in explaining the cross section of returns. We find that the CK factors do much better than characteristics-based factors, which contradicts what is found in the actual return data. We argue that this can be due

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<sup>2</sup>We choose to use Gordon growth model for the simulation because it is enough to clarify the relation between linear factor model and DDM/PVR without explicit modeling of the term structures of the variables.

<sup>3</sup>In this simple example, we assume that PBR is exogenously given, however, rational corporate investment behavior can be easily included: firms will always invest at the maximum or minimum level based on the relation between ROE and discount rate.

to time varying factor loadings. Therefore, we construct another economy with time variation in risk exposures. We find characteristics-based factors outperform the CK factors in that economy.

## B.1 Calibration

We construct a cross section of firms with expected perpetual ROE, PBR, and  $r$ . To calibrate discount rates, we first estimate the market beta for each firm with complete observations from 2013 to 2022 in the CRSP/Compustat merged sample. Such a filtering process biases the sample toward high quality firms but helps eliminate outliers which is particularly important when calibrating for perpetual ROE. The firms must also have a NYSE, AMEX, or NASDAQ primary exchange code and a share code of 10 or 11 to be included.<sup>4</sup> We take the mean (1.10) and standard deviation (0.51) of the market beta across firms, and then compute the discount rate by multiplying a 6% annual risk premium by the market beta, assuming, without loss of generality, that the riskless rate is 0. The (annual) expected returns  $r_i$  of the 4,000 firms are therefore drawn from a normal distribution with mean 6.62% and standard deviation 3.06%.

To calibrate for ROE, we compute the 10-year average ROE<sup>5</sup> for each firm and exclude firms with negative sum of book equity or ROE below 0% or above 60%<sup>6</sup>. The expected ROE of the 4,000 simulated firms are drawn from a normal distribution with mean 0.25 and standard deviation 0.13. The top and bottom panels of Figure B1 plot the empirical distribution of ROE and expected return.

The GGM is, obviously, a toy model, which assumes  $r_i$  and ROE <sub>$i$</sub>  and PBR <sub>$i$</sub>  are constant in perpetuity. In that setting certain combinations of parameters lead to implausibly high (infinite) valuations. In order to avoid these situations we place the following restrictions on the parameters. We exclude the draws of ROE <sub>$i$</sub>  outside the range [0%, 60%] and draw of  $r_i$  outside the range [2%, 12%]. Then we draw plowback rate from uniform distribution over  $[0, \frac{r_i}{2\text{ROE}_i}]$ , subject to the restriction that PBR <sub>$i$</sub>  < 1. Though we start with 4,000 firms, there are, on average, 3,413 firms (with a minimum of 3,349 and a maximum of 3,476) left after applying all the restrictions for the 100 independent cross sections we generated.

Imposing equation (3) to the randomly generated ROE, PBR, and the expected returns, we can compute the P/B for firms. In our setup, ROE and expected return are exogenous variables, and a ratio of them,  $\frac{r_i}{2\text{ROE}_i}$ , determines the upper bound of the PBR. This does not make PBR endogenous because firms with ROE greater (lower) than expected return will always want to invest at maximum (minimum) level. To keep the

<sup>4</sup>There are 1,853 out of total 6,016 firms with complete observations in the CRSP/Compustat merged sample from 2013 to 2022.

<sup>5</sup> $\text{ROE} = \frac{1}{10} \sum_{t=2013}^{2022} \frac{\text{REVT}_t - \text{COGS}_t - \text{XSGA}_t - \text{XINT}_t + \text{RD}_t}{\text{CEQ}_t}$ .

<sup>6</sup>This leaves a final sample of 1,586 firms

example easy to follow, we choose not to specify a model to endogenize the corporate investment behavior.

## B.2 Relation between Expected Returns and Characteristics

We rewrite GGM to derive the relations between expected return and ROE, P/B, and PBR respectively as follows:

$$r_i = \left( \text{PBR}_i + \frac{1 - \text{PBR}_i}{P_i/B_i} \right) \text{ROE}_i, \quad (4)$$

$$r_i = \text{ROE}_i \text{PBR}_i + \text{ROE}_i (1 - \text{PBR}_i) \frac{1}{P_i/B_i}, \quad (5)$$

$$r_i = \frac{\text{ROE}_i}{P_i/B_i} + \left( 1 - \frac{1}{P_i/B_i} \right) \text{ROE}_i \text{PBR}_i. \quad (6)$$

According to equations (4) and (5), there is a negative relation between  $r_i$  and P/B and a positive relation between  $r_i$  and ROE when controlling for other characteristics, however, the relation between  $r_i$  and PBR depends on whether P/B is above or below one (see equation (6)). A firm with  $P_i/B_i > 1$  has  $\text{ROE}_i > r_i$ , and therefore any extra investment earn positive net value for the equity holders. In this case, to get the same P/B ratio with a same ROE, a firm with higher expected return must have higher PBR. Therefore, we could expect a positive relation between the expected return and PBR (and thereby growth rate) when holding other characteristics fixed if the sample is dominated by firms with P/B above one.<sup>7</sup>

Though the potential characteristics are identified, whether all characteristic provide incremental explanatory power in explaining the cross section of expected returns remains a question. We use two methods for this question: (1) run regressions of expected returns on the characteristics (results are reported in Panel A of Table B1); (2) assign firms into groups based on characteristics and form long-short portfolios (results are reported in Panel B and C of Table B1). The DDM motivation for characteristics-based factors requires controlling for other characteristics which naturally relies on conditional sorting procedure to form portfolios while the conventional approach is to form portfolios independently. Therefore, we run both univariate and multiple regressions and apply both independent and conditional sorting in portfolio assignment. We can expect that the results will be similar when the correlation between characteristics is low.

In the univariate regressions, all characteristics are strongly associated with the expected returns except for ROE.<sup>8</sup> However, ROE turns significant in the multiple regres-

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<sup>7</sup>The expected return of a high-minus-low PBR portfolio can be different from the expected return of a high-minus-low growth rate portfolio if PBR and ROE do not have a zero correlation within each ROE group.

<sup>8</sup>We generate the cross section independently 100 times. The t-statistics of ROE in the univariate regression ranges from  $-0.8$  to  $4.0$  with a mean of  $1.2$  while the absolute value of the t-statistics of other



sions with P/B and PBR which shed light on the importance of the keeping other things fixed in the factor construction. Because the asset growth measure in Fama and French (2015) is equivalent to the growth rate measure,  $g_i = \text{ROE}_i \times \text{PBR}_i$ , in the GGM, we include the growth rate as an independent variable in the regression. We find that PBR becomes less significant when the growth rate,  $g$ , is added.<sup>9</sup> We obtain similar coefficients and t-statistics for remaining characteristics in the multiple regression without PBR.

For the portfolio sorts, we assign firms into groups based on (1) univariate decile sort, (2) 2-by-2-by-3 P/B, PBR, and ROE sort, (3) 2-by-2-by-3 P/B, ROE, and growth rate sort, and (3) 2-by-2-by-2-by-3 P/B, PBR, ROE, and growth rate sort of given characteristics. The return of each group is the equal weighted average return and the return for each long-short strategy is the mean return of the top groups minus the mean return of the bottom groups based on the given characteristic. Panel B and C in Table B1 apply independent sorts and conditional sort, respectively, and report the mean, minimum, and maximum of the strategy returns based on the 100 independently generated cross sections. Using the 2-by-2-by-3 P/B, ROE, and PBR conditional sort strategy as an example, we first assign firms into 2-by-2 groups based on median P/B and median ROE, and then assign firms with each group further into three groups based on the 30 and 70 percentiles of PBR. The monthly percentage expected return of this strategy is the average return of the four high PBR groups minus the average return of the four low PBR groups. We also report the performance of similar strategies using CRSP/Compustat merged data from July 1963 to June 2023 in Panel D in Table B1. To make comparison easier, we convert the annual expected returns into monthly percentage expected returns for the simulation results.<sup>10</sup>

The results using portfolio sorts are consistent with the findings in the regression analysis. A univariate sort on ROE performs poorly with a 0.02 monthly percentage expected return and the minimum and maximum values take opposite signs, while the other three characteristics show persistent results. With both independent and conditional sorts, the performance on ROE strategy improves and becomes persistent when controlling for other characteristics. We then use all four characteristics to form 2-by-2-by-2-by-3 portfolios. The return of the PBR strategy becomes weak and removing it does not attenuate the performance of the other three characteristics.<sup>11</sup> More interestingly, the strategy performance of the independent sorts is better than the conditional sort across all characteristics. However, it remains an empirical question whether the better performance of independent sort holds in the real data.

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characteristics is not less than 15.

<sup>9</sup>The t-statistics of PBR ranges from  $-3.1$  to  $1.1$  with a mean of  $-0.9$  while the absolute value of the t-statistics of other characteristics is not less than 24.

<sup>10</sup>The mean, minimum, and maximum of the equal weighted average expected (monthly percentage) return are 0.562, 0.555, and 0.571 respectively.

<sup>11</sup>The 2-by-2-by-2-by-3 independent sort does not always guarantee 24 groups with due to the high correlation between PBR and growth rate.

We then compute the performance of the similar strategies using CRSP/Compustat merged data from July 1963 to June 2023.<sup>12</sup> Note that P/B, ROE, and growth rate in the GGM corresponds to the (opposite of) book-to-market, operating profitability, and investment rate in Fama and French (2015) respectively. For all three characteristics, we compute the strategy performance using the mean return of the top groups minus the mean return of the bottom groups for the given characteristic. P/B and ROE are not significant with the univariate sort, and growth rate is not significant with 2-by-2-by-3 sort.<sup>13</sup> There is some evidence that independent sort performs better than the conditional sort for ROE, but no evidence for P/B and growth rate. More importantly, the sign of the growth rate strategy contradicts the simulation results.

To understand the relation between the expected return and growth rate, first note that when controlling for ROE and P/B, this relation is determined by the relation between the expected return and PBR.<sup>14</sup> There are two sources of ambiguity on the sign of this relation. First, whether  $P/B > 1$  is a dominant feature in the population. Second, the impact of PBR on the book value of equity. In the GGM, ROE is the expected perpetual return on the one-period lagged book value of equity. As the firm invests more at  $t + \tau$ , the firm will have more book equity at  $t + \tau + 1$  and thereby higher earnings at  $t + \tau + 1$  with that same ROE. In equation (2), however, future profitability is measured by future earnings divided by the book equity at time  $t$ . With such a profitability measure, the more a firm invests at  $t$  the higher the after-tax earnings at  $t + 1$  will be if ROE is constant and book equity grows with investment. Thus, one cannot vary  $dB_{t+\tau}/B_t$  while keeping  $Y_{t+\tau+1}/B_t$  fixed in equation (2). We will need additional assumptions beyond the present value relation to explain the empirically discovered negative relation between expected return and investment rate.<sup>15</sup>

### B.3 Linear Latent Factor Structures

We wish to determine whether, in a world in which asset returns follow a factor structure, whether the characteristic-based factors can mimic the true underlying factors. That

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<sup>12</sup>We measure book equity following Fama and French (2006), size as the market equity at the end of June in year  $t - 1$ , growth,  $g$ , as asset growth and ROE as the operating profitability following Fama and French (2015). We use Compustat data item CEQ to measure book equity when constructing intangible-adjusted profitability factor because its coverage is broad enough after 1975 when the R&D expense data became valid.

<sup>13</sup>Our results for the univariate sort and 2-by-3 Size-X sort are close to the results in Kenneth French's data library. In Kenneth French's data library, the time-series average (monthly percentage) return gaps between top 10% and bottom 10% for book-to-market, operating profitability, and investment are 0.34, 0.30, and -0.40, and the associate t-statistics are 1.86, 1.86, and -3.11 respectively from July 1963 to June 2023.

<sup>14</sup> $\frac{P_i}{B_i} = \frac{ROE_i - ROE_i PBR_i}{r_i - ROE_i PBR_i}$

<sup>15</sup>A mechanic fix to match the empirically found negative relation between expected return and investment rate is to assume higher expected returns for firms with lower investment rate potentially due to behavioral reasons.

is, we construct a world where the cost of capital for assets is determined by four latent factors. We study the performance of the characteristic-based factors (long/short portfolios) relative to latent factors estimated by asymptotic principal components Connor and Korajczyk (1986). We assume the following linear (latent) factor model for returns:

$$R_{i,t} = \underbrace{\sum_{k=1}^K \beta_{i,k} \lambda_k}_{r_i} + \underbrace{\sum_{k=1}^K \beta_{i,k} F_{k,t}}_{\beta_i F_t} + \epsilon_{i,t}, \quad (7)$$

where  $\lambda_k$  is the expected return of the  $k$ th factor,  $F_{k,t}$  is the realized value of the  $k$ th factor at time  $t$ , and it is normally distributed with mean zero and standard deviation  $\sigma_k$ ,  $\beta_{i,k}$  is the firm  $i$ 's loading on the  $k$ th factor, and  $\epsilon_{i,t}$  is the noise term for firm  $i$  at time  $t$ , and  $K = \{1, 2, 3, 4\}$  is the total number of latent factors. We generate 600 monthly observations for the factors and 600 monthly noise term independently. For simplicity of the notation, we assume a risk-free rate of zero. When  $K = 4$ , we refer the four factors as the market, ROE, P/B, and growth rate. The three characteristics, ROE, P/B, and growth rate, are chosen because they show strong correlation with the firms' expected returns. When  $K \leq 4$ , we calibrate for only the first  $K$  factors of the market, ROE, P/B, and growth rate factors.

We choose the first cross section (out of the 100 cross sections) generated in the last section to fix the firms' characteristics and expected returns.<sup>16</sup> The chosen cross section has 4,000 firms, in which 564 firms are removed due to the restrictions we impose. When  $K = 4$ , the true expected returns,  $\lambda_k$ , of the four factors, market, ROE, P/B, and growth rate, are naturally the expected returns of the corresponding factors for the given cross section, which are 0.56, 0.22, -0.40, and 0.19 in monthly percentage respectively. The standard deviations of the four latent factors,  $F_{k,t}$ , are set equal to the standard deviations of the Fama-French RMW, HML, and CMA in the last 50 years (from January 1973 to December 2022), which are 4.64, 2.33, 3.10, and 2.04 in monthly percentage respectively. The noise term  $\epsilon_{i,t}$  is assumed to be normally distributed with mean zero and the standard deviation 10 in monthly percentage, which is set equal to the empirical standard deviations of the CAPM residuals of firms from 2013 to 2022.<sup>17</sup>

The firms' factor loadings,  $\beta_{i,k}$ , are generated as follows. When  $K = 1$ ,  $\beta_{i,1}^{K=1} = r_i/\lambda_1$ , where  $r_i$  is the expected return of firm  $i$  and  $\lambda_1$  is the equal-weighted expected return of all firms. In this case, the factor loading  $\beta_{i,1}^{K=1}$  is perfectly correlated with the true expected return of firm  $i$ . When  $K = 2$ , we randomly draw  $\beta_{i,1}^{K=1}$  without replacement

<sup>16</sup>Randomly choosing a cross section should have no impact on the simulation results as the quantitative relations derived from the static cross sections are stable.

<sup>17</sup>For the firms that have complete monthly observations from January 2013 to December 2022, the mean/median of the standard deviation of the monthly return is 9.9/12.2. The mean/median of the standard deviation of the CAPM residual is 8.5/10.9. The Fama-French five factor model bring these numbers down to 7.9/10.2.

and set them as  $\beta_{i,1}^{K=2}$ . Then the firm  $i$ 's loading on factor 2 is  $\beta_{i,2}^{K=2} = (r_i - \beta_{i,1}^{K=2} \lambda_1) / \lambda_2$  where  $\lambda_2$  is the expected return of the ROE factor for the chosen cross section. In this case,  $\beta_{i,2}^{K=2}$  is perfectly correlated with the residual expected return of the first factor for firm  $i$ . Similarly, when  $K = 3$ , we set  $\beta_{i,1}^{K=3} = \beta_{i,1}^{K=2}$ ,  $\beta_{i,2}^{K=3}$  to be the randomly drawn  $\beta_{i,2}^{K=2}$  (without replacement), and  $\beta_{i,3}^{K=3} = (r_i - \beta_{i,1}^{K=3} \lambda_1 - \beta_{i,2}^{K=3} \lambda_2) / \lambda_3$  where  $\lambda_3$  is the expected return of the P/B factor. Finally, when  $K = 4$ , we set  $\beta_{i,1}^{K=4} = \beta_{i,1}^{K=3}$ ,  $\beta_{i,2}^{K=4} = \beta_{i,2}^{K=3}$ ,  $\beta_{i,3}^{K=4}$  to be the randomly drawn  $\beta_{i,3}^{K=3}$  (without replacement), and  $\beta_{i,4}^{K=4} = (r_i - \beta_{i,1}^{K=4} \lambda_1 - \beta_{i,2}^{K=4} \lambda_2 - \beta_{i,3}^{K=4} \lambda_3) / \lambda_4$  where  $\lambda_4$  is the expected return of the growth rate factor.

## B.4 Characteristic-based vs. Statistical Factors

We repeat the simulation 100 times. The firms' characteristics and expected returns are fixed for not only the 100 simulations but also the 600 months within each simulation. The realization of the factors and noise terms are randomly generated based on the same given distribution for each simulation. Within each simulation we construct the market factor and the characteristics-based factors, and examine their performance and explanatory power for the cross section of stock returns. The market factor in a month is the equal-weighted average realized return of all firms in the given month. To construct the ROE factors, firms are assigned into 2-by-2-by-3 P/B, growth rate, and ROE groups independently based on median P/B, median growth rate, and 30 and 70 percentiles of ROE. We then take the equal-weighted average return for each group, and then take the equal-weighted average return for the four high ROE groups and four low ROE groups. The return of ROE factor in a month is the return difference between the high ROE groups and the low ROE groups in the given month. The P/B and growth rate factors are constructed using the same procedure except that firms are assigned into 30 and 70 percentiles of P/B and growth rate respectively. We do not rebalance the groups through the 600 months in each simulation as the firms' characteristics do not change over time. We also take the negative of the P/B factor return so that it has a positive expected return.

Figure B2 plots the histogram of the time-series average return of the market factor and three characteristics-based factors for the 100 simulations. Columns one to four correspond to the cases where the total number of latent factors equal to one to four,  $K = 1, 2, 3, 4$ , respectively. All characteristic-based factors have positive expected returns even when the true model has a one-factor structure,  $K = 1$ . This is not surprising characteristics are conditionally correlated with the cost of capital, hence correlated with market beta when the economy is priced by a single factor. We run regressions of the three characteristics-based factors on the market factor and report the intercepts in Figure B3. Interestingly, the intercepts of the factors are clearly centered around 0 when there is only

one latent factor,  $K = 1$  (see the first column). This implies that the characteristics-based factors do not seem to provide independent information when controlling for the market factor when the true model has only one factor even though they all have expected returns significantly different from zero. When  $K > 1$ , however, the characteristics-based factor alphas become significantly different from zero against the market factor but the intercepts do not increase as the total number of factors grows.

We also construct the 3-by-3-by-3 conditionally sorted ROE, P/B, and growth rate portfolios and use them as test assets to examine the explanatory power of the factors.<sup>18</sup> Figure B4 plots the histogram of the p-value of the GRS test of the 27 test assets on the market and the characteristics-based factors for the 100 simulations. The total number of latent factors can be 1 to 4,  $K = 1, 2, 3, 4$  (for each row), and the total number of factors included in the factor model can be 0 to 4,  $J = 0, 1, 2, 3, 4$  (for each column). When  $J = 0$ , the GRS test is equivalent to test whether the test assets have a maximum squared Sharpe ratio of zero. When  $J > 0$ , we always add the factors into the model in the following order: market, ROE, P/B, and growth rate. We provide the frequency of p-values being less or equal to 10% for each plot. If the factor model can fully explain the returns of the test assets, we should expect this frequency to be near 10%. This is clearly not the case when no factor is included in the model,  $J = 0$  (see the first column). When the true model has one factor,  $K = 1$ , the market factor effectively lowers the rejection frequency from 44% to 13%, and adding more factors does not have any impact. This implies that the market factor can be good enough to describe the cross section of stock returns when  $K = 1$ . When  $K > 1$ , however, the market factor is no longer sufficient. Adding more factors into the model does lower the chance of rejection, but the improvement seems to stop after adding the ROE factor and remains high relative to the significance level of 10% level.

We then estimate statistical factors following Connor and Korajczyk (1986) and examine their performance relative to the characteristics-based factors. Figure B5 plots the average eigenvalues of the first six principal components across the 100 simulations for  $K = 1, 2, 3, 4$ . We regress the three characteristics-based factors on the first C-K factor when  $K = 1$ , on the first two C-K factors when  $K = 2$ , on the first three C-K factors when  $K = 3$ , and on the first four C-K factors when  $K = 4$ . Figure B6 shows that the intercepts of the characteristics-based factors are clearly not significantly different from zero. Moreover, we compare the maximum squared Sharpe ratios of the statistical factors with the characteristics-based models (with a market factor) in Figure B7. When  $K = 1$ , the C-K model has a higher Sharpe ratio than the four-factor characteristics-based model for 63% of the simulations. This frequency becomes 61%, 88%, and 99% when  $K = 2$ ,  $K = 3$ , and  $K = 4$  respectively. The C-K models are also tested against the same test

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<sup>18</sup>We do not use independent sort to construct the test assets because they do not guarantee 27 portfolios due to the high correlations among characteristics in extreme cases.

assets and the results are shown in Figure B8. The frequency of rejecting zero alphas for the test assets is generally close to, but slightly higher than, the stated 10% level of the test when the right C-K model is used.

This finding that C-K models can dominate the characteristic-based models contradicts what is typically found in actual data. We argue that this may be due to time-varying factor loadings in the actual data. In the next section, we show that characteristics-based factors have advantages over the C-K factors if the firms' factor loadings change over time.

## B.5 Time-varying Factor Loadings

We generate time variations in factor loadings by repeating the generating process for the factor loadings described in section B.3 every 24 months. When there is only one latent factor, the factor loadings will remain constant over the entire 600 months. When there are more than one latent factor, the factor loadings will change every 24 months. We construct both the characteristic-based factors and C-K factors over the entire sample following the same procedure as above. The performance of the characteristic-based factors and their intercepts against the market factor are very similar to those reported earlier in Figures B2 and B3. However, Figure B9 shows that the performance of the characteristics-based factors in describing the cross section improves substantially especially when the correct number of factors are chosen for a given number of latent factors.

Figure B10 shows that the eigenvalues of the C-K factors drop much faster after the first principal component when the factor loadings are time-varying relative to the simulations with static factor loadings. Figure B10 suggests choosing a two-factor model except when  $K = 1$ . For a more reasonable comparison to the static factor loading case, we use the correct number of C-K factors for a given  $K$  even though the third and fourth factors may not contribute to the performance of the model. The intercepts of the characteristics-based factors against the “correct” C-K model are given in Figure B11. As the number of latent factors grows, the C-K models become less capable of explaining the performance of the characteristics-based factors. Comparing the maximum squared Sharpe ratios, Figure B12 shows that the characteristics-based model can outperform the “correct” C-K model though when  $K > 2$ . The GRS test p-values of the test assets against the C-K models in Figure B13 provides additional evidence that the marginal benefit of adding the C-K factors become close to zero after the second principal component.

## B.6 Summary

In this section/appendix, we argue that the DDM/PVR provides a solid motivation to construct the Fama-French five factors. It can be given a “casual” interpretation when viewed as a valuation model though the results hold irrespective of rational or

irrational pricing because PVR is a mathematical identity. In the simulation, we take ROE, the discount rate, and PBR as given and let DDM formula determine the P/B ratio. We find that the relation between expected return and growth rate (investment rate in the context of the Fama-French five factor model) is not clear mainly due to the interaction between future investment rate and profitability. In our linear latent factor structure, the characteristic-based factors have positive expected returns and, hence, some explanatory power for the cross section of stock returns. When the factor loadings are static over the entire sample, the benefit of adding more than one characteristic-based factor, in addition to the market factor, seems limited whereas the asymptotic PCA factors dominate the characteristics-based factor models when the number of latent factors for simulated economies with multiple factors. However, when the factor loadings become time-varying, we find the opposite results: the benefit of having more than two asymptotic PCA factors seems limited and the characteristics-based factor models do better in describing the cross section of the stock returns and outperform C-K models.

## References

- Asness, Clifford, Andrea Frazzini. 2013. The devil in hml's details. *The Journal of Portfolio Management* **39**(4) 49–68. URL <https://doi.org/10.3905/jpm.2013.39.4.049>.
- Barillas, Francisco, Raymond Kan, Cesare Robotti, Jay Shanken. 2020. Model comparison with sharpe ratios. *Journal of Financial and Quantitative Analysis* **55**(6) 1840–1874.
- Campbell, John Y, Robert J Shiller. 1988. Stock prices, earnings, and expected dividends. *the Journal of Finance* **43**(3) 661–676.
- Connor, Gregory, Robert A. Korajczyk. 1986. Performance measurement with the arbitrage pricing theory: A new framework for analysis. *Journal of Financial Economics* **15**(3) 373–394. doi:[https://doi.org/10.1016/0304-405X\(86\)90027-9](https://doi.org/10.1016/0304-405X(86)90027-9). URL <https://www.sciencedirect.com/science/article/pii/0304405X86900279>.
- Eisfeldt, Andrea L., Edward T. Kim, Dimitris Papanikolaou. 2022. Intangible value. *Critical Finance Review* **11**(2) 299–332.
- Fama, Eugene F, Kenneth R French. 2006. Profitability, investment and average returns. *Journal of Financial Economics* **82**(3) 491–518.
- Fama, Eugene F, Kenneth R French. 2015. A five-factor asset pricing model. *Journal of Financial Economics* **116**(1) 1–22.

- Fama, Eugene F, Kenneth R French. 2018. Choosing factors. *Journal of Financial Economics* **128**(2) 234–252.
- Gebhardt, William R, Charles MC Lee, Bhaskaran Swaminathan. 2001. Toward an implied cost of capital. *Journal of Accounting Research* **39**(1) 135–176.
- Gulen, Huseyin, Dongmei Li, Ryan H. Peters, Morad Zekhnini. 2025. Intangible capital in factor models. *Management Science* **71**(2) 1756–1778. URL <https://doi.org/10.1287/mnsc.2022.01261>.
- Peters, Ryan H, Lucian A Taylor. 2017. Intangible capital and the investment-q relation. *Journal of Financial Economics* **123**(2) 251–272.



# Tables and Figures

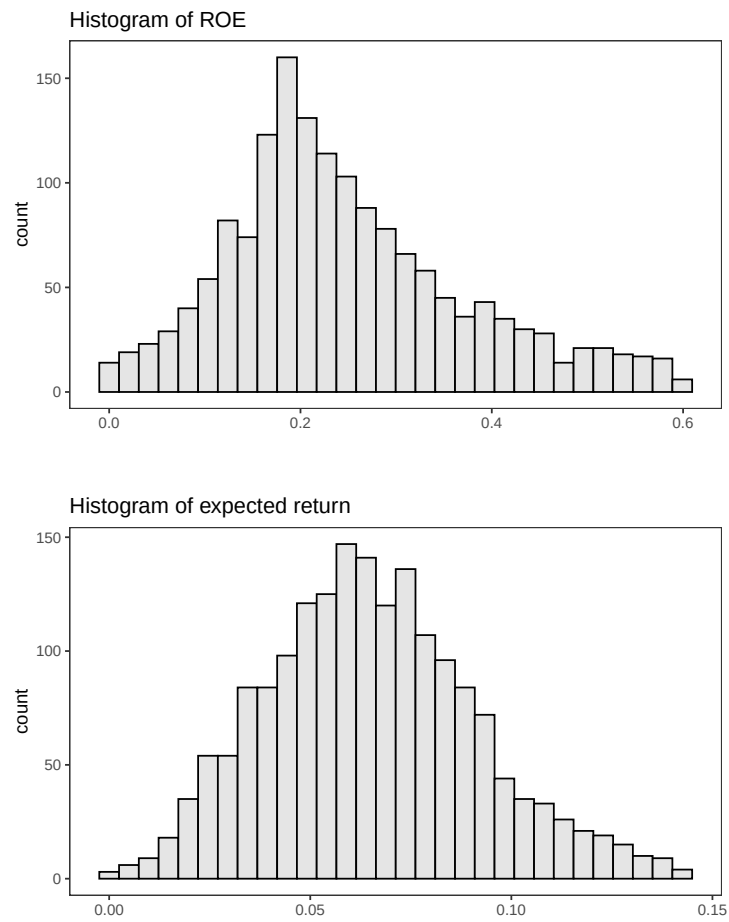


Figure B1: Sample of ROE and expected returns.

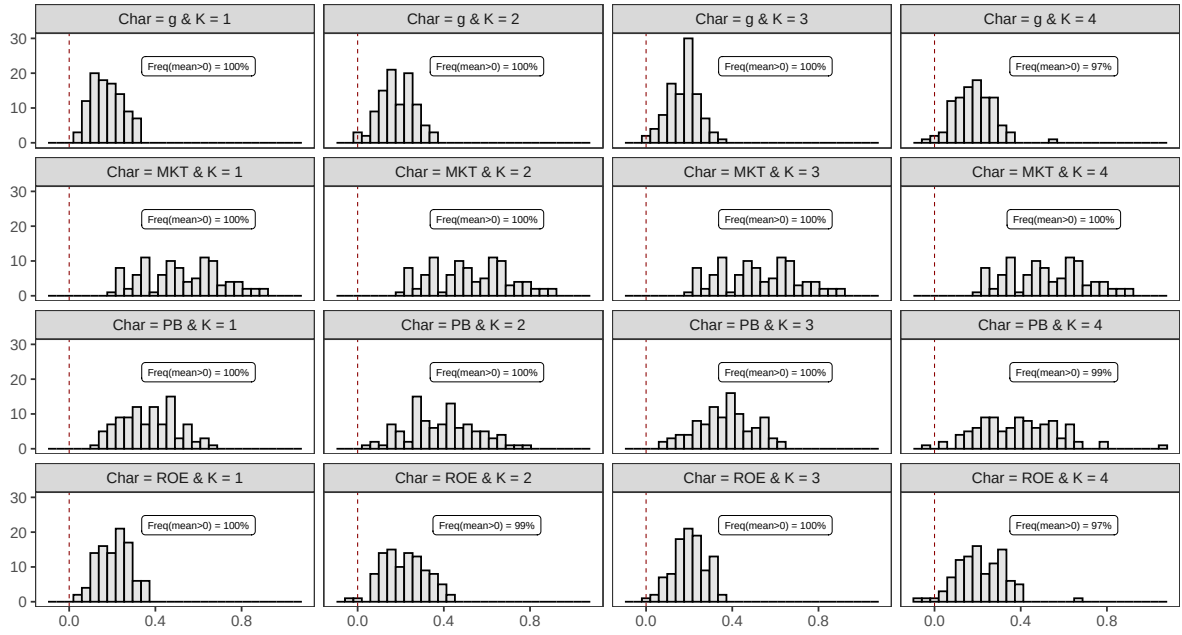


Figure B2: Histogram of time-series average returns of market and characteristics-based factors when  $K = 1, 2, 3$ , and  $4$ .

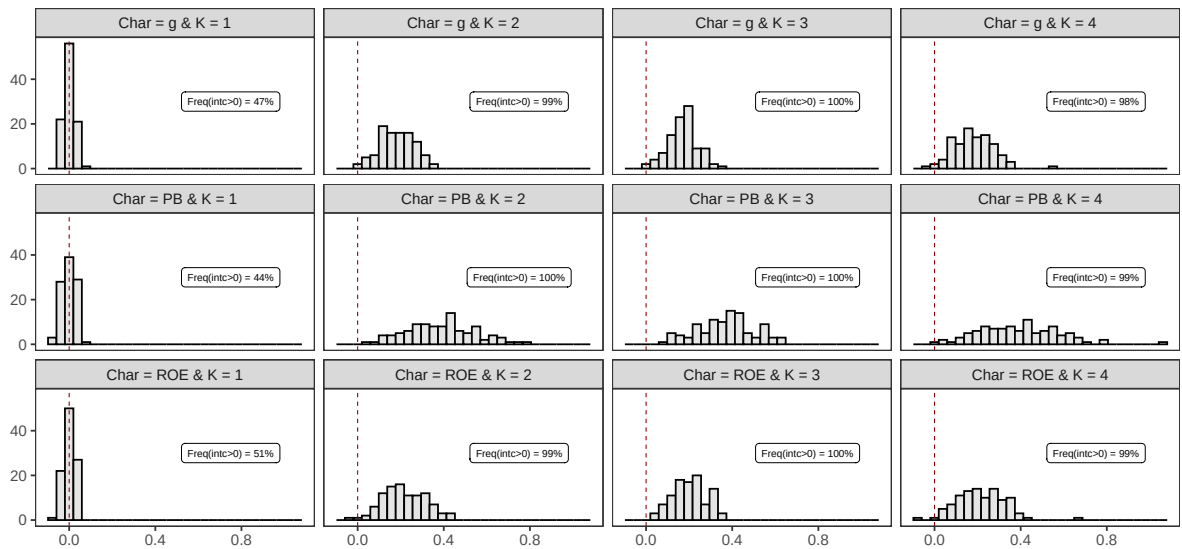


Figure B3: Histogram of intercept of characteristics-based factors on market factor when  $K = 1, 2, 3$ , and  $4$ .

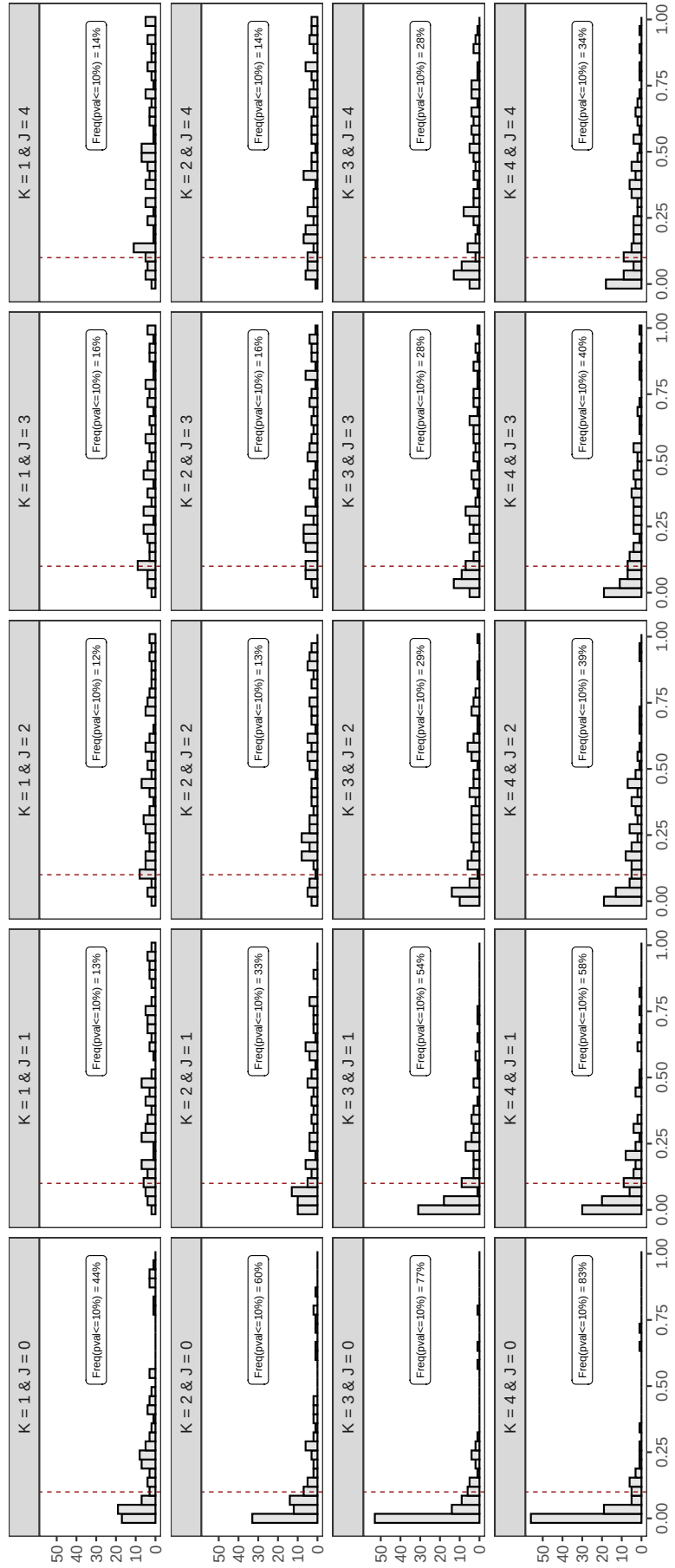


Figure B4: Histogram of GRS test p-value of the 3-by-3-by-3 ROE, P/B, and growth rate portfolios on market and characteristics-based factors when  $K = 1$ , 2, 3, and 4 and  $J = 1, 2, 3$ , and 4.

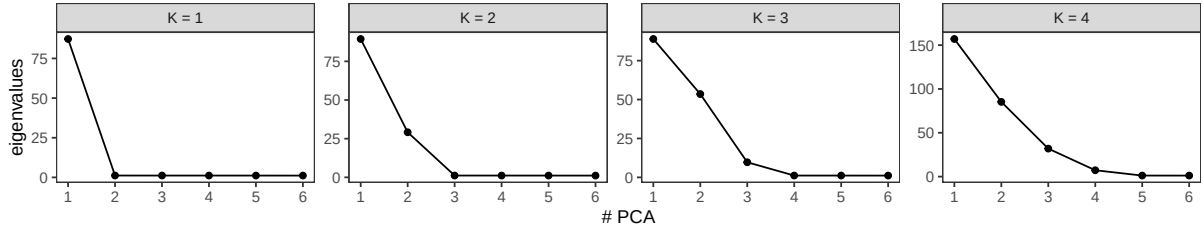


Figure B5: Eigenvalues of the Connor-Korajczyk factors when  $K = 1, 2, 3,$  and  $4$ .

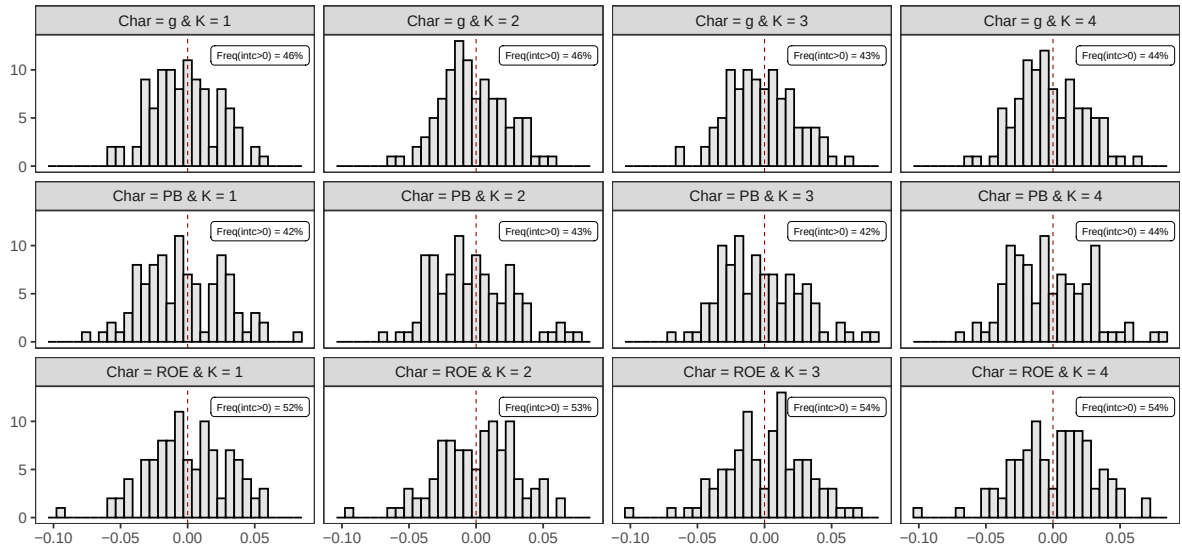


Figure B6: Histogram of the intercept of characteristics-based factors on the Connor-Korajczyk factors when  $K = 1, 2, 3,$  and  $4$ .

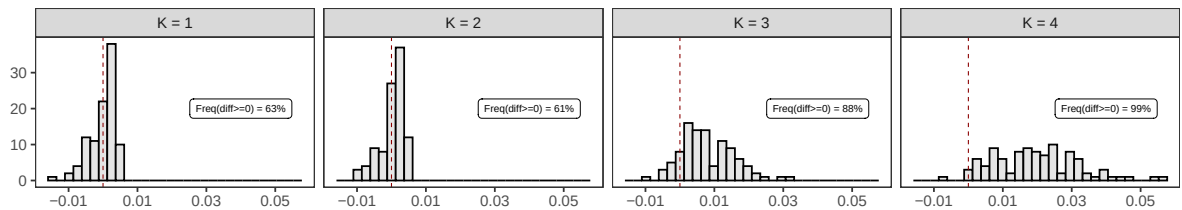


Figure B7: Histogram of the difference in squared Sharpe ratio between Connor-Korajczyk factors and market and characteristics-based factors when  $K = 1, 2, 3,$  and  $4$ .

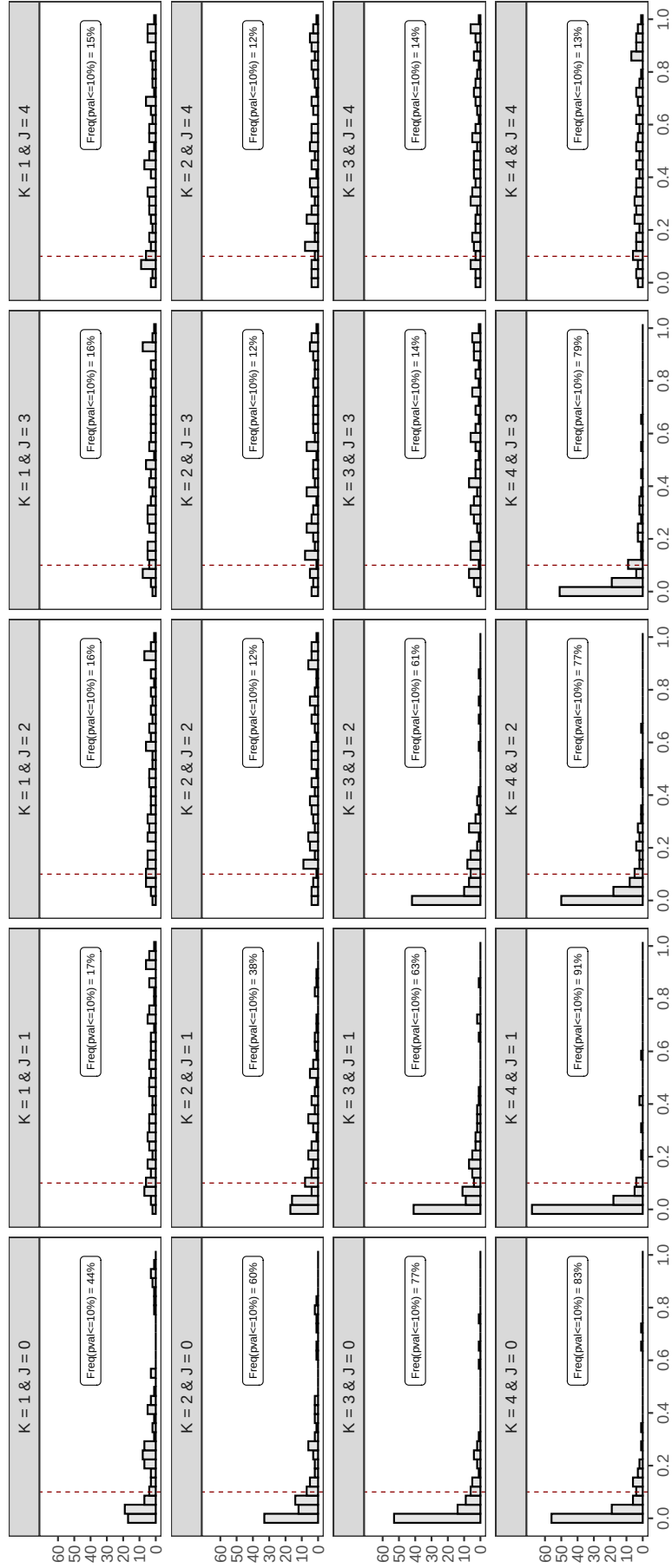


Figure B8: Histogram of GRS test p-value of the 3-by-3-by-3 ROE, P/B, and growth rate portfolios on Connor-Korajczyk factors when  $K = 1, 2, 3$ , and 4 and  $J = 0, 1, 2, 3$ , and 4.

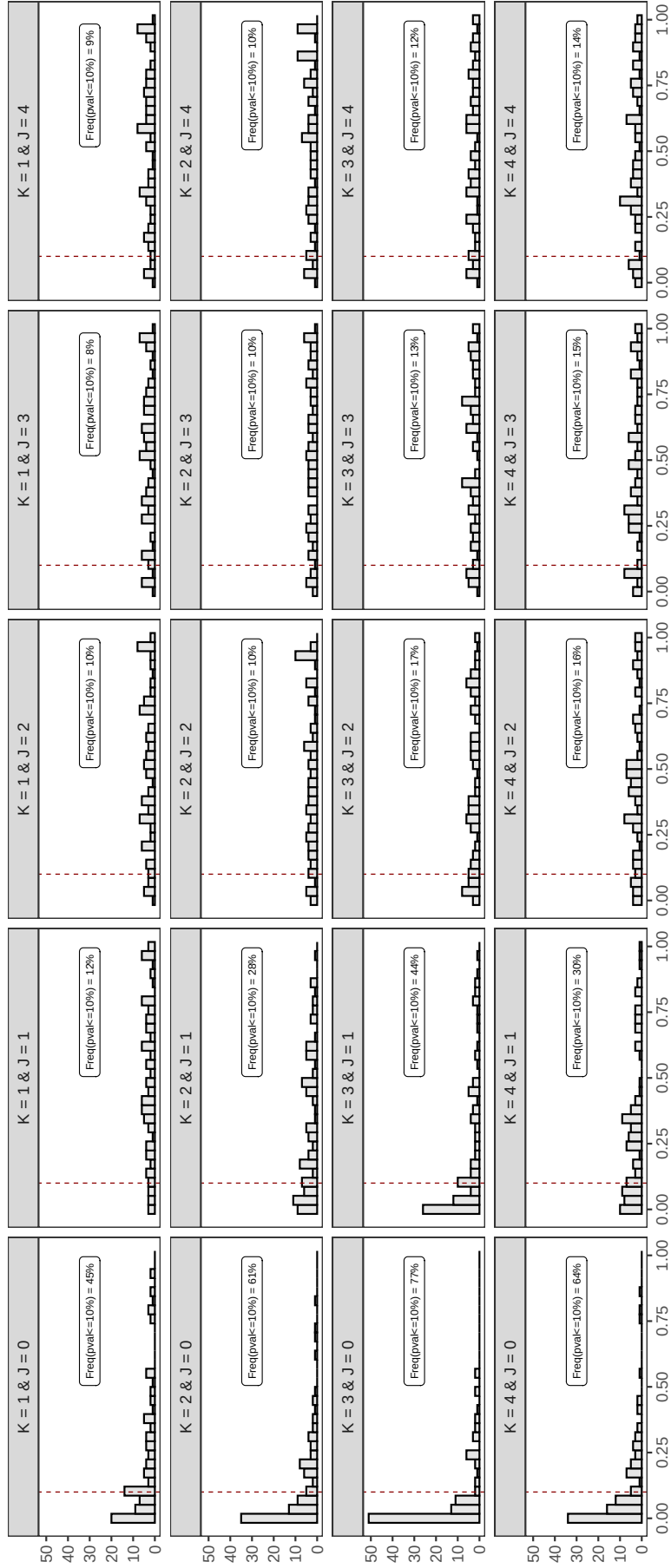


Figure B9: Histogram of GRS test p-value of the 3-by-3-by-3 ROE, P/B, and growth rate portfolios on market and characteristics-based factors when factor loadings are time-varying and  $K = 1, 2, 3$ , and 4 and  $J = 1, 2, 3$ , and 4.

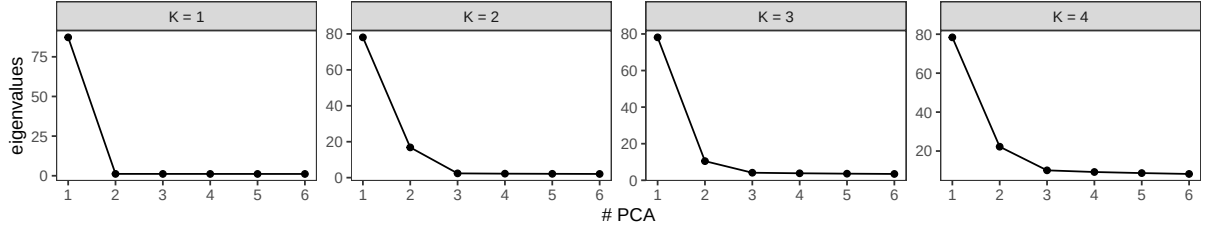


Figure B10: Eigenvalues of the Connor-Korajczyk factors when factor loadings are time-varying and  $K = 1, 2, 3$ , and  $4$ .

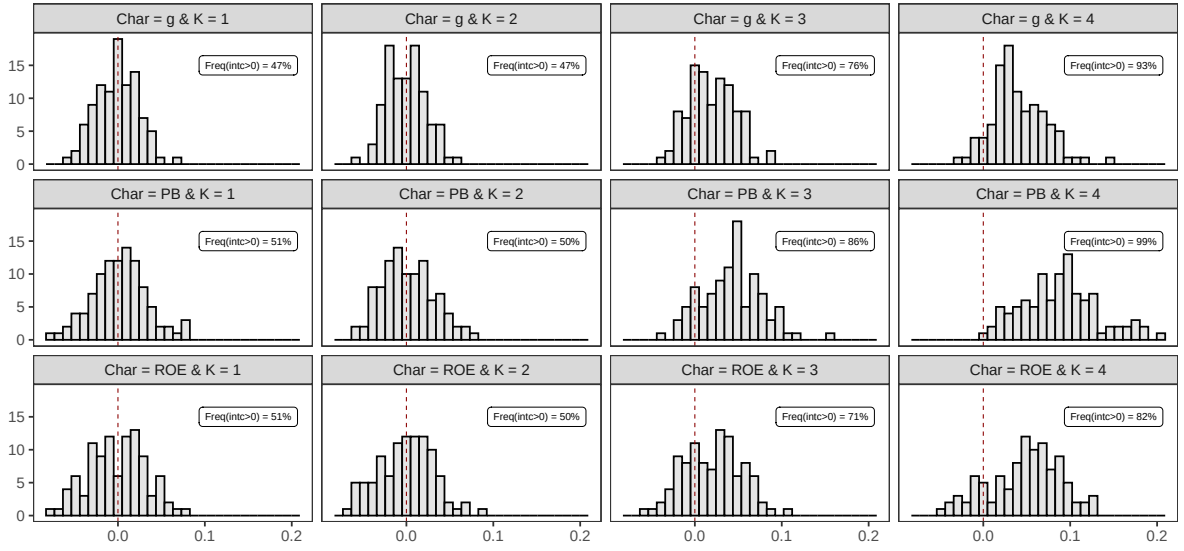


Figure B11: Histogram of the intercept of characteristics-based factors on the Connor-Korajczyk factors when factor loadings are time-varying and  $K = 1, 2, 3$ , and  $4$ .

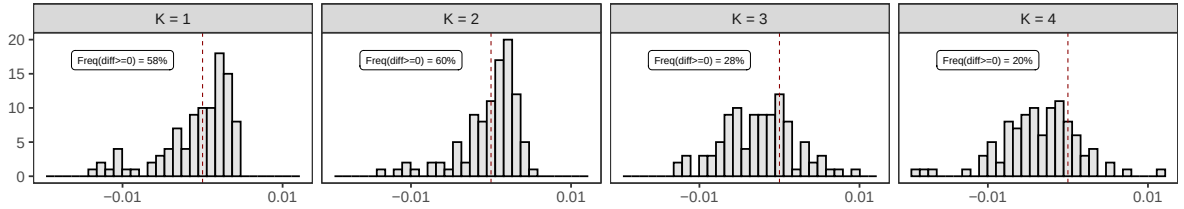


Figure B12: Histogram of the difference in squared Sharpe ratio between Connor-Korajczyk factors and market and characteristics-based factors when factor loadings are time-varying and  $K = 1, 2, 3$ , and  $4$ .

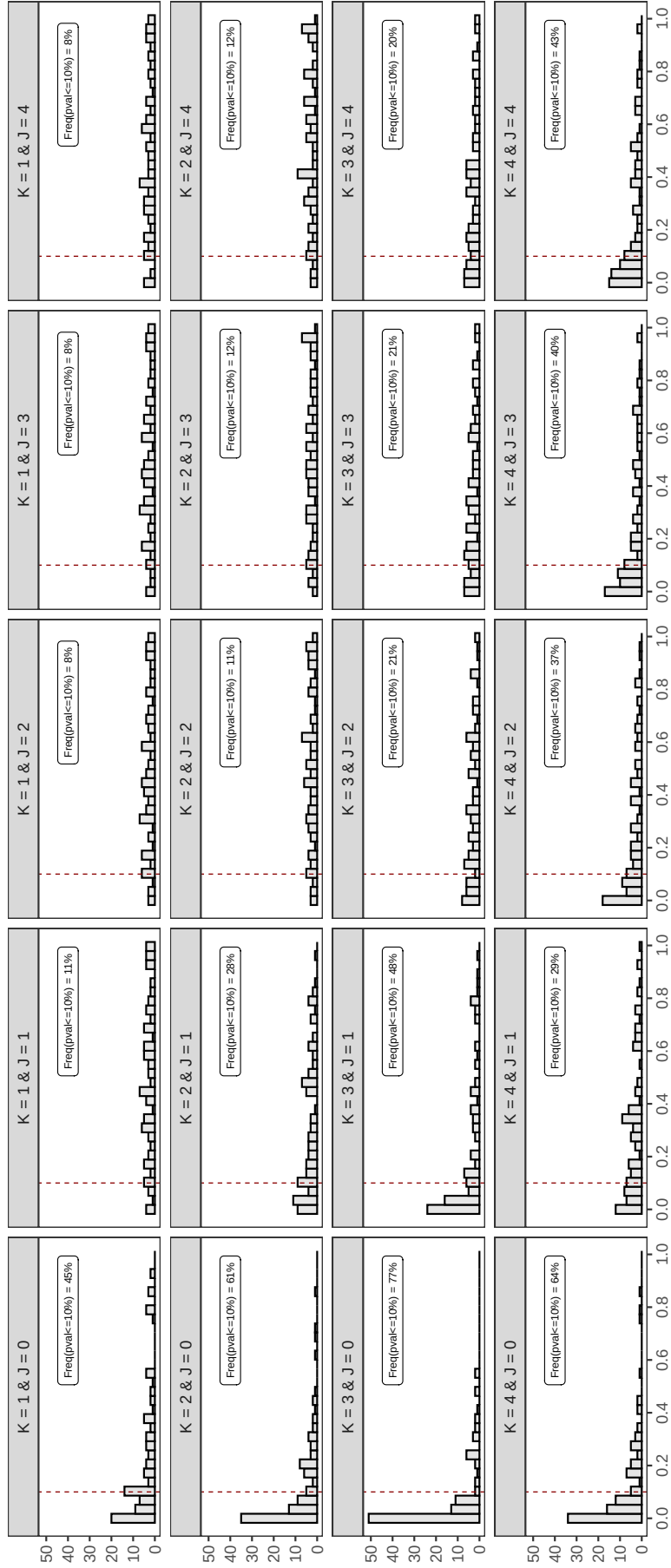


Figure B13: Histogram of GRS test p-value of the 3-by-3-by-3 ROE, P/B, and growth rate portfolios on Connor-Korajczyk factors when factor loadings are time-varying,  $K = 1, 2, 3$ , and  $4$  and  $J = 0, 1, 2, 3$ , and  $4$ .



Table B1: Static relation between (monthly percentage) expected returns and characteristics

Panel A, B, and C are based on the simulation with 100 independently generated cross sections. Each cross section has 4,000 firms with randomly generated characteristics including ROE, expected return, and PBR, where ROE and expected return are drawn from normal distributions with means and standard deviations taken from firms from 2013 to 2022 in the CRSP/Compustat merge dataset, and PBR is drawn from uniform distributions from 0 to one half of the expected return divided by ROE. We remove firms with ROE outside 0% to 60% range or (annual) expected return outside 2% to 12% range, and we further exclude firms with (annual) expected return greater than two times of its ROE. The price-to-book (P/B) is computed based on GGM and the perpetual growth rate is ROE times PBR. Panel A summarizes the univariate and multiple regressions of the expected returns on the characteristics. In panel B and C, the firms are then assigned into groups based on univariate decile sort, 2-by-2-by-3 P/B, PBR, and ROE sort, 2-by-2-by-3 P/B, ROE, and  $g$  sort, and 2-by-2-by-2-by-3 P/B, PBR, ROE, and  $g$  sort of characteristics. The return of each group is the equal weighted average return and the return for each sorting strategy is the mean return of the top groups minus the mean return of the bottom groups based on the given characteristic. Panel B and C apply independent sort and conditional sort respectively, and report the mean, minimum, and maximum of the strategy returns based on the 100 independently generated cross sections. Panel D computes the average monthly percentage returns of similar strategies based on CRSP/Compustat merge data from July 1976 to June 2023. Fama and French (2015) measure investment rate as the change of total assets divided by lagged total assets (asset growth) which is close to ROE times PBR, equal to growth rate  $g$ , in the GGM formula. We use  $g$  to represent the characteristic asset growth. The book equity is measured following Fama and French (2006).

| Panel A: Regressions of expected returns on the characteristics            |                        |      |       |                              |                  |       |                              |                  |       |                                |                       |       |      |      |
|----------------------------------------------------------------------------|------------------------|------|-------|------------------------------|------------------|-------|------------------------------|------------------|-------|--------------------------------|-----------------------|-------|------|------|
|                                                                            | Univariate             |      |       |                              | Multiple         |       |                              | Multiple         |       |                                | Multiple              |       |      |      |
|                                                                            | P/B                    | PBR  | ROE   | g                            | P/B              | PBR   | ROE                          | P/B              | ROE   | g                              | P/B                   | PBR   | ROE  | g    |
| Mean                                                                       | -0.03                  | 0.64 | 0.04  | 8.40                         | -0.04            | 0.66  | 1.34                         | -0.04            | 1.00  | 6.14                           | -0.04                 | -0.03 | 0.99 | 6.30 |
| Min                                                                        | -0.03                  | 0.57 | -0.02 | 7.77                         | -0.05            | 0.58  | 1.26                         | -0.04            | 0.94  | 5.62                           | -0.04                 | -0.11 | 0.93 | 5.87 |
| Max                                                                        | -0.03                  | 0.75 | 0.13  | 8.86                         | -0.04            | 0.75  | 1.43                         | -0.04            | 1.08  | 6.57                           | -0.04                 | 0.04  | 1.07 | 6.69 |
| Panel B: Independent sort                                                  |                        |      |       |                              |                  |       |                              |                  |       |                                |                       |       |      |      |
|                                                                            | Univariate decile sort |      |       |                              | 2-by-2-by-3 sort |       |                              | 2-by-2-by-3 sort |       |                                | 2-by-2-by-2-by-3 sort |       |      |      |
|                                                                            | P/B                    | PBR  | ROE   | g                            | P/B              | PBR   | ROE                          | P/B              | ROE   | g                              | P/B                   | PBR   | ROE  | g    |
| Mean                                                                       | -0.38                  | 0.20 | 0.02  | 0.32                         | -0.39            | 0.17  | 0.28                         | -0.40            | 0.23  | 0.19                           | -0.39                 | 0.06  | 0.26 | 0.25 |
| Min                                                                        | -0.41                  | 0.15 | -0.01 | 0.28                         | -0.42            | 0.14  | 0.26                         | -0.42            | 0.21  | 0.17                           | -0.41                 | -0.02 | 0.25 | 0.16 |
| Max                                                                        | -0.36                  | 0.23 | 0.06  | 0.36                         | -0.38            | 0.19  | 0.30                         | -0.39            | 0.24  | 0.21                           | -0.37                 | 0.11  | 0.28 | 0.30 |
| Panel C: Conditional Sort                                                  |                        |      |       |                              |                  |       |                              |                  |       |                                |                       |       |      |      |
|                                                                            |                        |      |       |                              | 2-by-2-by-3 sort |       |                              | 2-by-2-by-3 sort |       |                                | 2-by-2-by-2-by-3 sort |       |      |      |
|                                                                            |                        |      |       |                              | P/B              | PBR   | ROE                          | P/B              | ROE   | g                              | P/B                   | PBR   | ROE  | g    |
| Mean                                                                       |                        |      |       |                              | -0.32            | 0.15  | 0.23                         | -0.33            | 0.18  | 0.19                           | -0.32                 | 0.04  | 0.16 | 0.14 |
| Min                                                                        |                        |      |       |                              | -0.34            | 0.13  | 0.22                         | -0.35            | 0.17  | 0.17                           | -0.35                 | 0.03  | 0.14 | 0.12 |
| Max                                                                        |                        |      |       |                              | -0.31            | 0.17  | 0.25                         | -0.32            | 0.19  | 0.20                           | -0.27                 | 0.06  | 0.18 | 0.15 |
| Panel D: Results using CRSP/Compustat merged data (July 1963 to June 2023) |                        |      |       |                              |                  |       |                              |                  |       |                                |                       |       |      |      |
|                                                                            | Univariate decile sort |      |       | 2-by-2-by-3 independent sort |                  |       | 2-by-2-by-3 conditional sort |                  |       | 2-by-3 Size-X independent sort |                       |       |      |      |
|                                                                            | P/B                    | ROE  | g     | P/B                          | ROE              | g     | P/B                          | ROE              | g     | P/B                            | ROE                   | g     |      |      |
| Mean                                                                       | -0.27                  | 0.29 | -0.39 | -0.31                        | 0.42             | -0.14 | -0.27                        | 0.26             | -0.15 | -0.27                          | 0.30                  | -0.26 |      |      |
| SD                                                                         | 4.75                   | 4.18 | 3.46  | 3.35                         | 2.27             | 2.13  | 3.00                         | 1.90             | 2.15  | 3.02                           | 2.21                  | 2.05  |      |      |
| t-stat                                                                     | -1.53                  | 1.83 | -3.01 | -2.46                        | 4.94             | -1.82 | -2.45                        | 3.68             | -1.84 | -2.40                          | 3.65                  | -3.41 |      |      |